

MILITARY ENGINEER SERVICES
GARRISON ENGINEER EAST BAREILLY
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DESERT COOLER, VOLTAGE STABILIZER AND DEEP FREEZER AT MIL STN
SHAHJAHANPUR CANTT UNDER GE (EAST) BAREILLY**

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* These documents are not attached with the tender and can be seen in the office of GE (E) Bareilly during working hours on any working day.

(Signature of Contractor)
Dated.....

AGE (Contracts)
For Accepting Officer

Tele : 0581-2510068

Garrison Engineer East
Military Engineer Services
Shahjahanpur Road,
Bareilly Cantt.-243001

8529/ 04 /E8

23 May 2026

M/S
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**NAME OF WORK : REPAIR TO REFRIGERATOR, WINDOW/SPLIT TYPE AC, WATER COOLER,
DESERT COOLER, VOLTAGE STABILIZER AND DEEP FREEZER AT MIL STN
SHAHJAHANPUR CANTT UNDER GE (EAST) BAREILLY**

Dear Sir,

1. Tender documents in respect of above work are uploaded on the site www.defproc.gov.in. The tender is on single stage two cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.
2. Bids will be received online by ACCEPTING OFFICER up to the date and time mentioned in the **NOTICE INVITING TENDER (NIT)**. No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bonafide tender/bid.
3. Bid will be opened on line on a due date fixed for opening as per critical dates given in the portal. Cover 1 will be opened first. Tenderers may see the result of the opening of Cover 1 on the tender portal. Date of opening of Cover 2 shall be decided after technical evaluation of Cover 1 keeping in view the various eligibility criteria given in the NIT and the same will be intimated to the tenderers while uploading the technical evaluation on the tender portal.
4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents through e-mail and/or depute your technical representative for discussion on tender/drawings and to clarify doubts, if any, before the bid submission starts. You are requested not to write piece meal points and forward your points duly consolidated in one go. You may also attend the pre bid meeting on the date given in the tender.
5. Un-enlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix 'A' to **NIT** along with EARNEST MONEY DEPOSIT (EMD) and tender fee on e-procurement portal and submit the physical documents in the office of **GE (East) Bareilly** within time limit specified in **NIT**. Inadequacy / deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.
6. Contractor having not executed standing security bond and standing security deposit in any MES formation shall upload scanned copy of EARNEST MONEY DEPOSIT (EMD) mentioned in Notice of Tender and shall ensure receipt of hard copy of EMD in the office of tender issuing authority before date & time fixed for this purpose. In case of failure to abide by any of these two requirements, the finance bid will not be opened.
7. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appendix 'A' to **NIT** on e-procurement portal and submit physical documents in the office of **GE (East) Bareilly** before date & time fixed for this purpose.

Contd.....

8. The contractor must ensure that the tender/bid is uploaded on the tender portal using the authorised DSC in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder.

9. Keeping in view delays due to system failure or other communication related failures, it is suggested that tender/bid may be uploaded sufficiently in advance of the last due date and time fixed on the portal.

10. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of Minimum Fair Wages and MES SSR (Part-I and Part II) are not enclosed with these documents. These are available for perusal in the Office of GE/AGE concerned and this office. Tenderer is also requested to keep the copy of these documents with him as same are available in the market.

11. PERFORMANCE SECURITY DEPOSIT

After acceptance of the Tender, the contractor will be required to lodge with the Accepting Officer PERFORMANCE SECURITY DEPOSIT @ 5% of contract amount. The amount is required to be lodged within 28 (Twenty eight) days of the receipt by the contractor of notification of acceptance of tender/bid, failing which action as stipulated in Condition 19 of GCC shall be taken.

11. ANY TENDERER, WHICH PROPOSED ALTERATIONS TO ANY OF THE CONDITION, SPECIFICATIONS LAID DOWN IN THE TENDER DOCUMENTS OR ANY NEW CONDITION, WHATSOEVER, IS LIABLE TO BE REJECTED.

Yours faithfully,

Encls :- (As above)

AGE (Contracts)
For Accepting Officer

Signature of Contractor

Dated:.....

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER**1. EARNEST MONEY DEPOSIT (EMD)**

Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, along with their tender/bid:-

- (a) Deposit at Call Receipt from a Scheduled Bank in favour of **Garrison Engineer (East) Bareilly**.
- (b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of **Garrison Engineer (East) Bareilly**.

It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'EARNEST MONEY DEPOSIT' in any; other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE.

NOTICE: Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will not be accepted. NON-SUBMISSION OF EARNEST MONEY DEPOSIT (EMD) (scanned copy along with Technical Bid & hard copy before the date & time fixed for opening of BOQ) WILL RENDER THE BID DISQUALIFIED FOR OPENING OF COVER II (FINANCE BID).

2. **SECURITY DEPOSIT:-** Normally contractors whose names are on the MES approved list for the area in which the work lies, and within whose financial category the estimated amount would fall, may tender/bid but in case of term contracts, contractors of categories SS to E may tender/bid. In case, where the tender amount is in excess of the financial limit of the contractor and the Accepting officer decides to accept the tender/bid, in which event the successful contractor shall deliver to the Accepting Officer a performance security in any of the forms prescribed in condition 19 of IAFW 2249 for an amount equivalent to **5% of contract sum** within 28 days of receipt of the letter of Acceptance conditions of contract.

3. Blank

4. **CONTRACTORS ENLISTED IN MES FORMATIONS OTHER THAN CECC AND CWE BAREILLY:-** Any Contractor, borne on the approved list of contractors of any MES formation and who has executed a Standing Security Bond, may be permitted to tender for works under any other MES formation, without furnishing Earnest Money along with his tender. If the tender submitted by such a Contractor is proposed to be accepted, the Contractor will be required to lodge **to the Accepting Officer a performance security in any of the forms prescribed in condition 19 of IAFW 2249 for an amount equivalent to 5% of contract sum within 28 days of receipt of the letter of Acceptance conditions of contract.**

5. GENERAL INSTRUCTIONS FOR COMPLIANCE

5.1. The bids received only in the electronic form will be considered. All bids shall be submitted on 'eprocuremes.gov.in' portal, Documents should be scanned and forwarded in 'pdf' form and 'xls' form as indicated.

5.2. Bids shall be uploaded on 'eprocuremes.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like email/fax/by hand/through post will be considered.

5.3. Bid should be DIGITALLY signed using valid DSC. All pages of tender documents, corrections/alterations shall be signed/initial by the lowest bidder after acceptance.

5.4. Drawings, if issued in physical form, must be returned duly initialed by the tenderer/bidder in separate envelope indicating his name and address.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD/-)

5.5. The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initial. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

5.6. In the technical bid, a scanned copy of Power of Attorney in favour of the person uploading the bid using his/her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clause, shall be attached in 'pdf' form. In case of partnership concern or a limited company, digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Articles of Association of the Company). In all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the documents confirming of such authority shall be attached with the tender/bid in 'pdf' form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of a firm or company using his DSC shall upload with the tender/bid a scanned copy (in 'pdf' form) of power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be, in all matters pertaining to the contract including the Arbitration Clause.

5.7. Even in case of Firms or Companies which have already given power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh power of Attorney duly executed in his favour stating specifically that the said person has authority to bind such partners of the Firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in 'pdf' form with the tender/bid; unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

5.8. Hard copies of all above documents should be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

5.9. Bid (Cover 1 & 2) shall be uploaded online well in time.

5.10. The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to special condition 3 referred hereinafter and also conditions 24 & 25 of IAFW 2249 (General conditions of contract)

5.11. Tenderers/bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

5.12. The tenderer/bidder shall quote his rate on the BOQ file only. No alteration to the format will be accepted, else the bid will be disqualified and summarily rejected.

5.13. In case the tenderer/bidder has to revise/modify the rates quoted in the BOQ (excel sheet) he can do so only in the BOQ, through eprocuremes.gov.in site only before the bid closing time and date.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD/-)**6. REVOKATION/REVISION OF OFFER UPWARD/OFFERING VOLUNTARY REDUCTION, AFTER OPENING OF FINANCIAL BIDS BY LOWEST BIDDER**

In the event of lowest tenderer/bidder revoking his offer or revising his rates upward/offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the "Earnest money deposit (In case of un-enlisted contractor) **performance security deposited by him and retention money as per last RAR** shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer/bidder for depositing the amount through MRO. Bids of such Contractors/bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Govt Treasury. In addition, bids of such tenderer/bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tender/bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

7. CPM (Critical Path Method)

7.1. The project planning for work covered in the scope of tender is based on CPM.

7.2. The tenderer/bidder is expected to be fully conversant with the CPM technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literature on the subject are widely available in the market which the tenderer/bidder may make use of.

7.3. The tenderer /bidder's attention is drawn to special condition of the tender regarding preparation of the detailed network analysis and time schedule for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/bidder in using the technique will be taken as his technical inefficiency and will affect his class of enlistment and future prospect/invitation to tenders for future works.

7.4. Department may issue amendments/errata in form of **CORRIGENDUM** to tender/revised BOQ to the tender documents. The tenderer/bidder is requested to read the tender documents in conjunction with all the errata/amendments/corrigendum, if any, issued by the department.

8. These instructions shall form part of the contract documents.

AGE (Contracts)
For Accepting Officer

Signature of Contractor

Dated:.....

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

1. A tender is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT)
2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work cost more or less, a tenderer/bidder will have no claim on that account. The tender shall be based on as mentioned in aforesaid Appendix 'A'.
3. The work is to be completed within the period as indicated in aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of Acceptance of tender.
4. Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and unenlisted contractors may submit tender/bid subject to other criteria mentioned in Appendix A. However in case of term contracts, enlisted contractors of Class SS to E may submit tender. Not more than one tender shall be submitted/uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/director is common among both of them. A breach of this condition will render the tenders/bids of both the parties liable for rejection.
5. The Office of **Garrison Engineer (East) Bareilly** will be the **Accepting Officer** here in after referred to as such for purpose of the contract.
6. Not more than one tender/bid shall be submitted/uploaded by one bidder firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/bid for the same tender as separate competitors. A breach of this condition will render the tenders/bids of both the parties liable for rejection.
7. The Technical Bid and Financial Bid (Cover-I and Cover-2) shall be uploaded by the tenderer/bidder on or before the date and time mentioned in NIT. A scanned copy of DD with enlistment details/documents shall be uploaded as packet 1/cover-1 ('T' bid) of the tender/bid on e-tendering portal. DD is refundable in case T bid is not accepted resulting in non-opening of 'Q' bid. The applicant contractor shall bear the cost of bank charges for procuring and encasing the DD and shall not have any claim from Government whatsoever on this account.
 - 7.1 Tender form and conditions of contract and other necessary documents shall be available on defproc.gov.in/ eprocure.gov.in site for download and shall form part of contract agreement in case the tender/bid accepted.
 - 7.2 In case of contractor who has not executed the Standing Security Bond, the Cover-I shall be accompanied with by **Earnest Money** of amount as mentioned in Appendix 'A' in the form of deposit at call receipt in favour of **GE (East) Bareilly** by a scheduled bank or in receipted treasury Challan the amount being credited to the revenue deposit of the **GE (East) Bareilly**.
 - 7.3 A Contractor who is not enlisted for the area in which the work lies by whose name is in the MES approved list of any MES formation and who has deposited standing security and executed standing security Bond, may bid without depositing earnest money along with the tender, but if the Accepting officer accepts the tender/bid, the contractor shall deliver to the Accepting Officer a performance security in any of the forms prescribed in condition 19 of IAFW 2249 for an amount equivalent to **5% of contract sum** within 28 days of receipt of the letter of Acceptance conditions of contract.
 - 7.4 The **GE (East) Bareilly** will return the Earnest Money wherever applicable to all unsuccessful tenderers/ bidders by endorsing an authority on the deposit-at-call receipt for its refund, on production by the tenderer, bidder a certificate of the Accepting Officer that a bonafide tender/bid was received and all documents were returned.
 - 7.5 The **GE (East) Bareilly** will either return the Earnest money to the successful tenderer/ bidder by endorsing an authority on the deposit-at-call receipt for its refund on receipt of an appropriate amount of Performance Security or will retain the same in part or full on account of security deposit if such a transaction is feasible.

MILITARY ENGINEER SERVICES
NOTICE OF TENDER (Contd....)

7.6 Copies of the drawings and other document pertaining to the work signed for the purpose of identification by the Accepting Officer or his accredited representative, sample of materials and stores to be supplied by the contractor will also be available for inspection by the tenderer/bidder at the office of Accepting Officer and **GE (East) Bareilly**.

8. The tenderers/ bidders are advised to visit the site of work by making prior appointment with **GE (East) Bareilly** who is also executing Agency of the work. The tenderers/ bidders are deemed to have full knowledge of all relevant documents, samples, site etc., whether they have inspected them or not.

9. Any tender/bid which proposes any alteration to any of the conditions laid down or which proposes any other condition or prescription whatsoever, is liable to be rejected.

10. The uploading of bid implies that bidders has read this notice and the Conditions of Contract and has made himself aware of the scope and specification of work to be done and the conditions and rates at which stores, tools and plants etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

11. Tenderers/ bidders must be in possession of a copy of the MES Standard Schedule of Rates including amendments and errata thereto.

12. Invitation for e-tender does not constitute any guarantee for validation of 'T' bid and subsequent opening of finance bid of any applicant/bidder, even of enlisted contractors of appropriate class, merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the 'T' bid and not open the finance bid of any applicant/bidder. 'T' bid validation shall be decided by the Accepting Officer based on, inter alia, capability of the firm as per criteria given in Appx 'A' to this NIT. The applicant contractor/ bidder will be informed regarding non-validation of his 'T' bid assigning reasons thereof through the defproc website. The applicant contractor/bidder if he so desires may appeal to the next higher Engineer authority vix HQ Chief Engineer Bareilly Zone on email id **brlylz3-mes@gov.in** with copy of the Accepting Officer on email before the scheduled date of opening of Finance Bid. The decision of the Next Higher Engineer Authority (NHEA) shall be final and binding. The contractor/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

13. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking, giving a price preference over other Tender (s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible from such tenderer/bidder whose tender/bid is rejected.

14. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

15. This **Notice Inviting Tender (NIT)** including Appendix 'A' shall form part of the contract.

Signature of contractor
Dated :

AGE (Contracts)
for Accepting Officer

NOTICE INVITING TENDER (NIT) (CONTD/-)

1.	Name of Work	REPAIR TO REFRIGERATOR, WINDOW/SPLIT TYPE AC, WATER COOLER, DESERT COOLER, VOLTAGE STABILIZER AND DEEP FREEZER AT MIL STN SHAHJAHANPUR CANTT UNDER GE (EAST) BAREILLY																
2.	Estimated Cost	Rs. 13,50,000.00																
3.	Period of Completion	180 Days																
4.	Cost of tender documents	Rs. 500.00 in the form of DD/ Bankers Cheque from any Schedule Bank in favour of GE (East) Bareilly and payable at Bareilly (Note: in case of retendering, the contractor who had quoted in the previous call is not required to submit the cost of tender)																
5.	Website/ portal address	www.defproc.gov.in																
6.	Type of contract	Tender Shall be based on drawing and specifications (IAFW-2159)/IAFW-1779A and GCC (IAFW-2249) with Schedule 'A' (List of items of work) to be priced by contractor/pre-priced by MES. The contractor is required to quote the lump sum amounts for parts of Schedule 'A' and quote rates against items of other parts of Schedule 'A'.																
7.	Timeline details:	Please refer www.defproc.gov.in																
	(a) Published date																	
	(b) Bid submission start date																	
	(c) Bid submission end date																	
	(d) Date of bid opening																	
8.	Eligibility Criteria																	
	(A) For MES enlisted contractors	Contractor shall be enlisted with MES in Class 'D' and category c (i) subject to satisfactory remarks wrt performance in respect of wks in hand as reflected in work load return (WLR) or any other report circulated by competent engineer authority.																
	(B) For contractors not enlisted with MES	(i) Contractor not enlisted with MES should meet the enlistment criteria of 'D' Class & c (i) category contractor with regard to satisfactorily completion of requisite value works with Central/State Government/Central/State PSUs/AWHO/AFNHB/CGEWHO/DGMAP, annual turnover, bank solvency, working capital and other requirements given in Para 1.4 & 1.5 of section 1 of MES Manual of Contracts 2020 as available in all MES formations as well as MES website (www.defproc.in). (ii) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (iii) Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government Department of any Central/State Government PSU or any autonomous Body under Central /State Government or any Local Body as on the bid submission end date. (iv) Details of works completed and under progress in MES be submitted in the following format:- <table><tr><td>Srl No.</td><td>CA NO. & Name of Work</td><td>Value of CA</td><td>Date of Commencement</td><td>Date of Completion</td><td>Extended Date of Completion</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>					Srl No.	CA NO. & Name of Work	Value of CA	Date of Commencement	Date of Completion	Extended Date of Completion						
Srl No.	CA NO. & Name of Work	Value of CA	Date of Commencement	Date of Completion	Extended Date of Completion													

Contd..

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

		(v) Un-enlisted Contractor who have secured two works in MES should get themselves registered in the appropriate designated Class with any Registering Authority, else the firm will not be eligible for participation in the tender unless until the firm is enlisted with the MES.
9.	Tender issuing and Accepting Officer	Name : Lt Col Deepak Singh Bohra Address: Garrison Engineer East Bareilly Military Engineer Services Shahjahanpur Road, Bareilly Cantt. Contact Details:- Phone No. 0581- 2970509 email id- geeastbarelly@gmail.com
10.	Executing agency	GE (East) Bareilly
11.	Earnest Money	Rs 27,000/- in favour of GE (East) Bareilly in the form of Deposit at call receipt, FDR not acceptable.
12.	Performance Security and Defect Liability Period	Within 28 days of acceptance of tender contractor shall deliver to the Accepting Officer a Performance Security @ 5% of the contract sum in the form of Bank Guarantee/Govt Securities /FDR or any other Govt instrument's as per condition 19 of IAFW-2249 and Defect Liability Period of the work is 01 (One) years from the actual date of completion in terms of condition 46 (b) of IAFW-2249.

Notes:

1. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT is less than 7 (Seven), applications in respect of MES contractors of one class or two classes (In case of remote and difficult areas to be decided as per list circulated by CE commend / ADG) below the eligible class shall also be considered subject to fulfillment of other eligibility criteria given in the NIT. Therefore, MES contractors one class below (Two classes below in case of remote and difficult area) may also bid for this tender. Such contractors (contractors of one / two classes below the eligible class) shall not be considered in case their present residual work in hand is more than FIVE TIME their present tendering limit. However, in case such contractors fulfill the criteria of upgradation to the stipulated eligible class based on past experience of completed works (individual work experience and/ or average annual turnover, as applicable) and financial soundness (solvency/ financial soundness and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover-I bid details related to residual work in hand like details of works in hand showing names of work, names of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfill the criteria of upgradation shall also upload the requisite information/ documents in support of upgradation. These details shall be verified by the tender issuing authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening of Cover 1, the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7 (Seven) or more, applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of up gradation in respect of past experience of completed works (Individual work experience and/or average annual turnover as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in Manual on Contracts. Therefore, such contractors shall upload the requisite information/documents in the Cover-1.

3. Un-enlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign national / Indian nationals staying abroad / Indian national having taken foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities.

4. Contractor's enlisted with MES will upload following documents in Cover 1 for checking eligibility: -

- (a) Application for tender on Firm's Letterhead.
- (b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.

NOTICE INVITING TENDER (NIT) (CONTD/-)

- (c) Scanned copy of DD / Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.
 - (d) Any other documents required as described in this Appendix.
5. Contractors not enlisted with MES will be required to upload following documents in Cover 1 for checking eligibility.
- (a). Application for tender on Firm's letterhead.
 - (b). Scanned copy of DD/ Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.
 - (c). Copy of Police Verification Report/Police Clearance Certificate/ Character Certificate from the Police Authority of the area where the registered office of the firm is located/notarized copy of valid passport of Proprietor/each Partner/each Director.
 - (d). All documents required for enlistment in MES for the class mentioned in Para 8(b) Above and as per Para 1.5 of Section 1 of MES Manual on Contract 2020.
 - (e). Details of works being executed in MES, if any.
 - (f). Any other documents required as described in this Appendix.
6. Tenders not accompanied by scanned copies of requisite DD/Bankers Cheque towards cost of tender and earnest money (as applicable) in Cover I shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.
7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach in the office of Accepting Officer **within 07 days** from the date of bid submission end date, failing which following action will be taken against the defaulter bidders: -
- (a) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD / Bankers Cheque towards cost of tenderer have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non-submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of **six months commencing from the date of opening of Financial Bid** (Cover 2).
 - (b) In case of tenders from un-enlisted contractor, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover 1 but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened. Name of such contractors along with complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of Financial bid (Cover 2).
 - (c) In case of tenders from enlisted and un-enlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover I but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover 2).
8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However, a contractor can execute the work through power of attorney to sons / daughters / spouse of Proprietor/Partner/Director and firm's own employees, director, project manager **provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director**.
9. After opening of Cover I and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same.

NOTICE INVITING TENDER (NIT) (CONTD/-)

10. Invitation for e-tender does not constitute any guarantee for validation of technical bid and subsequent opening of financial bid of any applicant /bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the technical bid and not to open the financial bid of any applicant /bidder, technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given this Appendix. Tenderer/bidder will be informed regarding non-validation of his technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher Engineer Authority (NHEA) viz HQ Commander Works Engineer on email id brlyl3-mes@gov.in with copy of the Accepting Officer on email before the scheduled date of opening of Finance Bid (Cover-2). NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the Next Higher Engineer Authority (NHEA) shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

11. In case an un-enlisted contractor is already executing works in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES Class of contractor for which it is eligible. For this purpose, details of the works being executed by such a contractor shall be uploaded in the Cover-I of the bid and shall be checked/ verified by the Accepting Officer.

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (i.e he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

13. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover 2 shall be considered as a willful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an un-enlisted tenderer. Earnest Money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the Standing Security Bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary / administrative action shall be taken against such tenderers. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be restored to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related, if Proprietor / one or more Partners / Directors are common. Decision of Accepting Officer on issue /deny the tender to any one of the related firms shall be final and binding.

15. Court of place from where tender has been issued shall alone have jurisdiction to decide any dispute out of or in respect of this tender. After acceptance of tender, "Condition 72- Jurisdiction of Courts" IAFW-2249 shall be applicable.

Signature of Contractor

AGE (Contracts)
For Accepting Officer

File No. 8529/ /E8

Dated May 2026

Garrison Engineer East
Military Engineer Services
Shahjahanpur Road,
Bareilly Cantt.-243001

IN LIEU OF IAFW-1799 & 1799-A

(To be used in conjunction with General Conditions of Contracts)
(IAFW-2249 - 1989 PRINT)

Tele : 0581-2510068

Garrison Engineer East
Military Engineer Services
Shahjahanpur Road,
Bareilly Cantt.-243001

8529/ 04 /E8

23 May 2026

**NAME OF WORK : REPAIR TO REFRIGERATOR, WINDOW/SPLIT TYPE AC, WATER COOLER,
DESERT COOLER, VOLTAGE STABILIZER AND DEEP FREEZER AT MIL STN
SHAHJAHANPUR CANTT UNDER GE (EAST) BAREILLY**

The technical bid of subject tender is to be delivered through E- tender on our website www.defproc.gov.in for inviting electronic bids through NIC e- procurement solution only. E-tender will be will be received on date as given in website through and all connecting documents such as request application alongwith cost of tender will be addressed to Garrison Engineer (East) Bareilly and marked **GE (E) / BLY - OF 2026-27** on the subject of application

All required documents must be reached before bid opening date and time physically, otherwise technical bid as well as financial bid shall not be opened and no claim whatsoever on this account will be tenable by the department.

Any correspondence concerning this tender should be addressed as indicated at top of this sheet, quoting the reference.

**THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO
ACCEPT THE LOWEST OR ANY OTHER TENDER**

Signature of contractor
Dated :

Signature of Officer
Issuing the tender documents.
Appointment: AGE (Contracts)
For accepting Officer

SCHEDULE "A" NOTES(contd....)**NOTES: -**

1. **SCOPE OF WORK** : This contract comprises the following Job/work :-
 - (i) **Maint Porgramme - 2026-27**
2. The quantities given under column 3 are Approximate and are inserted only as guidance. While executing work, GE shall personally ensure that quantities of respective item as mentioned under col 3 of BOQ of tender shall not be varied beyond the limits laid down in condition 7 of IAFW-2249 (General Conditions of Contracts).
3. Unit Rates:--
 - (a) The contractor shall calculate his own details and insert unit rates **in column 5** hereinafter and extend **the amount in column 7** against each items of Schedule "A". The aforesaid unit rates shall be based on description of items in Schedule 'A', drawings (where mentioned in Schedule 'A' or specifications), specifications and other general specifications attached herewith. It is an express condition of the contract that the rates quoted by the Contractor for various items in Schedule 'A' shall be inclusive of all labour and material required for completion of work specified hereinafter in particular specifications/special conditions/requirements detailed in particular specifications and full and final completion thereto, irrespective of whether these are mentioned in the description of Schedule 'A' or not and the contractor will not be paid any extra for the same. The Govt. shall I not entertain any claim whatsoever on account of inaccuracies, misunderstanding, if any, in the aforesaid unit rates.
 - (b) Tendered rates shall be deemed to include for the provision of all material , processes, operations and special requirements detailed in the particular specification irrespective of whether these appear as specific items or not and the contractor will not be paid extra for the same.
 - (c) Unless specifically mentioned otherwise such as labour only/fixing only, the unit rates for all items of works shall be deemed to be including cost of all materials, labour, tools and plants, testing and commissioning work in position complete as required for entire completion of work.
 - (d) Special conditions given in SSR Part-II 2020 and preambles to items given in MES Schedule Part-II under respective trades/items shall be applicable. If any provisions in the description of Schedule 'A' and/or in particular specifications is/are at variance with the provisions of special conditions in MES SSR Part II 2020 and preambles to SSR items, the provisions in description of Schedule 'A' items and provisions in particular specifications shall taken precedence there over.
 - (e) Method of measurements for all items listed in Schedule 'A' shall be as given in MES Standard Schedule of Rates Part II 2020 unless specifically stated otherwise here-in-after for any particular item(s) of work.
 - (f) All materials as required for incorporation in work should be of reputed makes as given in these documents. In case of makes not mentioned in the contract agreement the item shall be ISI marked as approved by GE.
4. Unit rates against each item of schedule 'A' shall be quoted in words as well as in figures in column No 6 and extended the amount in column 7 by contractor and the total shall be carried over to general summary.
5. The amount in column 7 is not firm but will be treated as the "CONTRACT SUM" referred in IAFW-2249.
6. **PERIOD OF COMPLETION:-** The entire work under the contract shall be completed within **180 Days** from the date of handing over of the site as per WO No 1.

(contd....)

SCHEDULE "A" NOTES

7. Site for execution of entire work will be made available on the date of commencement of work as per W.O. No. 01. In case, however, it is not possible for the department to make the entire site available on the date of commencement of work, the contractor will have to arrange his working programme accordingly. No claim whatsoever, for not handing over the entire site on the date of commencement of the work & handing over balance site for execution gradually will be tenable.
8. The unit RM, Cum or CUM, Sqm or SQM, Per 10 RM, Qtl and Per 10 Sqm wherever mentioned in the tender documents denotes the unit running meter, cubic meter, square meter, per ten running metre, per quintal and per ten square metre respectively.
9. **DISMANTLING/DEMOLITION/TAKING DOWN**
- 9.1. An inventory of all the items to be demolished/ dismantled/ taken down / shall be prepared by the Engineer-in-charge before handing over of site to the contractor for demolition/ dismantling of items/materials/pipe lines etc indicating all the fittings / fixtures / assets etc and shall obtained receipt of contractor on ibid inventory in token of his acceptance.
- 9.2. The contractor shall take down/demolish/dismantle the work to the extent all as mentioned in Schedule 'A' and as specifically ordered by GE.
- 9.3. Dismantled/Demolished material indicated in schedule of credit shall become the property of the contractor and contractor shall give credit for it at the rates mentioned in Schedule of credit. The rates given for credit are fixed and no variation on this amount shall be admissible on account of quality /conditions of the materials obtained from dismantling/demolition. The contractor at his own expense shall carry out transportation of old materials. Materials retrieved from dismantling/demolition shall not be used in the new work.
- 9.4. All retrieved materials obtained from demolition/ dismantling which are not mentioned in schedule of credit shall become Govt. property and contractor shall deposit the same in MES store yard of GE. Such materials shall be stacked properly by contractor at his own arrangements and own cost at MES store all as directed and the same shall be recorded in MB for record purpose only duly signed by GE, Engineer-in-Charge and contractor and necessary RV thereof shall be obtained by the contractor from Engineer-in Charge.
- 9.5. Contractor is advised to 'visit' the site with Engineer-in-Charge/ GE before quoting the lump sum prices against credit Sch .
- 9.6. The Govt does not guarantee for the value of the materials that will naturally accrue to the contractor. The contractor shall be deemed to have visited the site of work and have made himself conversant with present state of materials before tendering for the purpose of assessing their quality, condition and money value of materials that would be obtained from dismantling/demolition etc. and no extra cost whatsoever shall be entertained on this account.
- 9.7. Refer condition 10,47 and 48 of IAFW-2249, General Conditions of Contracts and Para 21.1 to 21.3 stipulated on page 21-1 to 21-3 of MES SSR' 2009 Part I, . All materials shall be sorted out, stacked at site for joint measurements and recorded in the measurement book. Both parties will sign entries. Govt property shall be removed and handed over to Engineer-in-Charge through proper receipt in proper manner as described above at the MES store yard. Site shall be cleaned off rubbish/waste materials etc. to the entire satisfaction of Engineer-in-Charge.
- 9.8. Any damages caused to the existing structure and fitting/fixtures, due to the negligence and carelessness of the contractor or his workmen, shall be made good by the contractor at his own expense to the entire satisfaction of Engineer-in-Charge. Extent of materials/items damaged due to negligence and carelessness of the contractor or his workman due to bad workmanship will be assessed by the GE, whose assessment in writing shall be final and binding on the contractor and in case of contractor's failure in making good the damages inspite of GE's notice, cost thereof assessed by GE shall be recovered from the contractor's payment due to him and decision of GE in this respect shall be final & binding.

(contd....)

SCHEDULE "A" NOTES

- 9.9. The contractor shall be responsible for the safe custody of materials until handed over to MES which are Govt. property or are to be incorporated in the work and receipt obtained for the same.
- 9.10 For the purpose of calculating compensation in terms of condition 50 of IAFW-2249, deductions at source towards GST/Income tax etc. the amount of Credit for Old materials becoming contractor's property shall not be taken into account.
10. The work under this contract lies in **UNRESTRICTED AREA**.
11. The rates quoted by the Contractor shall include all taxes/duties i.e. sales tax, Sales Tax on works contracts, Excise duty, Custom duty, Octroi duty, cess, labour cess, Service tax and any other taxes/duties as applicable. Form 3D, 31 & 32 or any other form shall not be issued by MES. This is an express condition of this contract that no claim of the contractor shall be entertained by MES on this account.
12. With the implementation of GST (Goods and Service Tax), the effect of the same shall be deemed to be included in the quoted rates and no reimbursement/refund on this account shall be admissible.
13. M & L', 'S & F' and 'S&F in repairs' terms wherever occurs on tender documents, shall mean 'Material and labour', 'Supply and fixing' and 'Supply & fixing of new one item after demolition / dismantling of old unserviceable' respectively.
14. VISIT OF SITE: The contractor shall visit the site and ascertain the nature/ conditions/ locations of the works before quoting the rates. Whether or not visited the site, contractor's quoted rates shall be deemed to consider that he has visited the site.
15. In case of any deviation, mode of pricing shall be decided by the Accepting Officer in terms of Condition 62 of IAFW-2249.
- 15.1 In the event of a deviation order involving fixation of special (Star) Rate, Draft Rate shall be prepared by GE (within a maximum period of 30 days) while initiating the proposal for deviation seeking approval of the Accepting Officer and notified to the contractor. While notifying the Draft Rate, it will be clearly stipulated that the same is merely an estimated rate and firm rate shall be fixed based on actual and receipt of supporting documents from the contractor such as purchase vouchers, literature of product and test certificates etc (as applicable) on completion of the work involving Star Rate. Any objection to the method of fixing Star Rate will be dealt as per Condition 7 of IAFW-2249.
- 15.2 Draft Star Rate shall be made based on market enquiry through telephonic enquiry/quotations / email / rate lists / internet based sources, material & labour constants available in various 'Civil Engineering Books' and record available in respect of 'Star Rates' approved in the past for similar items of work etc. The contractor may also assist GE's Office in preparation of 'Draft Star Rate'.
- 15.3 The 'Draft Star Rate' shall be purely a draft rate and shall not be used for claiming final payment during execution of work. However, GE shall allow part payment to the tune of 80% during execution to avoid any financial hardship to the contractor.
- 15.4 After completion of the item of work involving Star Rate, the contractor shall submit the purchase vouchers, literature of products & test certificates (as applicable, decision of GE being final in case of any disagreement) for finalization of 'Star Rate'. The 'Star Rate' shall be technically checked by AGE (C) / ACWE (C) / DCWE(C) / Director (C) (as applicable) depending upon the financial effect & approval by the competent authority within a period of one month from submission of the relevant documents by contractor as mentioned above.
- 15.5 The 'Star Rate' as approved by the competent authority after technical check by AGE (C) / ACWE (C) / DCWE(C) / Director (C) (as applicable) depending upon the financial effect shall be referred as "the rate decided by GE" under Para 62(G) of IAFW-2249.

(contd....)

SCHEDULE "A" NOTES

16. In case of no makes specified in the tender documents for a particular item, BIS marked / approved items from BIS approved manufacturers shall be incorporated / provided at site. No extra payment shall be paid to the contractor on this account.

17. **Defect Liability period shall be of "Twelve calendar months" after the works handed over to Government.**

18. Prior to commencement of work contractor shall undertake to either obtain valid Electrical License of required class in its name or engage an agency having valid Electrical License of required class through MoU (Memorandum of Understanding) for execution of all electrical works as well as works under category (c) [including its sub categories (i) to (vii)] and the category (d)(v) to the extent Rule 29 under Part III of Central Electricity Authority (Measures Relating to Safety and Electric Supply) Regulations, 2010 is applicable either in full or part of the scope of work of contract awarded to him and that all such electrical works shall be executed by him through such agency only (either self or other agency) having valid Electrical License of required class from State /Union Territory Authorities where works are to be executed.

Signature of contractor
Dated

AGE (Contracts)
For Accepting Officer

SCHEDULE OF CREDIT**SCHEDULE OF CREDIT FOR OLD DEMOLISHED/DISMANTLED/TAKING DOWN MATERIALS**

Ser No	Description of old unserviceable materials	Unit	Approx quantities	Rate per Unit (Rs)	Amount (Rs)
1	Old unserviceable compressor of any description	Each	64	250.00	16000.00
2	Old unserviceable over load protector any of any description	Each	24	5.00	120.00
3	Old unserviceable relay assay of any description	Each	22	5.00	110.00
4	Old unserviceable Switch control knob of any description	Each	8	5.00	40.00
5	Old unserviceable Metal switch unit of of any description	Each	20	30.00	600.00
6	Old unserviceable Freezer door of any description	Each	14	20.00	280.00
7	Old unserviceable Gasket of any description	Each	8	10.00	80.00
8	Old unserviceable Thermostat Switch for AC	Each	15	10.00	150.00
9	Old unserviceable Thermostat Switch for Refrigerator	Each	10	10.00	100.00
10	Old unserviceable body of Refrigeration	Each Job	7	5.00	35.00
11	Old unserviceable Broken legs of Refrigeration	Each	15	5.00	75.00
12	Old unserviceable Condenser for water cooler	Each	8	1.00	8.00
13	Old unserviceable thermostat for water cooler	Each	9	10.00	90.00
14	Old unserviceable body of water cooler	Each	11	20.00	220.00
15	Old unserviceable fan blade of AC/Water cooler/Deep freezer	Each	27	5.00	135.00
16	Old unserviceable tab for water cooler	Each	18	5.00	90.00
17	Old unserviceable water lifting pump	Each	150	2.00	300.00
18	Old unserviceable capacitor Dry type 4 MFD	Each	110	0.50	55.00
19	Old unserviceable fan blade for Desert cooler	Each	55	1.00	55.00
20	Old unserviceable cooler parts	Each	20	750.00	15000.00
Total					33,543.00

Note:-

1. Material obtained from dismantling/demolished of any items not included in schedule of credit shall be the property of government and shall be returned to store yard of AGE.
2. Tenderers are advised to inspect the old material / item at site and ascertain their condition. It shall be deemed that the Contractor has made himself well aware and ascertained the condition of these items, whether he inspects them or not. No claim whatsoever on this account shall be admissible by the Government.
3. The items as indicated in this schedule shall be the property of the Contractor and shall not be used in new work. The total amount of credit shall be deducted from total of Schedule-'A' to arrive at Contract Sum
4. Amount of credit shall be recovered from the Contractor's RARs and then only retrieved material under this schedule will be allowed to be removed from site / store by the Contractor. Serviceable material / items other than shown here-in-below obtained from demolition /taking down if any, shall remain the property of the Government and the Contractor shall deposit the same with the Engineer-in-Charge as directed and debris / unserviceable material shall be disposed off the site as per his direction without any extra cost to the Government.
5. The Accepting Officer shall reserve the right to retain any or the full quantity of demolished /dismantled materials which in his opinion is serviceable. However, the recovery shall be effected only for the quantities issued to the Contractor under this schedule.
6. The unit rate for recovery under column No 5 is fixed by department. The tenderer shall not change the same at any account. The final amount of the recovery shall be worked out based on the actual quantities obtained from dismantling demolition and the rate indicated under column No 5.
7. For measurement purposes quantities of Credit items shall be measured same as per demolition / dismantling / taking down of concerned items of sch 'A' whatsoever obtained at site

(Signature of Contractor)

Dated : _____

AGE (Contracts)

for Accepting Officer

SCHEDULE 'B'**LIST OF MATERIALS TO BE ISSUED BY THE DEPARTMENT**
(SEE CONDITION 10 OF IAFW-2249)

Srl No	Description of item	Unit	Rate	Place of issue by name	Remarks
1	2	3	4	5	6

----- NIL -----

SCHEDULE 'C'
LIST OF TOOLS AND PLANTS (OTHER THAN TRANSPORT) WHICH
WILL BE HIRED TO THE CONTRACTOR
(SEE CONDITION 15, 34 & 35 OF IAFW-2249)

Srl. No.	Particulars	Details of MES Crew supplied	Hire Charges per unit per day	Stand by charges per unit per off day	Place of issue by name	Remarks
1	2	3	4	5	6	7

----- NIL -----

SCHEDULE 'D'
(TRANSPORT TO BE HIRED TO THE CONTRACTOR)
(SEE CONDITION 16 AND 35 OF IAFW- 2249)

Srl No.	Quantity	Particulars	Rate per unit per working day	Place of issued by name	Remarks
1	2	3	4	5	6

----- NIL -----

(Signature of Contractor)

Dated : _____

AGE (Contracts)
for Accepting Officer

TENDER

TO

THE PRESIDENT OF INDIA

Having examined and perused the following documents:-

1. Specifications signed by AGE (Contracts)
2. Drawings detailed in the particular specifications.
3. Schedule 'A', 'B', 'C', 'D' attached thereto.
4. MES Standard Schedule of rates Part I (Specifications)-2009 and Part II (Rates)-2020 (here-in-after and in IAFW-2249 referred to as the "MES Schedule".), together with errata/ amendments Nos. 1 to 3 for SSR Part-I (2009) and 1 to 122 for SSR Part II.
5. **General conditions of contracts, I.A.F.W.-2249 (1989 Print) together with errata 1 to 20 and amendments 1 to 49.**
6. **Water**-condition 31 of I.A.F.W.-2249 General conditions of contracts.
Water will be supplied by MES and the same shall be paid by the contractor @ Rs. 3.75 per Rs. 1000.00 worth of work done priced at contract rates.
7. Should this tender be accepted IWE AGREE:-

* (a) That the sum of Rs.....(Rupees.....only) forwarded as earnest money shall either be retained as part of the security deposits or be refunded by the Government on receipt of the appropriate amount of security deposit all as per condition 19 of IAFW-2249.

(b) To execute all the works referred to in the said documents upon the terms and conditions contained/referred to therein and as at the item rates contained in the aforesaid schedule 'A' or at such other rates as may be fixed under the provision of condition 62 of IAFW-2249 and to carry out such deviations as may be ordered vide conditions 7 of IAFW-2249 up to maximum **(±) 10% (TEN PERCENT)** and further agree to refer all disputes as required by condition 70 of IAFW-2249 to the Sole Arbitrator officer of MES not below the rank of Director/ Col/ Director (Contracts) (having done at least one executive tenure of minimum two years or having at least two years experience of handling contracts matters) to be appointed by Chief Engineer Central Command or in his absence the officer officiating as Chief Engineer Central Command if powers specifically delegated in writing by Chief Engineer Central Command subject to receipt of written consent from both the parties in terms of the provision under section 12 (5) of the ACA, whose decision shall be final, conclusive and binding.

* To be deleted where not applicable.

TENDER (CONTD....)

Signature.....in the capacity of Name of
signatory.....duly authorized to sign the tender for and on behalf of
M/S.....(IN BLOCK LETTERS)
Dated.....

Postal Address.....
Witness.....
M/S..... Telephone No.....
Address..... Telegraphic Address.....
.....
.....
Telephone No.....

ACCEPTANCE

.....Alterations have been made to the documents and as evidence that these
alterations were made before execution of the Contract Agreement; they have been initialed by the
contractor and Shri Anoop Kumar, AGE (Contracts). The said officer is hereby authorised to sign and
initial on my behalf the documents forming part of this contract.

The above tender was accepted by me on the.....day of.....**2026** on behalf of the
president of India at the item rates contained for the contract sum of Rs.....
(Rupees.....
.....only)

Signed on.....Day of..... **2026**.

Appointment: Garrison Engineer (East)
Bareilly
ACCEPTING OFFICER

SCHEDULE OF MINIMUM WAGES

It is, hereby, agreed that the "Schedule of Minimum wages" as published by Government of India/State Govt./Govt. Local authorities whichever is highest and which specifies minimum rates of wages for various categories of workmen as applicable on the last due date of receipt of this tender shall form part of these tender documents.

My/Our signature here under amounts to my/our having signed the aforesaid documents forming part of this tender.

(Signature of Contractor)
Dated.....

AGE (Contracts)
For Accepting Officer

GENERAL CONDITIONS OF CONTRACTS (IAFW-2249: PRINT - 1989)

FOR

*** LUMP SUM CONTRACTS (IAFW-2159) & *MEASUREMENT CONTRACTS
(IAFW 1779 & 1779A)**

A copy of the MES GENERAL CONDITIONS OF CONTRACTS (IAFW-2249: Print-1989) with errata and amendments has been supplied to me/us and is in my/our possession. I/We have read and understood the Provisions contained in the aforesaid GENERAL CONDITIONS OF CONTRACTS before submission of this tender and I/We agree that I/We shall abide by the terms and conditions thereof.

It is hereby further agreed and declared by me/us, that the MES General Conditions of Contracts (IAFW-2249: Print - 1989) including Condition 70 thereof pertaining to settlement of disputes by Arbitration containing 33 pages (Serial page Nos. 35 to 67) with errata 1 to 20 and amendment Nos. 1 to 49 containing 25 pages (Serial page Nos. 68 to 92) shall form part of these tender documents.

* DELETE WHICHEVER IS NOT APPLICABLE

AGE (Contracts)
For Accepting Officer

(Signature of Contractor)
Dated.....

SPECIAL CONDITIONS

1. **GENERAL:-** The following Special Conditions shall be read in conjunction with the General Conditions of Contracts, IAFW-2249 and IAFW-1779-A including Errata/amendments thereto. If any provision in these Special Conditions is at variance with that of the aforesaid documents, the former shall be deemed to take precedence there over. The term 'General specification' used in any of the documents forming part of the contract refers to the specification contained in the MES Standard Schedule of Rates.

2. **EMPLOYMENT OF PERSONNEL :-**

2.1. Contractor shall employ only Indian Nationals as his representatives, servants and workmen after verifying their antecedents and loyalty. He shall ensure that no person of doubtful antecedents and nationality is in any way, associated with the work. If for the reasons of technical collaboration or other considerations the employment of foreign national(s) is unavoidable, the contractor shall furnish full particulars to these effects to the Accepting officer at the time of submission of the tender. As a proof that the contractor has employed only Indian Nationals, he shall render a certificate to **GE** within one month from the date of acceptance of tender to this effect. In case the **GE** desires contractor will have the police verification done of personnel employed by him.

Verification of antecedents of Contractor's representatives/ labour deployed at site in connection with execution of work under the contract, as per security requirement of user unit/ installation shall be the responsibility of the contractor and all expenses in connection with verification of antecedents by Police Authority/ Security Agency shall be borne by the Contractor.

2.2. The **GE** shall have full powers and without giving any reason to order the contractor immediately to cease to employ, in connection with his contract, any agent, servant or employee whose continued employment is, in his opinion, undesirable. The contractor shall not be allowed any compensation on this account.

2.3. The contractor's attention is also drawn to condition **25 of IAFW-2249** in this connection.

3. **WATER (WHERE WATER IS TO BE SUPPLIED BY MES):-**

3.1. Refer to condition **31 of IAFW-2249**.

3.2. The contractor may buy water from MES and the water Points shown on site plan for piped system and shall be paid by the contractor @ Rs. **3.75 per Rs. 1000/-** worth of work done priced at contract rates. The contractor shall arrange at his own expense for storage of water and lifting pumping, carrying or conveying water to the site of work as required. In case water points are not marked on the site plan, the water shall be supplied at one point at decided by **GE** .

3.3. The supply of water may not be continuous. The contractor shall be deemed to have ascertained the hours of availability of water before submitting his tender. The MES do not guarantee the continuity of water supply and no compensation shall be allowed for intermittent or inadequate water supply and break down in the system. If the supply is not sufficient the contractor shall make his own arrangement to supplement the water supply at his own cost. For the purpose the contractor shall be allowed to install hand pumps at the site of work at places as directed by the engineer-in-Charge without any charges from the contractor on this account. The contractor shall remove the hand pumps as and when asked to do so by Engineer-in-Charge or **GE** and in any case on completion of work. NO compensation whatsoever shall be admissible to the contractor if the **GE** requires him to remove the pumps before completion of work. However, recovery of water shall be made as per special condition 4.2 here above.

4. **TESTING OF MATERIALS : _**

4.1. **Level of Tests :-**

4.1.1.1. **'A' LEVEL TESTS FOR WORKS COSTING UPTO 100 LAKHS**

:- The Contractor may set up site laboratory at his option for works upto 100 lakhs. The other stipulations will be same as specified in preceding para. However in case the contractor has not set up the site laboratory and tests are carried out in Zonal or any other laboratory approved/ setup by **GE**, the recovery shall be made at rates applicable i.e. as given in **APPENDIX 'D'**.

(Contd.../-)

SPECIAL CONDITIONS

- 4.1.2. **“B” & “C” Level Tests :-** For tests of ‘B’ and ‘C’ level as indicated in **APPENDIX 'D'**, the contractor shall provide all facilities for testing of materials at Zonal laboratory/ Government approved laboratories or test house/Engineering Colleges at his own cost. The lump sum / rates quoted by the contractor shall be deemed to be inclusive of these tests. The rates of various tests conducted in laboratory of MES are indicated in **APPENDIX 'D'**. The contractor shall bear the actual charges of “C” level tests (to be done in labs other than MES labs) irrespective of rates indicated in APPENDIX 'D'. Wherever it is convenient to get ‘B’ level test done at approved test house/ Engineering College, the same can be done at the cost of the contractor and no separate recoveries will be made by the Department for the same.
- 4.2. The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective tests or whenever the test is due whichever is earlier.
5. **INSPECTION OF SITE BY THE TENDERER :-**
- 5.1. The tenderer is advised to contact the Garrison Engineer for the purpose of inspection of site(s) and relevant documents other than those sent herewith, who will give reasonable facilities for the purpose. The tenderer shall also make themselves familiar with the working conditions, accessibility of site(s), availability of materials and other cogent conditions, which may effect the entire completion of work under this contract.
- 5.2. The tenderer shall be deemed to have inspected the site(s) and made themselves familiar with the working conditions, whether they have inspected the site(s) or not.
6. **CONDITIONS FOR WORKING IN UNRESTRICTED AREA :-**
- 6.1. All works lie in unrestricted areas as mentioned in forwarding letter. The restrictions for entry to work site and conditions of working in unrestricted area shall be as under:-
- 6.1.1. The work lies in unrestricted areas. However, the contractor, his agents, servants, workmen and vehicles may pass through the unit lines, in which case, the Engineer-in-Charge at his discretion has the right to issue the passes, control their admission to the site of work or any part thereof. The contractor on demand by the Engineer-in-Charge shall submit a list of personnel etc. concerned and any other information called for by the Engineer-in-Charge and shall satisfy the Engineer-in-Charge as the bonafide of such people. Passes shall be returned at any time on demand by Engineer-in-Charge and in any case on completion of work.
- 6.1.2. The contractor and his work people shall observe all the rules promulgated from time to time by authority controlling the area where the work is to be carried out e.g. prohibition of smoking etc. Any person found violating the security rules laid down by the authority shall be immediately expelled from the area without assigning any reasons whatsoever and the contractor shall have no claim on this account. Nothing shall be admissible for any man hours lost on this account.
7. **ACCEPTABLE QUALITY OF WORK AND FINISHES**
- (i) To determine the acceptable standard of materials and workmanship/final finishes and layout of fittings etc, the contractor shall execute stages of work viz excavation, foundation concrete, walling upto plinth/lintel/roof levels, roofing, flooring, joinery, built in items, finishes and the like and services i.e. internal electrification, water supply, plumbing, sanitary fittings as directed by the GE under the close supervision of Engineer-in-Charge and shall get it approved by the GE. The workmanship of various trades, finishes and fittings shall serve as guiding samples.
- (ii) Approval of the stages and workmanship of sample finishes and fittings shall be separately entered and approved in stage passing register giving reference for easy identification even at a later date.
8. **APPROACHES:** - The Contractors shall make arrangements for and provide at his own cost all temporary approaches, if required to the site(s), after obtaining approval in writing of the **GE** to the layout of such approaches.

(Contd.../-)

SPECIAL CONDITIONS

9. **CO-ORDINATION WITH OTHER AGENCIES:** - The Contractor shall permit free access and afford reasonable to other agencies or departmental workmen to carryout connected works of services under separate arrangements. The Contractor will not be allowed any extra payment on this account.
10. **RECORD OF MATERIALS AND PURCHASE VOUCHERS:-**
- 10.1. The quantities of materials such as cement, steel, paints, water proofing compound, chemicals for antitermite treatment and the like, as directed by the Engineer-in-Charge (the quantities of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.
- 10.2. Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in Measurement Book shall be suitably marked for identification.
- 10.3 Contractor shall produce vouchers/invoices from the manufacturers and/or their authorised agents for the full quantity of the material, as a pre-requisite before submitting claims for payment for advances on account of the work done and/or materials collected in accordance with condition 64 of General Conditions of Contracts IAFW-2249.
- 10.4 The contractor shall, on demand, produce to the GE, original receipted vouchers//invoices in respect of the materials other than as stated in sub Para hereinabove. Vouchers/ invoices so produced and verified shall be stamped by Engineer-in-Charge indicating contract number. The contractor shall ensure that the materials are brought to site, in original sealed containers/ packing, bearing manufacturer's marking except in the case of the requirement of material(s) being less than smallest packing.
11. **WATCH / LIGHTING:-** The contractor shall at his own cost take all possible precautions to ensure safety of life and property by providing necessary fencing, barrier, light, watchmen, etc., during the progress of work and as directed by the Engineer-in-Charge.
12. **CLEANING DOWN:** - Refer condition 49 of IAFW-2249. The contractor shall clean all floors, remove cement/lime/paint works drops, clean the joinery, glass panes etc., touch up all painters work and carryout all necessary items of work in connection therewith and leave the whole premises clean and tidy to the entire satisfaction of Engineer-in-Charge before handing over the items/ works. No extra payment shall be admissible to the contractor for this operation.
13. **MINIMUM WAGES PAYABLE :-**
- 13.1. Refer condition 58 of IAFW-2249. The contractor shall not pay wages lower than minimum wages for Labour as fixed by the Government of India/ State Government/ Union Territory, whichever is higher.
- 13.2. The Contractor's attention is also drawn, amongst other things to the 'explanations' to the schedule of minimum wages referred to above.
- 13.3. The fair wages referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages, referred to above as up to date from time to time.
- 13.4. Schedule of minimum wages are not enclosed along with tender documents. However contractor shall be deemed to have verified fair wages payable as on the last due date of receipt of tender.
- 13.5. The contractor shall have no claim whatsoever, if on account of local factor and or regulations, he is required to pay the wages in excess of minimum wages as described above, during the execution of work.
14. **SAMPLE OF MATERIALS :-**
- 14.1. Refer condition 10 of IAFW-2249 and clause 1.6 & 1.7 of MES Schedule Part-I.
- 14.2. Materials provided by the contractor for incorporation in the works, unless otherwise specified in the particular specifications, shall be ISI marked. IS means Indian Standards as issued by the Indian Standard Institute. Wherever in the specifications I.S is referred to, it means the edition with all amendments, current on the due date of receipt of the tender documents. The materials listed in Appx 'B' shall be of the makes as specified therein.

SPECIAL CONDITIONS

- 14.3. The tenderer is advised to inspect other materials which are displayed in the office of GE, before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfied himself, as to the nature and quality of materials, he is required to incorporate in the work irrespective of whether he has actually inspected them or not. The materials (not listed in Appendix 'A' and 'B') to be incorporated in the work by the contractor shall be IS marked or shall be equal or superior in quality to sample displayed and shall comply with the specifications given hereinafter. In case the contractor provides products of superior specification after getting it approved, it shall be at his own cost and nothing extra shall be payable on this account.
- 14.4 For obtaining sample approval, Contractor shall download list of BIS marked manufacturers from BIS site www.bis.org.in. They shall select manufacturer(s) meeting contract specifications criteria and handover the report with a request letter to Engineer-in-Charge with his signatures. The Engineer-in-Charge shall verify its correctness by visiting the BIS site and process it to GE with his recommendations. AE / JE (QS&C) of GE shall recheck the same from internet, endorse his recommendation and put up for approval of GE.
- 14.5 The printout having signature of all concerned including GE shall be kept on record and approval shall be conveyed within 7 days of Contractor's request.
- 14.6 After sample approval, GE shall send an e-mail to manufacturer (not the authorised dealer) intimating him, name of firm, approx quantity of materials being procured by the Contractor and request the manufacturer to ensure that his product only is purchased by the Contractor from proper authorised source. A printout of the e-mail sent to manufacturer shall be kept on record duly signed by GE and a copy thereof shall be sent by post to the manufacturer.
- 14.7 In cases involving suspected procurement of spurious material, purchase vouchers shall be verified through the manufacturer and proof of payment shall be obtained from the Contractor. The payment of such doubtful material shall be allowed only after GE satisfies himself about the genuineness of materials.
- 14.8. The contractor shall not procure materials unless the samples are first approved by GE.
- 14.9 The cost of testing of all materials including transportation of samples to the test lab shall be borne by the contractor. While quoting their lump sum, this aspect should be kept in mind and nothing extra shall be admissible on this account.
15. **HANDING OVER OF SITE**: - Site for execution of work will be available as soon as the work is awarded. In case it is not possible to make the entire site available on the award of work, the contractor will have to arrange his working programme accordingly. No claim whatsoever, for not giving entire site on award of work and for giving site gradually, will be tenable.
16. **SECURITY OF CLASSIFIED DOCUMENTS**: - Contractor's special attention is drawn to conditions 2-A and 3 of IAFW-2249 (General Conditions of Contracts). The contractor shall not communicate any classified information regarding the works either to sub contractors or others without prior approval of the Engineer-in-Charge. The contractor shall not make copies of the design/drawings and other documents furnished to him in respect of the works and he shall return all documents on completion of the works or on earlier determination of the contract. The contractor shall along with the final bill attach a receipt from the Engineer-in-Charge in respect of his having returned the classified documents as per condition 3 of IAFW-2249(General Conditions of Contracts).
17. **PERIOD FOR KEEPING THE TENDER OPEN** :- The tender shall remain open for acceptance for a period of **60 (Sixty) days**. Validity period of the bid shall commence from the **next day subsequent to bid submission end date**.
18. **NET WORK ANALYSIS:-**
- 18.1. The time and progress chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) shall consist of detailed net work analysis and a time Schedule. The critical path network will be drawn jointly by the **GE** and the contractor soon after acceptance of tender. The time scheduling of the activities will be done by the Contractor so as to finish the work with in the stipulated time. On completion of the time Schedule, a firm calendar date Schedule will be prepared and submitted by the contractor to the **GE** who will approve it after due scrutiny. The Schedule will be submitted in four copies within two weeks from the date of handing over of site.

(Contd.../-)

SPECIAL CONDITIONS

- 18.2. During the currency of the work, the Contractor is expected to adhere to the time Schedule and this adherence will be a part of contractor's performance under the contract. During the execution of the work the contractor is expected to participate in the reviews and updating of the network undertaken by the **GE**. These reviews may be undertaken at the discretion of **GE** either as a periodical appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or amendments. Any revision of the Schedule as a result of the review will be submitted by the contractor to the **GE** within a week he will approve it after due scrutiny.
- 18.3. The contractor shall adhere to the revised time schedule thereafter. In case of contractor disagreeing with revised Schedule, the same will be referred to the Accepting Officer, whose decision will be final, conclusive and binding. **GE**'s approval to the revised Schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of IAFW-2249 and separately regulated.
- 18.4. Contractor shall mobilize and employ sufficient resources to achieve the detailed schedule within the broad framework of the accepted method of working and safety. No additional payment will be made to contractor for any multiple shift work or other intensive methods contemplated by him in his work Schedule, even though the time schedule is approved by the department.

19. ELECTRIC SUPPLY:-

- 19.1. In case the contractor desires to buy electricity from the MES, he shall be charged for the electric energy consumed at the following rates:-
- 19.1.1. AT **Rs 10.40 (Rupees Ten and paisa forty only)** per unit for lighting and power.
- 19.2. Electric supply required for the work up to a maximum **5 KW** single phase/three phase shall be made available by the MES at the incoming terminal of the main switch marked on site plan. The main switch and KWH meters to register to electric energy supplied shall be provided and installed by the MES. The Contractor shall provide all necessary cables, fittings etc., from the main switch in order to ensure a proper and suitable supply of electricity for the execution of work.
- 19.3. MES do not guarantee continuity of supply and no compensation whatsoever shall be allowed for supply becoming intermittent or for break down in the system.
- 19.4. **GE** or his representative shall be free to inspect all the power consuming devices or any electric lines provided by the contractor. Any device or electric lines provided by the contractor, which is not to the satisfaction of the **GE**, shall be discontinued from the supply, if so directed by him.

ALLOTMENT OF LAND AND/OR ALLOTMENT OF QUARRIES SITUATED ON DEFENCE LAND:-

19.5 ROYALTIES :- Refer condition 14 of General Condition of Contracts (IAFW-2249). Quarries on defence land are not available.

20. LAND FOR TEMPORARY WORKSHOP, STORES ETC

- 20.1. Delete the following from lines 5 to 9 of sub Para 1 of condition 24 of IAFW-2249 reading "in the event of area of land..... land allotted to him" and insert "Separate land will not be allotted to the contractor for the storage of materials. The land for these purposes shall be arranged by the contractor at his own and the lump sum cost/ item rates (as applicable) quoted in the tender shall deemed to be included for the same.

21. ADMISSION TO SITE BY CONTRACTOR AND RESPONSIBILITY TO ASCERTAIN HIS OWN INFORMATION.

- 21.1. The tenderer shall contact the **GE** concerned for the purpose of inspection of site (s) and relevant documents other than those sent herewith, who will give reasonable facilities for this purpose. The tenderer shall also make themselves familiar with working conditions, accessibility of site(s), availability of materials and other cogent conditions, which may affect the entire completion of work under this contract.

SPECIAL CONDITIONS(Contd.../-)

- 21.2. The tenderer shall be deemed to have visited the site(s) before quoting rates and made themselves familiar with the working conditions, whether they actually inspect the site(s) or not.
22. **Indemnity:-** The contractor shall indemnify the Govt. from loss caused to him by the conduct of the contractor, or by the conduct of any other person.
23. **BLANK**
24. **REIMBURSEMENT/ REFUND ON VARIATION IN 'TAXES DIRECTLY RELATED TO CONTRACT VALUE':-**
- (a) The rates quoted by the Contractor shall be deemed to be inclusive of all taxes (including **GST** on materials/ **GST** on Works Contracts, Turnover Tax, Labour Welfare cess/Tax etc.), duties, Royalties, Octroi & other levies payable under the respective Statutes. No reimbursement/ refund for variation in rates of taxes, duties, Royalties, Octroi & other levies, and/or imposition /abolition of any new/existing taxes, duties, Royalties, Octroi & other levies shall be made except as provided in Sub para (b) here-in-below .
- (b)
- (i) The taxes which are levied by Govt. at certain percentage rates of Contract Sum/Amount shall be termed as "taxes directly related to Contract value" such as **GST**, Turnover Tax, Labour Welfare Cess/Tax and like but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all "taxes directly related to Contract Value" with existing percentage rates as prevailing on last due date for receipt of tenders. Any increase in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date for receipt of tenders shall be reimbursed to the contractor and any decrease in percentage rates of "taxes directly related to Contract value" with reference to prevailing rates on last due date for receipt of tenders shall be refunded by the contractor to the Govt./ deducted by the Govt. from any payments due to the contractor. Similarly imposition of any new "taxes directly related to Contract value" after the last due date of receipt of tenders shall be reimbursed to the Contractor and abolition of any "taxes directly related to Contract value" prevailing on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/ deducted by the Govt from the payments due to the Contractor.
- (ii) The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/ or imposition of any further "taxes directly related to Contract Value", give written notice thereof to the **GE** stating that the same is given pursuant to this Special Condition, together with all information relating there to which he may be in a position to supply. The Contractors shall submit the other documentary proof/ information as the **GE** may require.
- (iii) The Contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Govt. and shall further, at the request of the **GE** furnish, verified in such a manner as the **GE** may require, any documents so kept and such other information as the **GE** may require.
- (iv) Reimbursement for increase in percentage rates/ imposition of "taxes directly related to Contract value" shall be made only if the Contractor necessarily & properly pays additional "taxes directly related to Contract value" to the Govt. without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt Authority and submits documentary proof for the same as the **GE** may require."
25. **WORK ON HOLIDAY:-**The contractor shall not carryout any work on gazetted holidays except when he is specially authorized in writing to do so by **GE** . The **GE** may at his sole discretion declare any day as holiday or non-working day without assigning any reason for such declaration.
26. **RECORD OF CONSUMPTION OF CEMENT:-**
- 26.1. The contractor shall maintain a pucca bound register with serially numbered pages with all pages initialed by Engineer-in-Charge against numbering showing quantities of cement received, used in the work and balance at the end of each day. The form of record shall be as approved by Engineer-in-Charge.

(Contd.../-)

SPECIAL CONDITIONS

26.2. The register shall be signed daily by the representatives of MES and the contractor in token of verification of its correctness and will be checked by Engineer-in-Charge, at least once a week.

26.3. The register shall be kept at site in safe custody of the contractor's representative during the progress of the work and shall on demand, be produced for verification to the inspecting officer(s).

26.4. On completion of the work the contractor shall deposit the cement register with the Engineer-in-Charge for record in writing and take a receipt for the same.

27. **OFFICIAL SECRETS ACT**: - The contractor shall be bound by the Official Secrets Act -1923.

28. **BLANK**

29. **SAFETY PRECAUTION DURING EXECUTION**

29.1. The contractor shall take every precaution to control traffic on road keeping danger boards, necessary lighting arrangements, fencing and watchman to avoid any damage. In case due to excavation or others the road is to be blocked the contractor shall, without any extra cost to the Govt, provide separate by pass so that normal traffic is not disturbed.

29.2. Any damage to the existing road/building etc, shall be made good by the contractor with the same specifications as per existing work, without any extra cost to the Govt. In the event of contractor not fully complying with the above provisions to the satisfaction of the **GE**, the **GE** may provide the same for which the expenses incurred will be recovered from the contractor.

30.3 The contractor shall take precautionary measures to cover all the excavated drain as far as possible and also provide danger boards, necessary lighting arrangements and indicators to indicate the excavated pit along the road to avoid any mishappenings. The quoted rate shall deemed to be included the cost of safety precautions.

30. **ADVANCES ON ACCOUNT**

30.1. Refer condition 64 of IAFW-2249 (General Conditions of Contract).

32. **CONTRACTOR PLANT / EQUIPMENT AT SITE**

32.1 (a) The contractor shall furnish to the Engineer-in-Charge every morning distribution return of his plants / equipment on the site of work stating the following particulars :-

(i) Particulars of plants / equipment are their make, manufacturers Model No. if any, Registration No if any, capacity, year of manufacture and year of purchase etc.

(ii) Total No (Quantity) on site of work.

(iii) Location, indicating No, (Quantity) at each location on the site of work.

(iv) Purchase value on the date of purchase. For the purpose of the condition, plants /equipment, vehicle No, i.e of trucks and lorries but neither the workman's tools nor any manually operated tools / equipment shall be given. The Engineer-in-Charge shall record the particulars supplied by the contractor in the works diary and send the return to the GE for record in his office.

33. **LOSS OR DAMAGE ON ACCOUNT OF ENEMY ACTION**

33.1 If as a result of enemy action the Contractor suffers any loss or damage the Government shall reimburse to the Contractor such loss or damage to the extent and in the manner hereinafter provided :-

(i) The loss suffered by him on account of any damage or destruction of his plant / equipment [as defined in special condition 35.1 (a) above] or materials or any part or parts thereof. (The amount of loss assessed by the Accepting Officer of the contract shall be final and binding).

(ii) The compensation paid by him under any law for the time being in force to any workman employed by him for any injury caused to him or to be workman's legal successors for loss of the workman's life.

(Contd.../-)

SPECIAL CONDITIONS

(iii) Payment of compensation for loss or damage to any work or part of work carried out (The amount of compensation shall be determined in accordance with condition 48 of General Conditions of Contracts IAFW-2249).

33.2 No reimbursement shall be made nor shall any compensation be payable under the above provision unless the Contractor had taken Air Defence Precautions ordered in wiring by the GE / OC concerned or in the absence of such orders, reasonable precautions. No reimbursement shall be payable nor shall any compensation be payable for any plant / equipment or materials not lying on site of work at the time of enemy action.

34. **RECORD OF MATERIALS**

34.1 The quantity of materials such as paints, water proofing compound, chemicals for anti-termite treatment and the like, as directed by the Engineer-in-Charge (the quantity of which cannot be checked after incorporation in the works), shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

34.2 Materials brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in measurement book shall be suitably marked for identification.

34.3 The contractor shall, on demand, produce to the GE, original receipted vouchers/invoices in respect of the supplies. The vouchers/invoices shall be defaced and stamped by Engineer-in-Charge indicating contract no, name of work, under his dated signature. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing, bearing manufacturer's marking except in the case of the requirement of material(s) being less than smallest packing.

34.4 Contractor shall produce vouchers/invoices from the manufacturers and/or their authorised agents for the full quantity of the material, as a pre-requisite before submitting claims for payment for advances on account of the work done and/or materials collected in accordance with condition 64 of General Conditions of Contracts IAFW-2249.

35. **E-PAYMENT**

Refer condition 64 of IAFW 2249. The contractor may opt for E-payment system for receiving the payment for this work. If the contractor opt for E-payment, he shall provide the following details to the GE within 15 days from the commencement of the work as mentioned in the WO No. 01:-

- (a) Accounts Name
- (b) Bank account number
- (c) Bank IFSC code

36. **PERFORMANCE SECURITY AND REFUND OF PERFORMANCE SECURITY**

(A) PERFORMANCE SECURITY

36.1 Within 28 days of receipt of the letter of Acceptance, the successful contractor shall deliver to the Accepting Officer a performance security in any of the forms given below:-

- (i) A bank guarantee in the prescribed form as per Appendix 'E'.
- (ii) Government securities, FDR or any other Government instruments stipulated by the Accepting Officer.

36.2. If the performance security is provided by the successful contractor in the form of Bank Guarantee, it shall be issued by Nationalized /Scheduled Indian Bank but its confirmation shall be done only from Head Office of the Bank.

36.3 The period of validity of Bank Guarantee Bond against Performance Security shall be upto and including the date of expiry of the Defect Liability Period. In case of any delay in completion of work, Bank Guarantee shall be extended as required, the contractor shall be directed by the Accepting Officer of the contract to have the validity of BGB extended before expiry of defect liability period.

(Contd.../-)

SPECIAL CONDITIONS

36.4. Failure of the successful contractor to comply with the requirement of sub clause 36.1 shall constitute sufficient grounds for cancellation of award of work and forfeit of the Earnest Money. In case of MES enlisted contractors, amount equal to Earnest Money stipulated in the Notice Inviting Tender shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such contractors shall remain suspended till the aforesaid amount equal to Earnest Money is deposited in Government Treasury.

(B) REFUND OF PERFORMANCE SECURITY

(i) Refer conditions 19 & 68 of IAFW 2249.

(ii) The performance security deposit mentioned under Condition 19 of IAFW-2249 will be refunded to the contractor after the expiration of **Defect Liability Period** by the GE provided always that the contractor shall first have been paid the final bill and have rendered a No-Demand Certificate (IAFW-451).

(Signature of Contractor)
Dated.....

AGE (Contracts)
For Accepting Officer

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

1. GENERAL

- 1.1 The work under this contract shall be carried out in accordance with the description in Schedule 'A'(BOQ). Particular specifications, drawings (if any), general specifications and other provisions in the MES Standard Schedule of Rates shall be read in conjunction with each other.
- 1.2 The term general specifications referred to in clause 1.1 above and in IAFW-2249 shall mean the specifications contained in the MES Schedule 2009 Part-I and 2020 Part-II including erratas and amendments thereto, ISS and any other standard Engineering Books.
- 1.3 General rules, specifications, special conditions, method of measurements, preambles and all other provisions in the MES Schedule shall be deemed to apply to the works under this contract unless other wise mentioned in these tenders documents.
- 1.4 Particulars specifications given hereinafter shall be applicable to all the works covered in Schedule 'A' and they are in brief and emphasise the specifications given in MES Schedule which are not repeated here.
- 1.5 Where specifications/provisions given in these specifications are at variance with the specifications provisions given in the MES schedule, the specification/provisions given in these particular specifications shall be followed.
- 1.6 In the absence of any reference in particular specifications with regard to the item mentioned in Schedule 'A', the work shall be executed to the specification given in the MES Schedule in preference to the local trade practice.
- 1.7 Where specifications for any item of work is not given in MES Schedule or in these particular specifications, the specifications given in relevant Indian Standard specifications or code of practice shall be followed. Specification given for the items not included in this tender shall be deemed to have been deleted for the purpose of this tender.
- 1.8 The IS mentioned in the MES Schedule and in the other documents shall be deemed to be changed/modified as per latest amendments issued to IS upto date of receipt of tender.

2. MATERIALS TO BE SUPPLIED BY CONTRACTOR

- 2.1 Refer condition 10 of IAFW-2249 (General Conditions of Contracts).
- 2.2. The tenderer is advised to inspect materials which are displayed in the office of GE MES, before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials, he is required to incorporate in the work irrespective of whether, he has actually inspected them or not. The materials to be incorporated in the work by the contractor shall conform to or shall be superior in quality samples displayed and shall comply with the specifications given here-in-before/after
- 2.3 All materials shall be ISI marked. Where ISI marked materials/products are not available, these shall conform to relevant IS. Specifications or to BS Specifications, where IS is not existing. In absence of any IS/BS specifications, these shall be as approved by GE.
- 2.4 Sample of all materials, fittings & fixture, accessories etc to be supplied by the contractor shall be submitted to Engineer-in-Charge for approval of GE. The contractor shall not commence the work until the samples are approved in writing by GE. The contractor shall ensure that all the materials as incorporated in the work are identical in all respects. Samples approved and samples destroyed in testing shall be returned to the contractor after completion of contract. No payment shall be made for samples destroyed in testing.
- 2.5 All manufactured articles required for incorporation in the work shall be brought to site in the manufacturer's original packing with seal intact.
- 2.6 Approved names/brands of products to be incorporated in works are given in appendix 'A' to these particulars specifications. The product shall be any of the makes, subject to approval of the GE.

3. EXCAVATION AND EARTH WORK

- 3.1 The Contractor shall excavate to the specified depths and as directed by the Engineer-in-Charge. Excavation made to depth more than the required depth shall be made good by the Contractor at his own expense with lean 1:5:10 cement concrete No extra payment shall be made for extra depth excavated and filled by the Contractor. The Contractor shall familiarize himself with the site and the

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

cost of any shoring/benching and dewatering shall be deemed to included in the cost of excavation and no extra payment shall be made.

3.2 Also refer relevant clauses of section 3 of MES SSR – 2009, Part-I Specifications.

4. DISPOSAL OF SURPLUS SPOIL & RUBBISH ETC OBTAINED FROM DISMANTLING /DEMOLITION

4.1 All the surplus spoil, each, rubbish etc obtained from excavation/dismantling/ demolition shall be disposed off to outside Ministry of Defence land at any distance at Contractor's cost otherwise specifically not mentioned in Sch 'A'. Serviceable material if any obtained shall be deposited to MES store yard of Engineer-in-Charge.

4.2. HARD CORE

4.2.1 Hard core shall be of hard burnt broken bricks / broken stone aggregate etc not exceeding 63mm gauge and shall be provided all as per specification laid down in para 3.27 of MES Sch part-I. Thickness of hard core shall be as directed and shall be measured after consolidation. Hard core shall be blended with finer material of hard burnt broken brick aggregate. It shall be well watered and consolidated by hand ramming. Where thickness of hard core is not indicated, it shall be of 100mm consolidated thickness.

5. DEMOLITION/DISMANTLING/TAKING DOWN AND MAKING GOOD

5.1 Dismantling/demolition/taking down to the extent as specified shall be carried out all as specified in section 21 of MES Schedule. Particular precaution shall be taken to prevent damage to the part of structures to be left intact and adjacent structure. If any damage caused to the structures due to carelessness and negligence on the part of the contractor shall be made good by the contractor at his own expense.

5.2 Rubbish shall be disposed off to a distance not exc 100 meter as directed by the Engineer-in-Charge.

5.3 All works in any trade dismantled/demolished/taking down in excess of the requirements given in contract due to mistake/carelessness of contractor or his workman or any case whatsoever shall be made good by the contractor to the original specifications at his own cost.

6. CONCRETE WORK

6.1 CEMENT

6.1.1 GENERAL: Cement required for the entire work under the concrete shall be procured, supplied and incorporated in the works by the contractor under his own arrangement. Cement shall be of tested quality and shall comply with the requirements mentioned in the drawings, MES Schedules, IS specification as amended and particular specifications given here-in-after.

6.1.2. Type of cement for the subject work shall Ordinary Portland cement of Grade 43 (Forty Three) in accordance with IS-8112 of 1989 or Portland Pozzolana Cement (IS: 1489-1991) as mentioned in notes of Structural drawings. Mixing of OPC & PPC shall not be allowed in the work.

6.1.2.1 Blank

6.1.3. While using the PPC, following requirement shall be met by the contractor:-

(a) PPC shall meet the strength criteria of 43 Grade OPC as laid down in IS : 1489-1991.

(b) The minimum period before striking formwork given in clause 11.3.1 of IS : 456 of 2000 shall be increased at site by 03 days. The contractor shall not claim anything extra on this account.

(c) Wet curing shall be done for a period of 14 days, wet curing of columns shall be done through Hessians cloth wrapped on the columns.

(d) The contractor shall submit the following certificates from the manufacturer of the cement for every batch of cement :-

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- (i) The quality of fly ash is strictly as per IS : 1489 (Part-I)-2002.
- (ii) Fly ash is inter-ground with clinker and not mixed with clinker.
- (iii) Dry fly ash is transported in closed containers and stored in silos. Only pneumatic pumping has been used.
- (iv) The fly ash received from thermal power plant using high temperature combustion above 1000°C has only been used.
- (e) The compressive strength of cement at 03 days, 07 days and 28 days shall not be less than 23 Mpa, 33 Mpa and 43 Mpa respectively. Each lot of cement brought at site shall be tested for compliance by the GE before acceptance.
- (f) In cold climate regions, where temperature is lower than 15°C, OPC shall be used (However during working seasons in cold climate regions, where OPC is not available use of PPC may be permitted provided it shall be ensured that while using PPC, atmospheric temperature should be more than 15°C).
- (g) Mixing of OPC and PPC shall not be allowed in a work. However with the strict control of the ground executives, different buildings can have different types of cement.

6.1.4. SOURCES OF PROCUREMENT

- 6.1.4. (a) Cement shall be procured by the contractor directly from any of the following main producers of cement as given below :-

(i)	ACC	(vii)	Ramco	(xiii)	JK	All grades
(ii)	Ultra Tech	(viii)	Mangalam	(xiv)	JK Laxmi	
(iii)	The India Cement	(ix)	Birla	(xv)	Jaypee	
(iv)	Dalmia Infra Pro	(x)	Orient	(xvi)	Ambuja	
(v)	Century Cement	(xi)	Nuvoco			
(vi)	Saurashtra	(xii)	Shree			

(i)	Sartaj	(iv)	JSW OPC 43	(vii)	Wonder Cement	OPC 43
(ii)	Parasakti Cement	(v)	Sagar			
(iii)	Chettinad Cement	(vi)	Maha Cement			

(i)	JSW OPC 53	(iii)	Sanghi			OPC 53
(ii)	Sagar	(iv)	Wonder Cement			

(i)	Sartaj	(iv)	Sagar	(vii)	Wonder Cement	PPC
(ii)	Parasakti Cement	(v)	Sanghi			
(iii)	Chettinad Cement	(vi)	Maha Cement			

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(b) However, where the quantity of cement required is less than 1200 bags, same may be procured from authorised distributors/dealers of approved manufacturer's, if so, permitted by GE in writing. Testing of cement shall in variably be got done before incorporation in work. The contractor shall submit test certificate for each batch of cement issued by the manufacturer.

(c) The contractor shall furnish the particulars of the manufacturer of cement along with the date of manufacture to the GE for every lot of cement separately. The cement so brought shall be fresh and in no case older than 90 days from the date of manufacture. The GE shall verify the document in support of the purchase of cement. Before placing order for supply of cement by the contractor, he shall obtain written approval from the GE regarding name of manufacturer, quantity of cement etc. Cement shall be procured for minimum requirement of one month and not exceeding the requirements of the same for more than two months at a time. The cement shall be consumed in the work within three months after receipt. Cement shall conform to the requirement of IS specification and each bag of cement shall bear relevant ISI mark. The weight of each consignment shall be verified by the GE and recorded. The content of cement shall be checked at random to verify the actual weight of cement per bag. However, the content of cement per bag shall be 50 Kg only subject to tolerance given in Clause 9.2.1.1 and Annexure 'B' of IS-8112 and Clause 10.2.1.1 and Annexure 'B' of IS – 1489 for OPC and PPC respectively.

6.1.5. TESTING OF CEMENT

61.5. (a) The contractor shall submit the manufacturer's test certificate in original alongwith test sheets giving the results of each physical test as applicable in accordance with relevant IS provision and the chemical composition of the cement or authenticated copy thereof, duly signed by the manufacturer with each consignment, as per the following IS provision :-

- (i) Method of sampling hydraulic cement as per IS : 3535-1986.
- (ii) Method of physical test of hydraulic cement as per IS-4031.
- (iii) Method of chemical analysis of hydraulic cement as per IS-4032-1985.

(b) The test certificate and test sheet shall be furnished with each batch of cement. The Engineer-in-Charge shall record these details in cement acceptance register to be maintained by him which will be signed by JE (Civil), Engineer-in-Charge, GE and the Contractors as given in the format here-in-after for verification.

(c) The contractor shall however, organize setting time and a compressive strength test of cement through designated laboratory on samples collected from the lot brought at site before incorporation in work. The contractor will be allowed to use the cement only after satisfactory compressive strength of seven days. To meet this requirement contractor is required to keep minimum 10 days stock before any new lot is brought at site, which can be used, in the work. The contractor shall be required to remove the cement not meeting the requirement from site within 24 hours. Seven days strength test will be relied upon to accept the lot of cement to commence the work. 28 days compressive strength test will be the final criteria to accept/reject the lot.

(d) The random samples as per relevant IS shall be selected by GE before carrying out testing. The record of such samples selected by the GE for testing shall be properly maintained in the cement testing register giving cross reference to relevant consignment of cement and quantity received etc.

(e) Cost of transportation of samples to the approved laboratory/test house and all testing charges including cost of sample shall be borne by the contractor.

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(f) The contractor shall be required to set up adequate testing facilities at site to the entire satisfaction of GE for conducting setting time test and compressive strength test as per IS codes referred to here-in-before for the samples collected from the lot brought at site. These tests shall be carried out within 07 days of receipt of cement at site. The tests can alternatively be carried out at the Zonal laboratory, or any other recognized laboratory so designated by GE.

(g) The GE shall carryout independent testing as per the tests mentioned in the 'CEMENT SUPPLY AND ACCEPTANCE REGISTER' (as per **Annexure 'I'**) of random samples of cement drawn from various lots, if sample fails in 7 days compressive strength. The testing shall be carried out through National Test House, SEMT, CME, Regional Research Laboratories, Govt. approved Laboratories, Zonal laboratories/ IIT's as per IS : 3535-1986, (method of sampling hydraulic cement), IS-4031 (Method of physical test for hydraulic cement) and IS-4032-1985 (Method of chemical analysis of hydraulic cement) referred to above. The decision as to where the testing of cement is to be done shall be taken by GE. In case the cement is not of requisite standard despite manufacturer's test certificate, the contractor shall remove the total consignment from the site at his own cost after written rejection order of the consignment by the GE. The cost of test shall be borne by the contractor irrespective of the results of testing.

(h) The contractor shall submit original purchase vouchers for the total quantity of cement supplied under each consignment to be incorporated in the work. All consignment received at the work site shall be inspected by the GE alongwith the relevant documents to ensure the requirements as mentioned here-in-before, before acceptance, the original purchase vouchers and the test certificates shall be verified for subject contract and defaced by the Engineer-in-Charge and kept on record in the office of the GE duly authenticated and with cross reference to the consignment/ control number recorded in the 'Cement Acceptance Register'. The Cement Acceptance Register shall be signed by the JE (Civil), Engineer-in-Charge, GE and the contractor. The contractor shall maintain schedule of supply of cement for each consignment.

(j) The Accepting Officer may order a board of officers for random check of cement and verification of connected documents during the currency of contract.

6.1.6. STORAGE/ACCOUNTING/PRESERVATION OF CEMENT

(a) Cement shall be stored in covered godown over dry platform at least 20 cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of store room the stack should be at least 20 cm away from floors and 60 cm from walls. The stacking of cement shall be done as specified in relevant IS. The storage, accounting and preservation of cement supplied by the contractor shall be done as per standard Engineering practice till the same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate/ amount quoted by the tenderer. The Engineer-in-Charge shall inspect once a day to verify that cement lying at site is stored, accounted, preserved and maintained as per the norms. The cement shall be stored so as to differentiate each tested and untested consignment separately with distinct identification. If the GE is not satisfied with the storage/preservation of cement, he may order for any test (s) of cement as applicable for that consignment to ensure its conformity to the quality mentioned in the manufacturer's test certificate. The contractor shall bear the cost of necessary testing (s) in this regard and no claim whatsoever shall be entertained.

(b) Stacking of cement shall be done as per relevant IS and as under :-

(i) Each cement consignment shall be stacked separately and removal shall be made on the basis of First in First out.

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- (ii) Adequate top cover will be provided.
- (iii) Stacks in no case shall be higher than 12 bags. The maximum width of each stack shall be 3.0 m. If the stack is more than 07 or 08 bags high, the bags shall be arranged in header and stretcher fashion, i.e. alternatively lengthwise and crosswise so as to pile together and avoid topping over.
- (iv) Adequate space shall be kept between two stacks.
- (c) Cement godown shall be provided with two locks on each door. The key of one lock at each door shall remain with Engineer-in-Charge or his representative and that of the other lock with the contractor's authorized agent at site of works so that cement is removed from the godown only according to daily requirement with the knowledge of both the parties. During the period of storage, if any cement bag (s) is found to be in damaged condition due to whatsoever reason, the same shall be removed from the cement godown on written orders of the GE and suitable replacement for the cement bag (s) so removed shall be made and no claim whatsoever shall be admissible on this account.
- (d) In case more than one type of cement is used in the work i.e. Ordinary Portland cement or Portland Pozzolona Cement, both type of cement shall be stored separately as directed by the Engineer-in-Charge to avoid mixing of these type of cement. Separate record shall be maintained including the location/ items where this type of cement is used.
- (e) Cement shall be removed from the store only according to daily requirement with the knowledge of both the parties and the Engineer-in-Charge and the contractor shall record daily consumption of cement in cement consumption register, which shall be signed. Cement constants given in Appendix 'A' to E-in-C's branch letter No. 19280/E8 dated 03 May 1976 shall form the basis of consumption of cement for various items of works unless specifically indicated otherwise.
- (f) In case the consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to whatsoever reason, the contractor shall not have any claim whatsoever for such excess consumption of cement.

6.1.7. SCHEDULE OF SUPPLY

The Contractor shall procure the cement timely as required in accordance with CPM chart agreed between GE and the contractor. The contractor will forfeit his right to demand extension of time if the supply of cement got delayed due to his failure in placing order in time to the manufacturer.

6.1.8. MEASUREMENTS AND PAYMENT OF CEMENT

- 6.1.8. (a) The entire quantity of cement shall also be suitably recorded in the measurement book for record purposes as 'Not to be abstracted' before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.
- (b) The payment shall only be allowed after production of original purchase voucher, certified copies of test certificates from manufacturer for each consignment and results of testing carried out in laboratory on receipt of cement (07 days compressive test) are found satisfactory as per condition 64 of IAFW-2249. Rate of payment given in SSR shall be applicable for cement irrespective of type and grade of cement specified for use in the work.

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Annexure 'I'**CEMENT SUPPLY AND ACCEPTANCE REGISTER**

01. CA No. & Name of work :
02. Control No.* :
03. Name of Manufacturer/Brand Name/Grade of Cement :-
 (a) Manufacturer :
 (b) Brand :
 (c) Grade :
04. Qty of cement & Lot No./Week No. (in Bags) :-
 (a) Quantity :
 (b) Lot No./Week No. :
05. Manufacturer's test Certificates No. :
06. Random Test Details :-
 (a) Physical test report from _____ vide their letter No. _____
 (Name of approved Lab/Engineering College)
 (b) Chemical test report from _____ vide their letter No. _____
 (Name of approved Lab/Engineering College)
07. **Details of Physical & Chemical properties :-**

	Physical Requirements (as per IS-4031)									Chemical Requirements (As per IS-4032)								
	Specific Surface (Sqm/Kg)	Soundness by Le Chatelier	Soundness by Auto Clave	Initial Setting Time (Minutes)	Final Setting Time (Minutes)	Compressive Strengths (Mpa)			Temp during testing °C	Standard Consistency (%)	Lime Saturation Factor (Ratio)	Alumina Iron Ratio (Ratio)	Insoluble Residue (%)	Magnesium (%)	Sulphuric Anhydride (%)	Loss on Ignition (%)	Alkalies (%)	Chlorides (%)
						03 Days	07 Days	28 Days										
As per relevant IS																		
As per manufacturer's test certificate																		
As per random test certificate																		

Remarks with Signature

Contractor**Junior Engineer****Engineer-in-Charge**

Accepted / Rejected

Garrison Engineer

Remarks of BOO / Inspecting Officer / CWE

* To be allotted serially by GE consignment wise

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- 6.2. **Fine Aggregate.** Fine aggregate for concrete work shall conform to material specifications and gradings within the limits of Grading Zones II to III all as specified in clause 4.4 of MES Schedule 2009 Part-I.
- 6.3. **Coarse Aggregate.** Refer Clause 4.4 of MES Schedule Part I. Coarse aggregate for all plain cement concrete shall be graded crushed stone at contractor's options. For all RCC work, the coarse aggregate shall be graded crushed stone.
- 6.3.1. **Grading of Coarse Aggregate.** Graded Aggregate of nominal sizes given herein under shall be used, unless specified otherwise hereinafter : -
- (a) Plain or reinforced cement concrete except in foundation of Brick or stone Walls/Pillars, Floors and Sub base to Floors.
- (i) For structural elements of depth/thickness more than 50 mm - 20 mm.
- (ii) For structural elements of depth/thickness less than 50 mm - 12.5 mm.
- (b) Plain Cement concrete in Foundation of Brick/stone Walls, Pillars, Floors and Sub Base to Floors.
- (i) Under 30 mm thickness : 12.5 mm
- (ii) 30 to 80 mm thickness : 20 mm
- (iii) Exceeding 80 mm thickness : 40 mm
- 6.4. **Water.** Water shall conform to the requirement stipulated in IS 456 and as per clause 4.9 of MES Schedule Part I.
- 6.5. **Mix of Concrete.** Mix of cement concrete in various situations shall be as under : -

Srl No	Situation	Mix of Concrete
<u>Type of concrete</u>		
(a)	Foundation concrete for all buildings, under brick and lean concrete under plinth/beam and steps, in gaps between plinth/Column under footing, sunken portion filling, foundation for ramps/steps.	: PCC (1:5:10) type E-2 (By volume)
(b)	Foundation concrete under column footing	: PCC (1:4:8) type D-2 (By volume)
(c)	PCC in plinth protection, drain and channel, PCC cills and PCC block for hold fasts, back of pressed steel frames and plugging for scaffolding holes.	: PCC (1:3:6) type C-1 (By volume)
(d)	All RCC work except water retaining structure	: Grade M 25 (Design mix as per IS 456-2000)
(e)	PCC in bed plate, benching, coping, DPC and pre-cast articles and plain cement concrete not mentioned above.	: PCC (1:2:4) type B-1 (by volume)
(f)	All RCC work for water retaining structure	Grade M-30 (Design) mix as per IS 456-2000

6.6 Blank

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6.7 FORM WORK

- 6.7.1 Form work shall comply with requirements of clause 4.11.6, 4.11.6.1 to 4.11.6.5 and 7.15 of MES Schedule part-I.
- 6.7.2 Form work shall be of steel or plywood of adequate strength at the option of the Contractor. However deformed steel sheet shall not be permitted for use as form work. In case of any deviation involving form work to surfaces exposed to view the pricing shall be done at the rates of timber form work for fair finish.

6.7 FINISHING TO EXPOSED SURFACE OF CONCRETE

Refer to clause 4.11.16.1 4.11.16.2(a) to (e) and 4.11.16.3 of MES Schedule Part-I. If thickness of plaster in cement mortar (1:3) as specified in clause 4.11.16.2(b) of MES Schedule Part-I, is required to be increased in excess of 5mm to achieve fair and even surface it shall be provided by the Contractor without any price adjustment.

7 BRICK WORK**7.1 MATERIALS:-**

- 7.1.1 Cement: - Refer Para 4.3 of SSR Part –I.
- 7.1.2 Sand for masonry mortar:- Refer Para 5.4 of SSR Part I. It shall be natural conforming to IS-2116. Sea sand shall not be used in any case.
- 7.1.3 Common Burnt clay building Bricks:- Brick shall be described in BOQ Refer Para 5.6 of SSR Part I. The compressive strength of bricks shall be not less than 100 Kg/Sq cm for class A and 75 Kg/Sq cm for Class B. The rates given in SSR shall be deemed for the same. These shall be well burnt old size bricks. The nominal size of bricks shall be 230x113x75 mm. The tolerance shall be not more than +/- 8%. Water absorption shall not exceed 20 %.
- 7.1.4 Bituminous Bonding Materials:- Refer Para 5.16 of SSR Part-I. Blown bitumen shall be conforming to IS-712.
- 7.1.5 Bitumen Primer: - Refer Para 5.17 of SSR Part I. It shall be conforming to IS-3384.

7.2. WORKMANSHIP :-Refer Para 5.19 to 5.56 of SSR part I.

- 7.2.1 Proportioning: - Refer Para 5.19.1 of SSR Part I.
- 7.2.2 Preparation of cement Mortar:-Refer Para 5.19.2 of SSR Part I. unless specified otherwise cement mortar shall be of mixes not leaner than 1:6. Mixing shall be done in mechanical mixer.
- 7.2.3 Laying of bricks:- Refer Para 5.23 of SSR Part I. Top courses of all plinths, Parapets, steps and top of walls below roof/ floor slab shall be laid with brick on edge. Bearing of floors, roofs etc.: - Refer Para 5.34.14. Of SSR Part I. Tops of walls bearing, the edges of RCC floors, roof slabs, beams or lintels shall be finished with 15 mm thick rendering of cement and sand mortar 1:4 mix.
- 7.2.4 Half brick thick walls:- Refer Para 5.37 of SSR part I. Mix of mortar for half brick thick walls shall be 1:4 and shall be reinforced with 2 No 8 mm dia. Deformed bars at every 4th course , securely embedded in adjoining structure.
- 7.2.5 In the event of deviations, brick work as specified above, shall be priced at the applicable rates in MES schedule (Part-II) for M&L with sub class 'B' bricks, subject to concrete percentage, as applicable irrespective of strength of bricks specified.
- 7.2.6 Width of concrete lintels beams cills columns and the like coming in contact/connection with brick walls / pillars shall be kept to the actual width of brick work of that places unless of sets have been specifically shown, in which case, the width as shown on drawings shall be maintained.
- 7.2.7 : **FLY ASH BRICKS** : Brick shall be described in BOQ. Fly ash bricks shall be of Class A and designation 7.5. The specification approved manufactures or can be produced at work site with prior approval of accepting officer. The old size bricks shall be used. Fly ash bricks shall be conforming to IS 12894 – 2002 with technical specifications as under :-

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7.2.8 **Compressive Strength.** The minimum average wet compressive strength of any individual brick shall not fall below the minimum average wet compressive strength specified for the corresponding class of bricks by more than 20 percent as mentioned in table 1 in IS 12894 when tested as described in IS 3495 (Part I).

7.2.9 **Drying Shrinkage.** The average drying shrinkage of the bricks when tested by the method described in IS-4139, being the average of three units, shall not exceed 0.15 percent.

7.2.10 **Efflorescence Test.** The bricks when tested in accordance with the procedure laid down in IS 3495 (Part 3) shall have the rating of efflorescence not more than 'moderate' upto class 12.5 and slight for higher classes.

7.2.11 **Water absorption.** The bricks when tested in accordance with the procedure laid down in IS 3495 (Part 2) after immersion in cold water for 24 hour, shall have average water absorption not more than 20 percent by mass upto class 12.5 and 15 percent by mass for higher classes.

7.2.12 **Cement.** Refer Annexure I to Appendix 'B' to Particular specifications (List of manufacturers).

7.2.13 Fly ash bricks shall be procured from the following Make/Sources.

- (a) M/s Aditi Infra works/resources Bareilly.
- (b) Power bricks corporation, Ghaziabad.
- (c) Any other make approved by Accepting Officer.

7.2.14 SAND FOR MASONRY MORTAR :- Refer Para 5.4 of SSR Part I. It shall be natural conforming to IS-2116. Sea sand shall not be used in any case.

7.2.15. **TESTING OF BRICKS** : - The contractor shall make all arrangements for testing of bricks as per IS and shall bear all costs including transportation of bricks to Govt approved Lab and get it tested in the presence of his representative and the EIC. The results shall be recorded by the EIC and signed by both the parties.

8 WOODWORKS AND JOINERY

8.1 MATERIALS

8.1.1 Type of timber shall be all as described in Sch 'A'/as directed and as specified in section 7 of MES Schedule 1991 (Part-I). and other material viz ply wood, block board, veneere particle board, nails & screw, glue etc shall be as per para 8.6 to 8.10 on page 8-1 of MES Schedule 1991 (Part-I).

8.1.2 Unless otherwise specified else wherein Sch 'A' or specification, timber shall be second class hard wood (SAL) for chowkhat doors, windows and ventilators while timber for shutter of doors, windows and ventilators shall be of 2nd class hard wood (Except Mango) as ordered by the Engineer-in-Charge/ mentioned in Sch 'A' or specifications. However the Contractor may use 1st class hard wood shisham for shutters at his own without any price adjustment after written approval from GE.

8.2 WORKMANSHIP

8.2.1 Surface of wood and joinery in contact with or embedded in concrete work shall be clean sawn. All other surfaces of joinery and wood work shall be wrought on all faces.

8.2.1 Surface of wood and joinery in contact with or embedded in concrete work shall be clean sawn. All other surfaces of joinery and wood work shall be wrought on all faces.

8.2.2 Use of nails in joinery is prohibited. Bamboo/hard wood pins or appropriate size shall be used.

8.2.3 Tolerance for wrought faces of carpenter's work and joinery shall be allowed as indicated in the MES Schedule with this exception of the following for which no tolerance shall be allowed :-

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- (a) Thickness of planks and panels in paneled joinery.
- (b) Overall sizes of doors.

8.2.4 Bolt head of hold fasts shall be stopped with tight fitted wooden plugs finished flush with surrounding surfaces.

8.2.5 Exposed edges of timber frames, styles, bars, boards and the like head not be rounded but these shall be neatly finished.

8.2.6 The joints in timber frames shall be either mortice and tenon or devetos joints at contractor's option.

8.3 PRESERVATION OF TIMBER

9.3.1 Preservation anti termite treatment shall be carried out to all wood work and joinery fabricated by the Contractor at site. Factory made ply boards and factory made wooden shutters are deemed to be provided with anti termite treatment.

8.3.2 Chemical used for anti termite treatment to wood work and joinery shall be copper NAPHTHANATE, ASCU or any other chemical specified in IS 401, applied in any one of the manners specified in ibid IS.

8.4. PANELLED/GLAZED / GAUGED/FLUSH SHUTTERS

8.4.1 Unless otherwise mentioned in Sch 'A' shutters shall be made locally and fixed in locations shown on drawings/ or as directed by Engineer-in-Charge. Shutters shall be as specified in para 8.21 on page 160 & 161 of MES Schedule 2009 (Part-I). The size of rails and stiles shall be as specified in para 8.24 on page 161 of MES Sch 2009 (Part-I). The moisture contents of wood for any type of shutters to be provided under the contract shall not be more than 12%.

8.4.2 Unless described otherwise in Sch 'A' glazed shutters for windows and ventilators shall be manufactured locally as specified in MES Schedule 2009. (Part-I)

8.5. SOLID PVC JOINERY

8.5.1 Solid PVC joinery work shall be carried out as described in Sch 'A' and as specified in clause 8.34.1 to 8.34.7 of SSR Part-I on page 165 to 170

8.6. ANODISED ALUMINIUM DOOR WINDOW ETC

8.6.1 Refer Para 10.37 of SSR Part I. The shape and size of Aluminium sections shall be as directed / shown in drawings. The overall sizes of doors/ windows/ ventilators/ fixed glazing etc. shall be as directed/ shown in drawings.

8.6.2 Glazing clips, rubber packing, aluminium snap beadings, screws etc. shall be provided to all doors/windows/ventilators, fixed glazing etc. as required. However, where Aluminium section are specified to be powder coated, these shall be epoxy powder coated wet thickness not less than 70 micron.

8.6 Workmanship:-Generally:-

8.6.1 Refer clause 10.9, 10.10 to 10.16 of SSR Part I. Straightening of steel by hammer blows is not permitted. Unless otherwise indicated, welding shall be done by metal arc process. The type of welding shall be as indicated in Schedule "A"/ shown in drawings, failing which it shall be as directed by EIC.

8.6.2 Fixing/Placing of reinforcement:-

8.6.2.1 Refer Clause /Para 10.18, 10.19, 10.20 and 10.22 of SSR Part I.

8.6.2.3 Fixing of windows etc.:-Refer relevant clause of SSR Part I. These shall be fixed in the manner as directed/ shown in drawings/ specified elsewhere.

8.7 STEEL DOOR WINDOW AND VENTILATOR

8.7.1 Steel door window and ventilator shall comply with IS 1038 – 1983. for fabrication fixing etc refer clause No 10.25.1 to 10.25.12 of SSR Part-I on Page No 209 to 2011.

8.8. PARTICLE/MDF BOARDS/PRELAMINATED PARTICLE BOARDS.

8.8.1. Particle boards shall be flat pressed three layers or multi layers or graded veneered particle board bonded with BWP grade synthetic resin adhesive and shall conform to exterior grade of IS-3097. Medium density fiber (MDF) boards where indicated on drawings shall be of flat pressed single layer of exterior grade conforming to IS 13806. Veneered particle/MDF Board used on exposed surfaces of built-in furniture shall be of decorative type provided with laminated decorative surfaces and shall conform to IS 12823. Prelaminated particle board shall be BWP type, exterior grade, phenol

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formaldehyde bonded, flat pressed, 3 layered, pre-laminated melamine faced, grade I quality having desired decorative shade on one face and balancing white lamination on other face or both faces with desired decorative shade as shown on drawings and shall be ISI marked with IS-3087 (1985) or IS-12823 (1990) having bond strength not less than 12 Kg/Cm. Sq.

8.8.2. Shutters, shelves and partitions of built-in furniture, where indicated on drawings of particle/MDF boards, shall be of decorative type boards laminated on both side all as specified here-in-before.

8.8.3. Edges of shutters/partitions/shelves/tops pelmet boxes and the like where indicated on drawing to be of particle/MDF boards, shall be provided with 6 mm thick edging of wood specified hereinbefore.

8.8.4. Ply Wood. All ply woods where indicated on drawings shall be BWR grade conforming to IS-303. Decorative plywood where indicated in drawing shall conform to IS-1328. The thickness of plywood wherever not indicated in drgs shall be 6 mm.

8.8.5. Decorative laminates. Decorative laminates where indicated on drawings shall be 1 mm thick plastic laminated sheet type I as specified in clause 12.7 of MES Schedule. The laminates shall be fixed to base with approved adhesive as per manufacturer's instructions.

9. BUILDERS HARDWARE:-

9.1 MATERIALS:

9.1.1 Refer Para 9.2 of SSR Part I. Articles of builders hardware, unless specified otherwise in Schedule "A"/ particulars specifications herein after shall be of aluminium anodized of required size as directed by EIC. Mild steel articles may be stove enameled.

9.1.2 Unless specifically mentioned otherwise, all builder hardware shall be ISI marked and shall be an approved product of Bureau of Indian standard.

9.1.3 Tower Bolts: - Refer Para 9.3 of SSR Part I. Tower bolts up to 125 mm size shall have 10 mm dia. Bolt and 150 mm and above shall have 12 mm dia. Bolt.

9.1.4 Sliding door bolts:-Refer Para 9.5 of SSR Part I. The rates given in SSR Part II for mild steel sliding door bolts shall be deemed for the same and no price adjustment shall be made on this account.

9.1.5 Hinges:-Refer Para 9.7 of SSR Part I. Irrespective of whatever is specified elsewhere Mild steel butt hinges shall be Medium weight.

9.1.6 Double Acting door springs: - Refer Para 9.7.8 of SSR Part I. These shall be conforming to relevant IS.

9.1.7 Hasp and staples:-Refer Para 9.10 of SSR Part I. These shall be conforming to relevant IS. Irrespective of what ever is specified, hasp and staple shall be of safety type

9.1.8 Handles:-Refer Para 9.11 of SSR Part I. These shall be conforming to relevant IS.

9.1.9 Door closers (Hydraulically Regulated) :- Refer Para 9.16 of SSR Part I. These shall be ISI marked an approved product of BIS and shall have ISI certification Mark- ISI-3564. It shall be of designation 2 or 3 as required. It shall be of Aluminium anodized body. It shall be universal type.

9.1.10 Floor door stoppers :-Refer Para 9.15 of SSR Part I.

10 CORRUGATED AND SEMI-CORRUGATED ASBESTOS CEMENT SHEETS: -

10.1 AC sheets shall comply with IS 459-1992; Specification for unreinforced corrugated and semi corrugated asbestos cement sheet and shall not be pigmented . AC sheets shall have a rectangular shape, smooth surface on the exposed side, a good appearance and shall be free from visible defects. The corrugations shall be true and regular. The edges of the sheets shall be straight and clean. Ridge covering and other AC accessories shall be of the same quality as of the sheets

11. GALVALUME STEEL SHEET

11.1 PROFILED BARE GALVALUME (ER-32-251) STEEL SHEETS

0.47/55/60mm (TCT), high strength (550 MPA yield stress minimum) anti-corrosive Bare Galvalume steel sheets coated with alloy of 55% aluminium, 43.5% zinc and 1.5% silicon of 150 grams per sq. meter mass of coating (As per ASTMA792) shall be used. The profiled sheets shall have overall width of 1090mm and laid width of 1006m, with five crests of 32mm spaced at 251mm centers

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and with two stiffening ribs of 20mm in the gap of 44mm in each valley of the sheet with anti capillary flute at end lap full return leg at end lap.

11.2 COLOR COATED PROFILED GALVALUM E(ER-32-251) STEEL SHEETS

0.45 (TCT), high strength (550 MPA yield stress minimum) anti-corrosive Galvalume steel sheets coated with alloy of 55% aluminium, 43.5% zinc and 1.5% silicon and finished with resin coat on both surfaces to 150 grams per sq. meter mass of coating (total both side as per Australian standard 1397 & ASTM-A792) shall be used. The sheets will have the coating of 20 micron of Regular modified polyester coating system over the primer of 5 micron on top side of sheet & coat of 5-7 micron of paint over primer on rear side of the sheets. The profiled sheets with anti capillary flute at end lap full return leg at end lap manufactured by Everest Industries Ltd. Ridge / Flashing / Gutter providing and fixing of formed ridges, gutters, flashings, etc. Also available in size of 8 ft, 10ft, 12 ft and 14 ft respectively.

11.3 COLOR COATED PROFILED GALVALUM E(ER-32-251) STEEL SHEETS

0.50/0.55/0.60mm (TCT), high strength (550 MPA yield stress minimum) anti-corrosive Galvalume steel sheets coated with alloy of 55% aluminum, 43.5% zinc and 1.5% silicon and finished with resin coat of both surfaces to 150 grams per sq. meter mass of coating (total both side as per Australian standard 1397 & ASTM – A792) shall be used. The sheets will have the coating of 20 micron of Regular modified polyester coating system over the primer of 5 micron on top side of sheet & coat of 5-7 micron or paint over primer on rear side of the sheets. The profiled sheets shall have overall width of 1090 mm and laid width of 1006mm, with five crests of 32mm spaced at 251 mm centers and with two stiffening ribs of 20mm in the gap of 44mm in each valley of the sheet with anti capillary flute at end lap full return leg at end lap Ridge / Flashing / Gutter Providing and fixing of formed ridges, gutters, flashings, etc.

11.4 COLOR COATED PROFILED GALVALUM E(ER-32-251) STEEL (IPPGI) SHEETS

0.50/0.55/0.60mm (TCT), formed out of CR steel as per IS-513 having the galvanization of 120 GSM (Total on both side) as per IS-277, The sheets will have the coating on top side with 5-7 micron primer and min 20-22 micron of Regular Modified Polyester (RMP) and on rear side of the sheet 5-7 micron Primer and 5-7 micron Polyester Paint, having overall width of 1090mm and laid width of 1006mm, with five crests of 32mm spaced at 251 mm centers and with two stiffening ribs of 20mm in the gap of 44mm in each valley of the sheet, with anti capillary flute at end lap full return leg at end lap. Ridge / Flashing / Gutter Providing and fixing of formed ridges, gutters, flashings, etc.

.11.5 Fasteners for steel sheet, Al sheets and AC sheets:- L-types and J-type hook bolts and nuts, mushroom head roofing bolts and nuts, and bituminous felt and steel washers shall conform to IS 730-1978. Specification for Hook-bolts for corrugated sheet roofing. Steel bolts, nuts and washers shall be galvanised. However washers for aluminium sheet shall be of plain aluminium sheet of same quality as that of corrugated sheet.l.

12 . WATER PROOFING COMPOUND

Water proofing compound shall conform to IS-2645. Specifications for integral water proofing compound. The quantity of water proofing compound shall be as recommended by the manufacturer. However, in the event of deviations the quantity of water proofing compound shall be considered 3% by weight of cement. 11.

13 . WATER PROOFING TREATMENT TO RCC ROOFS

13.1.1 Primer for Roof treatment:- It shall be polymeric bitumen primer of brand/ make as approved by EIC/GE.

13.1.2 Polymerized modified bitumen:-It shall be polymerized bitumen modified of brand/ make as approved by EIC/GE.

13.2 Polymeric water proofing membrane:-

13.2.1 Standard Membrane:- It shall be of grade of 3 mm, 3Kg/Sqm. The membrane shall meet the following characteristics and specifications:-

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Sl No.	Characteristics	
01	Softening point not less than	150 ^o Centigrade
02	Penetration at 25 ^o Centigrade, 100g , 5 sec, d mm	20-35
03	Pliability	2 ^o Centigrade to 5 ^o C Does not Break
04	Heat Resistance	Does not drip at 125 ^o C
05	Tensile strength Lengthwise, N/5 cm Crosswise, N/5 cm	> 115 > 105
06	Elongation Lengthwise,%Crosswise, %	> 300 > 300
07	Tear Strength Lengthwise, N Crosswise, N	> 20 N > 25 N
08	Reinforcement	HMHDPE Film
09	Thickness of Center core	100 micron +/- 10 Micron

13.3 WORKMANSHIP:-

13.3.1 Roofing Generally:-RCC roof slab shall be laid 1:20 slope, unless specified otherwise.

13.3.2 Water Proofing Treatment to RCC Roof:-

13.3.3 Hyperplas Water Proofing Treatment to RCC Roof - Refer clause No 11.43 and 11.44 of SSR Part I (2009) and as described in succeeding paras :-

13.3.4 Preparation of top of concrete surfaces:- Surfaces of RCC roof slab shall be wire brushed cleaned of all dirt, dust & foreign materials.

13.3.5 APPLICATION OF RENDERING COAT :-After surfaces are thoroughly cleaned and dried , surfaces of RCC roof /concrete shall be applied with one coat of 15 mm thick rendering in cement and sand mortar 1:3 mixed with water proofing compound (Integral) and top surfaces finished even and smooth without using extra cement.

13.3.5.1 Application of primer and tack coat :- This shall be followed by one coat of hyper plus primer @ 0.30 Kg/sqm and a tack coat of hot blown bitumen 85/25 @ 1.20 Kg/sqm.

13.3.5.2 Laying of polymeric membrane shall be done as per manufacturers instructions and laying procedure as under:-The membrane shall normally be laid in length at right angle to direction of the runoff gradient, commencing at the lowest level and working up to the crest. The membrane shall be first cut to required lengths, brushed cleaned of dusting materials and laid out flat on the roof. Each length of membrane prepared for laying shall be laid in position and rolled up for a distance of half of its length. The hot binding material shall be laid on the roof across the full width of the rolled membrane as the later is steadily rolled out and pressed. The excess bonding squeezed out in the ends shall be removed as laying proceeds. When the first half of the membrane has been bonded to the roof, the other half shall be rolled up and then unrolled on to the hot bonding material in the same way as per Para above. The minimum overlaps of 100 mm shall be given at the ends and sides of strips polymeric membrane. All the overlaps shall be firmly bonded by heating and fusing the layer of asphalt and melting the protective plastic film, but taking care not to overheat this, as too much heat can harm the central polyethylene film/core, which essential for water proofing function. Keep the overlap open with the help of trowel and the direct the flame uniformly inside an towards both membrane. Working the flame in opposite direction to the advance of the applicator. When the asphalt starts to flow, the applicator should press both membranes with his foot, on the opposite side to the hand holding the torch, until a thin string of fluid asphalt appears, indicating correct fusing and bonding. During these phases, the trowel should be worked with a back and forth, lengthwise motion, to homogenize the bonded area. Rest procedure may follow as per manufacturer's instructions.

13.3.5.3 Laying topping:- Roof treatment shall be carried out under strict supervision of the EIC. To ensure that the treatment is leak proof, it shall be tested by ponding just after completion of the curing period of each building. Mud mortar fillets bunds shall be made longitudinally one meter apart over entire roof to form ponds of suitable size as directed by EIC. These ponds shall be filled with potable

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water so that an average 50 mm (Min 25 mm and Max 75 mm) height of water is maintained during the test period. Test shall be carried out for 72 Hours. The contractor shall rectify any seepage noticed by making good the defective portion to the entire satisfaction of EIC/GE. Mud mortar fillets bunds shall be removed and surface made clean and tidy after completion of satisfactory testing. Satisfactory completion of test shall not absolve the contractor from his responsibility of rectification of defects, which may arise during defects liability period. .

14. GUARANTEE OF WATER PROOFING TREATMENT OF ROOF:-

14.1 The Contractor shall furnish guarantee in favour of GE for effectiveness of WPT during the guarantee period. The guarantee period shall be 10(Ten) years from the certified date of completion of the work as per contract. Any leakage/Seepage in the roof noticed during this period shall be carried out by the Contractor forthwith on demand in writing from the GE specifying the roof of building (s) effected. Notwithstanding that the water proofing treatment to the roof might have been inadvertently passed, certified & paid for, the Contractor shall carry out the rectification of the leakage, if any in roof at his own expense during the guarantee period to be specified by the GE in this demand aforesaid. The GE shall undertake such rectification work at the risk & cost of the Contractor in case of default on rectification by him.

14.2 The amount of security deposit shall be calculated as per scales laid down for calculating Individual Security Deposit on total amount of relevant item of Sch 'A' for WPT or **Rs 5,000.00 whichever is more** shall be retained out of Contractor's final bill amount towards security deposit & shall be refunded to the Contractor after expiration of the period of 10(Ten) years from the certified date of completion by the GE provided there are no leakage/ seepage in the buildings & the Contractor first have been paid final bill & have rendered a "No demand certificate" (IAFW-451) Conditions 16, 46 & 68 of IAFW -2249 (General Conditions of Contract) shall be deemed to be amended to the extent mentioned above. Alternatively, the Contractor may give a separate interest bearing fixed security deposit valid for 10 (Ten) years for this amount duly pledged in favour of GE.

14.3 The Contractor shall provide a plaster plate of required size in situation as decided by Engineer-in-Charge on the wall of bldgs. The plate shall be 10mm thick in CM 1:4 to indicate the CA No., Name of Contractor, the date of completion of the work and the probable date of expiry of guarantee for 'WPT' by engraving & painting (black). The cost plaster is deemed to be included in quoted by the Contractor under respective item of Sch 'A'.

14.4 WATER PROOFING TREATMENT TO SUNKEN FLOORS

14.4.1 The RCC slabs of sunken portion shall be laid flat having slope of 1:20 at top by increasing thickness of slab at ends and decreasing at centre. The thickness of slab shown in drawing shall be deemed at centre of slab. Further lateral slope in central portion shall also be provided. At the lowest portion a PVC relief pipe (spout) of 40mm of dia shall be provided as explained here in after. The concrete in sunken portion floor slabs shall be mixed with liquid WPC @200 ml per 50 Kg (bag) of cement or as per manufacturer's instructions. However, deviation if any shall be priced at the rate of 2% by weight of cement with dry WPC.

14.4.2 The existing surfaces both at horizontal & vertical shall be cleaned by removing all the loose mortar, laitance, dirt dust etc by chisel hammer/wire brush/jute brush and air blower machine. Two coats of elastomeric compound viz beco bond WPC- 30 (Make- Beck India Ltd/ Master coat IRC (Make-Conchem India) over one coat of neat primer i.e Super Latex (Make- Dr Fixit)/ Master bond (Make – Conchem) shall be applied on the cleaned surfaces.

14.4.3 Application of polysulphite sealant (Make- Dr Fixit/ STP/Chowskey) around any outlet pipe of 75mm bore (Approx 25mm wide and 10 to 12 mm deep) in toilet/kitchen/washing areas and the like shall be given to make water tight.

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14.4.4 Application of polysulphite sealant (Make- Dr Fixit/ STP/Chowskey) around any outlet pipe of 100 mm bore (Approx 25mm wide and 10 to 12 mm deep) in toilet/kitchen/washing areas and the like shall be given to make water tight.

14.4.5 15mm thick plaster in cement mortar 1:3 using liquid water proofing compound @ 200 ml per 50 kg (bag) of cement or as per manufacturer's instructions shall be applied at both horizontal & vertical surface. However, deviation if any, shall be priced at the rate of 2% by weight of cement with dry WPC.

14.4.6 The sunken portion shall be filled with lean concrete 1:5:10 using brick aggregate as approved by GE including fixing of sanitary items requiring embedding in cement concrete.

14.4.7 The sunken floor shall then be tested for leakage/seepage by ponding it with water for minimum 72 hours. There shall not be any seepage/dampness in the slab as seen underneath and sides.

15. STEEL AND IRON WORK :

15.1 TYPE OF STEEL : All steel to be incorporated in the work shall be procured & supplied by the Contractor under his own arrangements and at his own cost. Steel for various types of work shall be of following quality / grade: -

15.1.1 REINFORCEMENT STEEL : High strength deformed steel bars produced by Thermo Mechanical Treatment Process (TMT steel bars of grade Fe-500 D) meeting all other requirements of IS-1786.

15.1.2 STRUCTURAL STEEL:

(a) Definition of structural steel as given in Clause 10.4 of SSR Part-I (2009) shall be applicable.

(b) Following types of structural steel shall be used in the work:-

(i) Standard quality steel Fe-410-W conforming to IS-2062 for all types of steel structures including those subject to dynamic loading.

(ii) Ordinary quality steel Fe-310-S conforming to IS 1977 for frame of gauged/panelled doors, windows, guard bars, grills, hand railing, fencing posts etc.

15.1.3 GALVANISED STEEL SHEETS: ISI marked GI sheets (Plain and corrugated) of grade 'O' with 120 gm/sqm galvanized coating conforming to IS-277-1992 shall be used where ever shown on drawings.

15.1.4 FABRIC REINFORCEMENT FOR CONCRETE : It shall be conforming to IS-1566.

15.2 SUPPLY OF STEEL : All steel to be incorporated in the work shall be procured & supplied by the Contractor under his own arrangements and at his own cost.

15.3 PROCUREMENT OF STEEL:-

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(A) REINFORCEMENT STEEL

1.	RINL	TMT Bars all grades/ sizes
2	TATA	
3	SAIL	
4	Balaji Shakti	TMT Bars grade Fe 500 & Fe 500D
5	Shyam	
6	Simhadri TMT	
7	Jindal Panther	
8	SRMB	
9	Goel TMT (8-32 mm)	
10	GK TMT (8-36 mm)	
11	Super Shakti	
12	Gallant TMX (8-32 mm)	
13	Rashmi TMT(8-25 mm)	
14	SEL (Fe 500D only) (8-32 mm)	
15	Elegant TMT	
16	Nakoda TMT(8-32 mm)	
17	Neosteel (8-40 mm Fe 500D only)	
18	Kamachi (8-40 mm)	
19	ET TMT (8-36 mm)	TMT Bars grade Fe 550 & Fe 550D
20	Jindal Panther (8-40 mm)	
21	SRMB (8-32 mm)	
22	Super Shakti (8-32 mm Fe 550 only)	
23	Rashmi TMT (8-25 mm Fe 550D only)	
24	Elegant TMT (Fe 500D)	
25	Neosteel (8-40 mm Fe 550D only)	
26	Kamachi (8-40 mm)	TMT Bars grade CRS
27	Shyam	
28	Gallant TMX (8-32 mm)	
29	Neosteel (8-40 mm)	
30	ET TMT (8-36 mm)	TMT Bars grade HSCRM
31	Simhadri TMT	
32	Kamachi (8-40 mm)	

(B) STRUCTURAL STEEL

Structural steel shall also be procured directly from the main producers of structural steel viz. SAIL / RINL / TISCO / IISCO etc. In case of non availability with main producers, **the structural steel can be procured from approved secondary producers with a reduction of 5% (Five Percent) of the accepted rate** of structural steel. In case the desired section of structural steel is not rolled / manufactured by the main producers, there shall be no price adjustment in use of structural steel procured from approved secondary producers.

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LIST OF APPROVED SECONDARY PRODUCERS FOR STRUCTURAL STEEL

List of secondary producers :-

- (i) M/s Kashi Vishwanth Steel Ltd. Narain Nagar, Bazpur Road, Kashipur - 244 713, Distt : U.S. Nagar (Uttaranchal)
(Ph: 05947-262109, 262138, Fax: 05947-262103)
(E-mail kvsl_ksi@datainfosys.net, kvscorporate@kvsl.com)
Visit us at : www.kvsl.com
- (ii) M/S K L Steel Pvt Ltd Post Box No. 61,
Lal Kuan, Bulandshahar Road,
Ghaziabad (UP), Tele : 0120 – 2867911, 2867915
Fax : 0120 – 2867917
- (iii) M/S Shri Badrinarain Alloys & Steels Ltd, 95, Stephen House,
4 B B D Bag, Kolkata 700001 Tele : 033 2220 5381 / 2248 1601
FAX : 033 2248 8664
- (iv) M/S Pushpak Steel Industries Pvt Ltd, Gate No. 119, Alandi Markai Road,
Dhanore, Tah Khed, Pune, Tele – Fax 020-26444700 / 070
- (v) M/s Amba Shakti Ispat Ltd. Plot No. 06, Phase-II Industrial Area,
Kala Amb, Distt – Sirmour (HP) Tele : 01734-309983, 309986
- (vi) M/s Shree Parashnath Re-Rolling Mills Ltd.
4C, Maharshi Devendre Road 3rd floor
Kolkata – 700 007 Phone : 033-2274 0045/4475, Fax : 033-2274 0039
- (vii) M/S Tata Steel Industries, Tata Steel – Tube Division
Jeevan Tara Building, 1st floor, 5, Sansad Marg, New Delhi – 110001
Tele : 99111 2334, 264601734, 309983, 309986
- (viii) M/S K L Concast Pvt Ltd
Z-18, Naraina, New Delhi
- (ix) M/S Karam Steel Corp, Nasrali, Road, PO Box No. 56,
Mandi Gobind Garh – 147301

FABRIC REINFORCEMENT FOR CONCRETE

(C) These shall be procured directly from primary producers as approved by E-in-C's Branch and shall be ISI marked.

(D) Steel sections for railings, gates, fencing, guard bars, grills, steel chowkhat, holdfasts etc., which do not constitute structural members, can be procured from main producers / secondary producers / BIS marked manufacturers or their authorised dealers at the option of Contractor without any minus price adjustment. Tests will not be insisted upon for such steel sections.

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NOTES

(i) All finished steel shall be well and clearly rolled to the dimensions, sections and weights specified. The finished materials shall be reasonably free from cracks, surface flaws, laminations, rough, jagged and imperfect edges and any other harmful defects and shall be finished in a proper manner. Tolerance on size and weight of reinforcement bars shall not be more than that specified in clause No. 10.17.4 and 10.17.5 of SSR Part-I and as specified in IS-1786 and other relevant IS Codes.

(ii) TMT bars shall have embossing / engraved marking of trade name of producer at regular intervals.

(iii) Reinforcement Steel, Structural steel and Galvanized Steel Sheets and Fabric reinforcement for concrete may be permitted from authorized dealers of main producers in case the total requirement of steel is less than 5 tonnes.

15.3.1 TESTING OF STEEL :-

(a) The manufacturer is to carry out inspection and testing of steel in accordance with the relevant BIS provisions. The Contractor shall submit the manufacturer's test certificate in original alongwith the test sheet giving the results of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in Steel Acceptance Register as given in **Annexure IV** after due verification.

(b) (i) Independent testing structural steel by the GE shall be optional at the discretion of GE, in case of procurement of steel from main producers and testing charges shall be borne in accordance with condition 10 of IAFW-2249 i.e. testing charges shall be borne by the department, if the test results are found in order otherwise these shall be borne by the contractor.

(ii) Independent testing of reinforcement and structural steel by the GE shall be mandatory in case of procurement of steel from secondary producers and testing charges shall be borne by the contractor irrespective of the outcome of test results. In both the cases the contractor at his own cost shall provide all facilities required for the testing and cost of material consumed in tests which shall be borne by the contractor.

(iii) In case structural steel are procured from secondary producers, independent testing shall be got carried out by GE on random samples of steel drawn from various lots from a National Test House, SEMT CME, Regional Research Labs, Govt. approved Labs, Zonal labs etc. as per frequency given below :-

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**FREQUENCY FOR NORMAL MASS, TENSILE, BEND AND
REBEND TESTS OF STEEL**

Srl No.	Nominal Size	Quantity
---------	--------------	----------

Steel for Reinforcement

- | | | | |
|---|--------------------------|---|--|
| 1 | Bar size less than 10 mm | : | 1 Sample (3 specimens) for each test for every 25 tones or part thereof. |
| 2 | Bar size 10 mm to 16 mm | : | 1 Sample (3 specimens) for each test for every 35 tones or part thereof. |
| 3 | Bar size over 16 mm | : | 1 Sample (3 specimens) for each test for every 45 tones or part thereof. |

<u>Structural Steel</u>		
Tensile Test	:	for every 25 tonnes of steel or part thereof.
Bend Test	:	for every 10 tonnes of steel or part thereof

NOTE :-

(i) For various tests, acceptance criteria, tolerance etc, refer to Annexure 'B' and relevant BIS Codes.

(ii) In case of procurement of structural steel from secondary producers, testing by GE as per above frequency is mandatory. In such case, no payment shall be released to the Contractor for any consignment or part thereof and steel shall not be incorporated in the work unless satisfactory test results are obtained and the consignment is passed by GE. Schedule of procurement shall be prepared keeping in view the time lost in activities related to testing. Any items of steel, not meeting the requirements, shall be rejected and the particular consignment removed from the site by the Contractor at his own cost. The Contractor will have no claim on this account.

(iii) Three samples (3.00 m long) of each section of each consignment shall be retained at the project site till completion of the work. These samples shall be suitably marked and properly preserved.

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Annexure 'IV'**STEEL SUPPLY AND ACCEPTANCE REGISTER**

1. CA No. and Name of Work : _____ :
2. Contract No. : _____
3. Name of Manufacturer's T.C. No. : _____
4. Manufacturer : _____
5. Random Test details (a) Physical test report from _____ vide their letter No.
(Name of approved Lab / Engg College)
- (b) Chemical test report from _____ vide their letter No.
(Name of approved Lab / Engg College)
6. Types of steel, Dia & Qty
 - (a) Type - TMT / CRS
 - (b) Dia – mm
 - (c) Actual Wt-MT
 - (d) Conversion Wt - MT

	Chemical test						Mechanical test						
	Carbon%	Sulphur%	Phosphorous%	Manganese%	Silicon%	Corrosion resistant element	Wt per metre	Sress (N/mm2) 0.2% proof	Tensile strength (N/mm2)	Elongation %	Bend Test	Rebend Test	Remarks
As per IS 1786													
As per manufacturer's test certificate													
As per independent test													

Remarks with Signature

Accepted /Rejected

ContractorJunior EngineerEngineer-in-ChargeGarrison Engineer

**PARTICULARS SPECIFICATIONS (CONTD...)
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15.3.2 DOCUMENTATION:-

- (a) The Contractor shall submit original machine numbered purchase vouchers from the manufacturer for the total quantity of steel supplied under each consignment to be incorporated in the work. All consignment received at the work site shall be inspected by the GE alongwith relevant documents before acceptance. The original vouchers and test certificates shall be defaced by the Engineer-in-Charge duly indicating the CA No. and particulars of work for which used and shall be kept on record in the office of GE duly authenticated and with cross reference to the control number recorded in Steel Acceptance Register. Two separate files may be maintained in GE office duly serially numbered for maintaining original vouchers and manufacturer's test certificates in original alongwith test sheets in original or an authenticated copy thereof.
- (b) The Steel Acceptance Register will be signed by JE, Engineer-in-Charge, GE and Contractor and shall be maintained by GE.
- (c) A Steel Test Register containing results of independent and additional testing done by GE, field testing done by GE /CWE, random check done by CWE etc shall be maintained by GE.
- (d) The Accepting Officer may also order a Board of Officers for random check of steel and verification of connected documents. Hence a register containing records of surprise checks and checks done by Board of Officers shall also be maintained by GE.
- (e) The entire quantity of steel items shall also be suitably recorded in the Measurement Book for record purposes as "Not to be abstracted" before incorporation in the work and shall be signed by Engineer-in-Charge and the Contractor duly checked by GE.
- (f) Certified true copies of vouchers and test certificates including test sheets shall also be maintained by Engineer-in-Charge after defacing, with cross reference to the control No. recorded in the Steel Acceptance Register.

15.3.3 (a) STORAGE:

- (i) Steel of different grades and sizes shall be stacked separately. For each class of steel separate areas shall be earmarked. Steel shall be marked with distinct painting marks for easy identification and it shall be always stacked at least 15 cm above the GL.
- (ii) Steel shall be stored in a manner so as to prevent distortion and corrosion. Any section that has deteriorated and corroded or if considered defective for any other reason, the same shall be removed from site by Contractor at his own cost.
- (iii) Steel sections which are not likely to be used before onset of monsoon shall be given cement slurry wash so as to ensure scale and rust free steel. Also steel sections which are procured during monsoon and are not likely to be used within a week from the date of procurement shall be given cement slurry wash immediately.

(b) SAFETY OF STEEL

It shall be the responsibility of Contractor to make sure that all possible arrangements are made for safe custody of the steel. In case of any loss of steel, the Contractor only will be responsible and the loss shall be made good by him without any delay and no claim whatsoever would be entertained on this account.

15.3.4 SCHEDULING AND SUPPLYING

Schedule of supply of steel will be finalised by GE in consultation with Contractor and the same shall be incorporated in CPM chart so that supply of steel is monitored in a way to avoid any delay in completion of work. GE shall ensure that Contractor places the demand(s)/requisition(s) of steel with adequate time lead. The schedule of supply of steel shall be vetted by CWE from time to time.

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15.3.5. PROCEDURE FOR MAKING PAYMENTS FOR STEEL INCLUDING MEASUREMENTS, CONVERSION WEIGHT ETC.

(a) The requirement of steel shall be worked out section wise and shall be recorded in a separate Register jointly maintained by Contractor and Engineer-in-Charge. Day to day record shall also be signed by the Contractor as well as Engineer-in-Charge. The register should contain different sheets for each steel section indicating reference to drawing number, location, number of bars, sketch of each length of bar with dimensions, length of waste and off cuts and the quantity thereof by multiplying the length of bars with the conversion factors given in MES schedule Part-II. For sections not listed in MES schedule, the I.S. conversion table shall be followed. The contractor shall not have any claim in case the actual weight of steel items works out to be more than the weight obtained by standard conversion factor. For making adjustment, arising out of reinforcement bars, the length of each bar for the purpose of calculation of laps shall be taken as 10 meters.

(b) On completion of work, the Register will be in the custody of the Engineer-in-Charge and Contractor may keep a copy for his record, if he so desires.

(c) Payment for the steel brought by the Contractor shall be released by GE only after following actions are taken by GE and Engineer-in-Charge:-

(i) Physical verification of steel to confirm the actual quantity of steel as well as to verify the aspects brought out in clause 7.3 above.

(ii) Receipt of original machine numbered purchase vouchers of manufacturer(s) from Contractor.

(iii) Receipt of manufacturer's test certificates (in original) alongwith test sheet giving result of each mechanical test and chemical composition of steel or authenticated copy thereof duly signed by manufacturer.

(iv) Receipt of satisfactory test results for independent testing on random samples of steel in terms of clause 7.3.1(b) above.

(v) Satisfactory test result in case of field testing done by GE/CWE in case of TMT steel manufactured by secondary producers.

15.3.6 ACCEPTANCE/REJECTION OF STEEL: The Contractor shall keep a separate stack of steel brought at site for inspection, away from the accepted stack of steel. In case, the consignment does not meet any of the requirements of the relevant IS codes, the steel shall be rejected by the GE, and it shall be removed from the site within 24 hours at the cost of the Contractor.

15.4. WELDING.

(a) Refer clause 10.15 of MES Schedule Part I and shall be carried out by metal arc process all as per IS-816 and IS-823.

(b) Unless, otherwise specified here-in-after or shown on drgs welding shall be 6 mm thick continuous fillet weld and shall cover the joint.

16. FLOOR FINISHES AND PAVING :

16.1 Materials:-

16.1.1 White cement: - Refer Para 13.2.1 of SSR Part I. IT shall be an approved product of Bureau of Indian Standard (BIS) and shall be of make specified hereinafter.

16.1.2 Pigments:- Refer Para 13.8 of SSR Part I.

16.1.3 Ceramic tiles:- Ceramic tiles shall be conforming to relevant IS. Unless specified otherwise in Schedule "A"/ shown in drawings the size of tiles shall be 200x300 mm (Classic/premium of Somany or equivalent of other make specified) for walls and 300x300mm (Somany express or equivalent of other make specified) for floors. Ceramic tiles shall be of make as specified hereinafter. These shall be conforming to IS-13753 for flooring and IS-13755 for walls. Tiles shall meet the requirement of IS-13630 or relevant EN.

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16.1.4 Vitrified tiles :- These shall be fully vitrified , having water absorption less than 0.50 %, modulus of rupture $\geq 35 \text{ N/mm}^2$ and resistance to deep Abrasion $\leq 175\text{mm}^3$ as required for group B 1 a of International standard for ceramic tiles ISO 13006:1998(E) when tested as per BS EN 15045. Colour / shade and sizes of tiles shall be as specified. Tiles shall be of Granite series (group-III).

16.2 Workmanship:-

16.2.1 Cement concrete sub floors, base or sub base:-Refer Para 13.27 of SSR Part I. **Mix of cement concrete** shall be 1:5:10 type E2, unless specified otherwise in Schedule "A". In case contractor uses cement slurry before laying floors, nothing extra shall be paid to contractor for the same. The thickness of concrete for the sub base shall be 75 mm if not specified otherwise else where.

16.2.2 Cement Concrete flooring Cast in situ:-Refer Para 13.32 of SSR Part I. Floor and sub floor shall not be laid monolithic. 200 mm wide border shall be provided in all rooms. The size of panel shall not exceed 2.00x2.00 m in any case. In case glass strips are provided, wooden forms are not required and no price adjustment shall be made on account of non-provision of wooden forms. Glass strips shall be paid separately at S&F rates. PCC floors shall be 40 mm thick, 1:2:4 type B1 in residential buildings and 50 mm thick in other nonresidential buildings. Top shall be finished even and smooth using extra cement, unless specified otherwise in Schedule "A". Glass strips shall be 3 mm thick x full thickness of floors.

16.2.3 Ceramic tiles flooring/dado/skirting:-Refer Para 13.40 of SSR Part I. Ceramic tiles in flooring shall be laid/bedded in neat cement slurry @ 3 Kg/sqm over 15 mm thick screeding of mix 1:6, applied over 30 mm thick PCC type B0. Sub base shall be 75 mm thick in PCC type E2, unless specified otherwise in Schedule "A" specifically or elsewhere. In case of dado/ skirting, tiles shall be smeared with cement slurry @ 3 Kg/sqm and bedded over 10 mm thick screeding of Mix 1:4. The joints of tiles shall be finished in cement to match with the colour of tiles. Unless specified otherwise, tiles shall be white colour (First quality) of make specified hereinafter. . The tiles provided at corners shall be mitred and fixed and no unglazed portion shall be visible. The rates quoted by contractor /or given SSR for respective items shall be deemed included for the same.

16.2.4 KOTA STONE SLABS:-The slab shall be of selected quality, hard, sound, dense and homogeneous in texture free from cracks, decay, weathering and flaws. These shall be hand or machine cut to the requisite thickness. These shall be of colour as directed. The slabs shall have the top (exposed) face polished before being brought to site. The sample shall be got approved before incorporation in work. The tolerance $\pm 2\%$ shall be allowed for the thickness. Unless specified elsewhere otherwise/ shown in drawings kota stone slab shall be of size not less than 550mmx550mmX20to25 mm thick, unless stone slab of lesser sizes required due to site requirement (vary specified size is not accommodating).

16.2.5 GRANITE STONE SLABS:-The granite slab shall be of selected quality, hard, sound, dense ,durable and homogeneous in texture and free from cracks, decay, weathering and defects like cavities ,crack flaws,sand holes ,patches of loose/ soft materials and other similar defects that may adversely affect its strength and appearance. It shall be of uniform colour and texture as directed . These shall be hand or machine cut to the requisite thickness. The slabs shall have the top (exposed) face polished before being brought to site. The sample shall be got approved before incorporation in work. The tolerance $\pm 2\%$ shall be allowed for the thickness. Granite stone slab shall be of size not less than 550mmx550mmX18 to 20 mm thick, unless stone slab of lesser sizes required due to site requirement (where specified size is not accommodating).

16.2.6 KOTA STONE FLOORING: - Refer Para 13.47 of SSR Part I .It shall be laid in the manner as specified in SSR Part I Para 13.47. The edges of the slabs to be jointed shall be buttered with grey cement, with admixture of pigment to match the slab. Kota stone slab be granite finish (Mirror) polished (07 process of grinding). Kota stone flooring shall be constructed

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with providing 16 to 18mm thick, 100mm wide white marble stone border (if specified in Schedule 'A') to the pattern and design as directed by the GE and contractor's quoted rates for kota stone flooring in any location shall be deemed to include this provision without any extra cost to the government irrespective of the provision is specified or not specified, elsewhere in these tender documents.

16.2.7 GRANITE / MARBLE STONE SLAB FLOORING

18mm-20mm thick polished granite / marble stone slab shall be laid over 20 mm thick bedding layer of CM 1:4 over base as specified hereinbefore .Unless specified otherwise in Schedule "A",

16.2.8 PRECAST INTERLOCKING PAVER BLOCKS. Precast concrete Paver blocks shall conform to IS 15658:2006, Specification for Precast concrete blocks for paving. Paver blocks shall be sound and free from cracks or other visual defects. The tolerance on length or breadth of paver blocks shall be +2mm and tolerance on thickness of tiles shall be +3mm. Water absorption shall not be more than 6 percent by mass. Shapes shall be triangular, Zigzag, Haxagon or other shape as indicated. Colour of paver blocks shall be indicated or as decided by GE. Thickness and grade of concrete of paver blocks shall be indicated in BOQ.

17. PLASTERING AND POINTING:-

17.1 Materials:-

17.1.1 Sand:- Refer Para 14.5 of SSR Part I. It shall be conforming to IS-1542.

17.1.2 Workmanship:-

17.1.3 Rendering:- Refer Para 14.12,14.14 and 14.16 of SSR Part I. Unless specified otherwise in Schedule "A", mix of mortar for rendering shall be 1:6, single coat , top surfaces finished fair and even all as directed.

17.1.4 The rates of SI-2051 and 2052 or SI-2057 and SI-2058 shall be payable proportionally in cases where the plaster thickness is less than "10mm" and '12 mm ' respectively subject to contractor's percentage . This clause is applicable for measurable contracts (except item rate contracts)

17.1.5 Repairs to Plastered surfaces:- Refer Para 14.12,14.16,14.14 and 14.28 of SSR Part I. Unless otherwise specified in Schedule "A"mix of mortar shall be 1:6, single coat work, top surfaces finished fair and even .

17.1.6 The rates of SI-2051 and 2052 or SI-2057 and SI-2058 shall be payable proportionally in cases where the plaster thickness is less than "10mm" and '12 mm ' respectively subject to contractor's percentage . This clause is applicable for measureable contracts (except item rate contracts)

18 WHITE WASHING/COLOUR WASHING/ DISTEMPERING :-

18.1 MATERIALS:-

18.1.1 Lime:- Refer Para 15.2 of SSR Part I. It shall be conforming to IS-712.

18.1.2 Yellow and red Ochre:- Refer Para 15.3 of SSR Part I.

18.1.3 Blue ochre:- Refer Para 15.4 of SSR Part I.

18.1.4 Dry distemper:- Refer Para 15.5 of SSR Part I. It shall be of first quality, any of the make specified hereinafter.

18.1.5 Oil Emulsion distemper:- Refer Para 15.6 of SSR Part I. It shall of first quality and shall be one of the brands specified hereinafter.

18.1.6 Cement base paint:- Refer Para 15.7 of SSR Part I. It shall be of make as specified hereinafter.

18.1.7 Wall Care putty:- It shall be white cement based one of the Make specified hereafter for internal / external use. Compressive strength at 28 days, tensile adhesion strength at 28 days and surface hardness shall be 13.46 N/mm², 3.0N and 2.26 N/MM² respectively.

18.2 Workmanship:-

18.2.1 General:- Refer Para 15.10 and 15.11 of SSR Part I.

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- 18.2.1.1 White washing:-**Refer Para 15.12 of SSR Part I. The quantities of lime, water, gum Sodium chloride and blue etc. shall be as specified in Para 15.12 of SSR Part I. Unless specified otherwise in Schedule "A", new plastered surfaces / unplastered surfaces shall be treated with three coats of White wash White wash shall strictly be carried out as per Para 15.12.5 of SSR Part I. Old surfaces shall be treated with one or more coats as indicated in Schedule "A".
- 18.2.1.2 Colour washing:-**Refer Para 15.12 of SSR Part I. The quantities of lime, water, gum, Sodium chloride, blue and mineral colour etc. shall be as specified in Para 15.12 of SSR Part I. Mineral colour shall be added to extent required to achieve required shade/tint. Unless specified otherwise in Sch"A: new plastered /un-plastered surfaces shall be treated with two coats of colour wash over one coat of White wash and old surfaces shall be applied with one or more coats as indicated. Colour washing/white washing shall be applied strictly as per Para 15.12.5 of SSR Part I.
- 18.2.2 Wall Care Putty: -** Wall care putty shall be applied to concrete/ rendered surfaces where specifically specified.
- 18.2.2.1 Surface Preparation and application of putty:-** Remove all the loosely adhering material from the wall/ concrete / rendered surfaces with the help of emery stone, putty blade or wire brush. Moisten the surfaces with sufficient quantities of clean water.
- 18.2.2.2** Wall care putty shall be mixed with water (35-40%) slowly to make paste. Continue the mixing of material for 10-15 minutes till a uniform paste is formed.
- 18.2.2.3** Apply first coat of "wall care putty" uniformly on already moistened surfaces with the help of putty blade or trowel. Allow the surface to dry for at least three hours and then apply a second coat of putty. Leave the surface to dry completely. Always prepare the required quantity of 'wall care' Putty. And use it within 03 Hours of mixing with water. The total thickness of both the coats should be 1 to 1.5 mm (. After complete drying of the surface, if needed, the surface should be rubbed with fine water proof emery paper. The surface is ready for further application of required finishes.
- 18.2.3 Distemper with oil emulsion:-** Refer Para 15.14 of SSR Part I. Unless otherwise specified in Schedule "A" all new plastered /unplastered surfaces shall be treated with one coat of primer followed by two coats of oil emulsion and old surfaces shall be treated with one or more coats of oil emulsion as indicated. Newly Plastered surfaces e distempered, before a period for 06 Months, shall be treated with a coat of alkaline resistant priming paint. 1st coat on old distempered surfaces shall be treated as "**Subsequent coat**" for the purpose of payment under the contract.
- 18.2.4 Cement Painting:-** Refer Para 15.15 of SSR Part I. To prevent algae growth and moss growth and efflorescence, silicon base water repellant compound shall invariably be added as directed by EIC. Unless otherwise specified in Schedule "A", new surfaces shall be applied with two coats of cement base paint over one coat of cement base primer whereas old surfaces shall be applied with one or more coats as indicated in Schedule "A". 1st coat on old distempered surfaces shall be treated as "**Subsequent coat**" for the purpose of payment under the contract.
- 18.2.5 Cement wash:-** Refer para 15.17 of SSR Part I. Cement for cement wash shall be white and mixed with 1 Kg of Nacl per 160 litres of slurry.
- 19 GLAZING:-**
- 19.1 Materials:-**
- 19.1.1 Sheet Glass:-** Refer Para 16.2 of SSR Part I. Sheet glass shall be conforming to IS-2835. Unless specified otherwise it shall be of OQ, 3mm nominal thickness.
- 19.1.2 Putty:-** Refer Para 16.5 of SSR Part I. It shall be conforming to IS-419.
- 19.2 Workmanship:-**
- 19.2.1 Glazing with putty:-** Refer Para 16.6 to 16.9 and 16.13 of SSR Part I.
- 19.2.2 Glazing with beads:- :-** Refer Para 16.6 to 16.8 and 16.13 of SSR Part I. Beads shall be fixed with screws.

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19.2.3 Glazing in steel surrounds:- Refer Para 16.6 to 16.8, 16.10 and 16.13 of SSR Part I. Beads shall be fixed with screws.

19.2 PAINTING:-

19.2.1 Materials:- Refer Para 17.2 of SSR Part I.

19.2.3 Plastic emulsion Paints:- It shall be conforming to IS 5411, Out of one of the makes /brand specified hereinafter.

19.2.4 Synthetic Enamel Paint:- It shall be conforming to IS-2932 and shall be one of the make specified hereinafter.

19.2.4 Red oxide Zinc Chrome Primer:- It shall be conforming to IS-2074. It shall be one of the makes/ brands specified hereinafter. It shall be of the same make as that of finishing / under coat.

19.2.5 Coal tar Black:- It shall be of grade "A" conforming to IS 290 and shall be one of the make specified hereinafter.

19.3 Workmanship:-

19.3.1 Painting Generally:- Refer Para 17.3 to 17.5 of SSR Part I. Unless specified otherwise new surfaces of wood/wood based materials /steel shall be applied with two coats of synthetic enameled paint over one coat of primer.

19.3.2 Painting to wood work:- Refer Para 17.6 of SSR Part I.

19.3.3 Painting steel and iron work:-Refer Para 17.8 of SSR Part I. Mordant solution made with composition of 64 parts of water and copper chloride, copper nitrate, aluminium chloride and HCL acid, each one part all by weight, shall be applied to GI surfaces @ 5 litres /100 sqm before application of red oxide etc. where GI surfaces ordered to be painted. All steel surfaces unless specified otherwise in Schedule "A" shall be applied with two coats of synthetic enameled paint over one coat of primer. However old surfaces shall be applied with one or more coats as indicated.

19.3.4 Tarring or Painting with coal tar:- Refer Para 17.12 of SSR Part I. Tarring shall be done with coal tar black paint type "A" conforming to IS 290 of make as approved.

20 WATER SUPPLY AND SANITARY:-

20.1 Materials:-

20.1.2 Cast Iron Pressure pipes and fittings :- Refer Para 18.2 of SSR Part I.

20.1.3 Centrifugally Cast (spun) Iron pipes :- These shall be conforming to IS-1536. Unless specified in Sch'A' pipes shall be of class 'B'.

20.1.4 Vertically Cast iron Pressure Pipes :- These shall be conforming to IS-1537. Unless specified otherwise in Sch'A' , these shall be of class 'B'.

20.1.5 Cast iron fittings for pressure pipes:- These shall be conforming to IS-1538.

20.1.6 Rubber Gasket for jointing:- Refer Para 18.11 of SSR Part I.

20.1.7 Sluice Valve :- Sluice valves shall be of pressure rating PN1.6 unless specified otherwise in schedule "A".These shall be an approved product of BIS having ISI certification Mark i.e ISI-780. These shall be of cast iron with stainless steel spindle non rising type with iron wheel head , both end flanged.

20.1.8 Non return Valves /Reflux or check valves : These shall be conforming to IS-5312 with out bypass arrangements. These shall be of Pressure rating PN 1.6 , unless specified otherwise in Sch'A'.These shall be of cast iron with rings and stem of AISI 410 or as requirement of IS.

20.1.9 Column pipe :-These shall be of mild steel of dia as specified in Schedule 'A', both end threaded and fitted with cast iron flanges at both ends and welded as directed by EIC. Flange shall be conforming to IS-1538 . Pipe shall be un-galvanized and an approved product of BIS and shall have ISI certification mark as ISI-1239 embossed on it. It shall be of heavy grade.It shall be of 3.00 M lengths unless specified otherwise in schedule 'A'.

20.1.10 Mild steel galvanized pipes (Tubes) and fittings:-Refer Para 18.4 of SSR Part I. Pipes shall be of any make an approved product of Bureau of Indian standard and ISI certification mark. Unless specified otherwise in Schedule "A"these shall be of medium grade. Fittings shall also be ISI marked of Make – NMC or RAM as approved.

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20.1.11 Caulking lead: - Refer Para 18.9 of SSR Part I. Irrespective of whatever is specified else where, pig lead shall be used in jointing, conforming to IS 782.

20.1.12 Spun Yarn: - Refer Para 18.10 of SSR Part I.

20.1.13 Bib Taps and stop valves: - Refer Para 18.14 of SSR Part I. These shall be ISI marked. The handles shall be of crutch type. These shall be polished bright. Stop valves shall be externally threaded and the weight of bib taps and stop valves shall be not less than given in Para 18.14 .1 of SSR Part I.

20.1.14 Fancy bib taps and stop valves: - Refer Para 18.15 of SSR Part I. These shall be ISI marked.

20.1.15 Pillar taps: - These shall be of brass, ISI marked, chromium plated.

20.1.16 Fancy pillar taps:- Refer Para 18.16.1 and 18.16.2 of SSR Part I.

20.1.17 Ball valves:- Refer Para 18.19 of SSR Part I. These shall be of brass, high-pressure type and shall conform to IS-1703, horizontal plunger type. The float shall be of polyethylene, high-pressure ball valves. The minimum weight shall be as specified in Para 18.19 of SSR Part I.

20.1.18 Gate Valves: - These shall be of dia as specified in Schedule "A" and shall be conforming to IS-778.

20.1.19 Waste fittings: - Refer Para 18.20 of SSR Part I. Waste fittings for wash basins and sinks shall be 32 mm dia. and 40 mm dia. respectively.

20.1.20 Waste plugs: - Refer Para 18.21 of SSR Part I.

20.1.21 CI soil, waste and vent pipes: - Refer Para 18.23 of SSR Part I. These shall be of CI, sand cast, ISI marked, ear type, spigot and socketed. CI fittings shall be conforming to IS-3989, medium weight.

20.1.22 CI cowls:- Refer Para 18.24 of SSR Part I. Unless otherwise indicated cowl shall be suitable for 50 mm nominal bore pipes.

20.1.23 Nahani Traps:- Refer Para 18.24 of SSR Part I. These shall be consisting of gratings.

20.1.24 SGSW pipes and fittings:- Refer Para 18.28 of SSR Part I. These shall be of commercial grade. These shall be an approved product of BIS and shall have ISI certification mark. The rates given in SSR shall be deemed amended for the same.

20.1.25 Concrete pipes:- Refer Para 18.29 of SSR Part I. Unless otherwise indicated these shall be of NP3 , non pressure type, ISI marked.

21 UPVC SOIL , WASTE AND RAIN WATER (SWR) DRAINAGE PIPES:-

21.1 Refer Para 18.27 A of SSR Part I. These shall be of type 'B' unless specified otherwise.

21.2 These shall be an approved product of BIS and shall have ISI certification mark as ISI-13592 for work above ground and ISI-15328 for underground drainage & sewerage work Rubber rings shall be conforming to IS-5382. Pipe fittings shall be ISI-14735.

21.3 UPVC pipes for Potable water supplies: - These shall be ISI marked and approved product of bureau of Indian Standards having ISI-4985. Unless specified otherwise these shall be class 3. PVC fittings shall be an approved product of BIS having ISI certification mark as ISI-7834.

21.4 Sanitary fittings and appliances:-Refer Para 18.32 of SSR Part I. These shall be of first quality.

21.5 Wash down water closet: - It shall be of pattern 2 and of colour as specified.

21.6 Squatting pans:- These shall be Orrisa pattern ,580x440 mm, of approved colour.

21.7 Semi pedestal pattern water closet :- These shall be conforming to relevant IS.

21.8 Foot Rest:- Refer Para 18.32.4 of SSR Part I.

21.9 Wash basins:- Refer Para 18.22.5 of SSR Part I. Unless otherwise specified specifically in Schedule "A", these shall be of size 630x 450 mm. Waste fittings of brass (CP) ,32 mm dia. shall be provided.

21.10 Sinks:- Refer Para 18.32.6 of SSR Part I. These shall be of vitreous china white.

21.11 Stainless steel Sinks: - These shall be of stainless steel of shape and size as specified in Schedule "A" made out of 1.00 mm thick sheet and fixed as directed by EIC. These shall be of make as specified herein after.

21.12 Urinals bowl type: - Refer Para 18.32.7.1 of SSR Part I. These shall be flat back type/ angle back type as indicated.

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21.13 Flushing cistern: - Refer Para 18.34 of SSR Part I. Flushing cistern shall be low level type. These shall be of moulded polypropylene virgin. The capacity of cistern shall be 10 litres. Flush pipe shall be of brass, 40 mm nominal dia.

21.14 Plastic water closet seat and cover: - Refer Para 18.36 of SSR Part I. These shall have recessed bottom. Seats and covers shall be of any of the materials as directed by EIC.

22 HDPE Water tanks:- These shall be conforming to IS-12701. These shall be of multilayer.

22.1 Mirror:- The size of mirror shall be 600x450mm, unless specified otherwise. It shall be of SQ. The thickness of sheet glass shall be not less than 6 mm, with beveled edges. Back shall be fitted with plywood, 12 mm thick BWP type, 5 ply and provided with aluminium "C" channel all-round to suit the thickness of glass (mirror) including plywood. Mirrors shall be of make Atul or Swastic. The rates given in SSR shall be deemed for the same.

22.1.1 WORKMANSHIP:-

22.1.2 Water supply: - CI pipes:- Refer Para 18.4 to 18.47 and 18.54 to 18.56 of SSR Part I.. Jointing of CI pipes (spigot and socketed) shall be done in any of the following manner as specified in Schedule "A", in the manner given in Para 18.48 of SSR Part I. **(a) Run lead joints (b) Lead wool joints (c) Rubber gaskets joints (d) Cement joints (e) Flanges joints. The quantity of lead wool /lead shall be as specified. In case, there is any variation (less lead) , it shall be adjusted through necessary price adjustment i.e. @ Rs. 80.00 per Kg of less lead provided..** The jointing materials used between flanges of the pipes shall be compressed fibre board 1.50 to 3.00 mm thick.

22.1.3 Galvanised pipes (external work):- Refer Para 18.41 to 18.47, 18.50 and 18.54 to 18.56 of SSR Part I.

22.1.4 Galvanised pipes (Internal work):- Refer Para 18.41 to 18.47, 18.51 and 18.54 to 18.55 of SSR Part I. The pipes which are embedded in walls, ceilings or floors shall be provided with bituminous paint.

22.1.5 CI soil waste and vent pipes:- Soil waste and vent pipes shall be of cement jointed or as specified in Schedule "A" and shall be fixed in manner as specified in Para 18.67 of SSR Part I.

22.1.6 Laying of SGWG pipes /RCC Hume pipes:- Refer Para 18.68 to 18.76 and 18.79 of SSR part I. Unless specified otherwise, PCC for bedding , haunching and surrounding or encasing shall be of mix type D2. Bedding, haunching and encasing as specified in Schedule "A" shall be provided. Testing shall be done by smoke test, for pipe above GL and by water test if pipe is below GL.

22.1.7 Laying and jointing of UPVC Soil , waste and rain water pipes (SWR) drainage pipes:-

22.1.8 Refer Para 18.67.7A of SSR Part I.

22.1.9 Fixing Sanitary fittings:- Refer Para 18.80 to 18.93 of SSR Part I.

22.1.10 Water closet (Indian type):- Refer Para 18.86 of SSR part I. These shall be embedded in PCC 1:5:10 type e2 using brick aggregate. The pan shall be provided with a 100 mm dia. CI "P" or "S" Trap. The rates quoted by contractor in Schedule "A"/ given in SSR shall be deemed as that for CI'P' or "S" traps.

23 Testing:-

23.1 After the joints are complete, the trenches shall be partially refilled between the joints and sides of trenches to provide anchorage. All specials shall be adequately supported to prevent movement, when the test pressure is applied.

23.2 The testing shall be carried out in sections. Testing of all GI Pipes /CI mains and fittings shall be carried out in accordance with stipulated requirements in SSR Part I and IS 1592. Testing shall be carried out by the contractor in presence of the EIC and approved by the GE. The pipe lines shall be tested in sections with the help of hydraulic pump which shall be arranged by the contractor under his own cost and the rates quoted in Sch'A' shall be deemed to include the same. The returning, filling in and ramming of the trenches shall than be carried out by contractor after the results are declare as pass by GE.

23.3 The pipes lines shall be tested in sections by subjecting a test pressure equal to 1.50 tomes the actual operating pressure. The test pressure shall be increased gradually rising about 1 Kg per minute.

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The pressure gauge must be placed at the lower level of section.. It must have 1.10 Kg calibration scale. The pressure must be maintained in mains till a careful check is carried out to pipe and joint to joint. The lines to be tested must not exceed 500 meters at a time. The lines to be tested shall be allowed to remain idle and filled with water for about 48 hours before the actual test is done, so that it has absorbed the specified amount of water as per its properties before the test is done. The closing arrangement at the end of the section under tests shall be much so as to resist the end thrust. The hydraulic test pressure may be applied through a sluice valve tee with the help of hydraulics pump.

23.4 Any failure of pipeline connections etc. to the specified test and any defects, which may be detected while the pipes and fittings, are under test must be made good by the contractor. New test shall than be carried out and the repeated until whole of the work is thoroughly found satisfactory and water tight and approved by the GE. The contractor shall arrange for testing and provide all facilities at his own cost.

23.5 Rates of pipes, fittings and accessories etc. shall be deemed to include for providing apparatus, materials, labour etc. for testing pipes under pressure for which no extra claim whatsoever shall be entertained.

23.6 The results of the same shall be recorded and signed by the contractor and EIC and shall be countersigned by GE.

23.7 Water for testing purposes shall be supplied by the department on payment, if so desired by contractor.

23.8 Valve pits:- Valve pits shall be built in locations as shown in drawings / directed by EIC. These shall be as shown in drawing for respective dia. of valve to be provided. These shall be constructed with the specifications given hereinafter in Annexure "G".

24 Making good Roads/paths etc.:-

24.1 Roads /paths, drains, culverts etc. excavated/ dismantled by the contractor for laying pipes etc. shall be made good by him to the original specifications and to the entire satisfaction of EIC. Rates inserted in Schedule "A" shall be deemed to include for such making good (unless items of making good catered for in Schedule "A" specifically).

24.2. In internal water supply work, GI bends will be provided in lieu of elbow, if not possible due to site conditions, prior permission shall be obtained in writing from Engineer-in-charge.

25 **RECORD DRAWINGS.** Three copies of line plan of completed work, indicating line of pipes, size, positions of fittings etc shall be submitted by the contractor on tracing cloth to the Engineer-in-Charge on completion of work on tracing cloth.

26 ANTITERMITE TREATMENT:-

26..1 Post construction ant termite treatment shall be carried out as described in BOQ and as specified in clause 3.26 of MES schedule Part-I. Ant termite treatment shall be got done by the main contractor through specialized agency holding valid license as per clause 13 of Insecticides Act-1968. The contractor shall furnish particulars of agency to the GE for his approval and obtained approval prior commencement of ATT work.

26.2 Should the GE any time during construction or reconstruction or prior to the expiration of period of ten years after the ant termite treatment has been completed as per contract, finds that the buildings have been infected with termites, the Contractor shall, on demand in writing by the GE specifying the building(s) complained of, notwithstanding that the same may have been inadvertently passed, certified and paid for, forthwith undertake to carry out such treatment as may be necessary to render the said building(s) free from termite infestation at his own expenses for a period of 10 years from the certified date of completion, and in the event of his failing to do so within a period to be specified by the GE in his demand aforesaid, the GE may undertake such treatment at the risk and expense in all respects of the contractor. The liability of the contractor under this condition shall not extend beyond the period of

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10 years from the certified date of completion unless the GE had previously given notice to the Contractor.

26.3 The Security deposit referred to be held from the contractor against the guarantee period for ant termite treatment shall be calculated as per scales laid down for calculating "Individual Security Deposit" on the amount of ant termite treatment at contract rates which treatment is to be done and may be refunded to the contractor after the expiration of the period of ten years from the certified date of completion by the GE, provided always that the Contractor shall first have been paid the final bill and have rendered a No- Demand Certificate (IAFA-451) . The facility of furnishing fixed deposit receipt/BGB in lieu of the sum to be retained as security may be accepted

27 INTERNAL ELECTRIFICATION:-

27.1 **MATERIALS AND SAMPLE BOARDS**. All materials shall be ISI marked. In case ISI marked product are not available these shall conform to relevant IS specifications or to BS specifications where ISI is not existing. In absence of any IS specifications these shall be as approved by GE.

27.1.2. Sample of all materials, fittings and fixture to be supplied by the contractor shall be submitted to GE for his approval. The contractor shall not commence his work until the samples are approved in writing by the GE. The contractor shall ensure that all the materials incorporated in the work are identical in all respect. Samples approved and samples destroyed in testing shall be returned to the contractor after completion of contract. No payment shall be made for samples destroyed in testing.

27.1.3. Of the makes specified (if any) materials shall be arranged of the best available make as approved by the GE. Out of different makes as specified in CA, the makes approved by the GE shall only incorporated in the work.

27.1.4. All manufactured articles required for incorporation shall be brought to site in the manufacturer's original packings with the seal intact. Incorporation shall be done when approved by Engineer-in-Charge.

27.1.5 GENERAL:-

27.1.6 Refer Para 19.24 to 19.49 of SSR Part I.

27.1.7 Cables:- These shall be of make as specified here in after. Unless specified otherwise, aluminium conductor cables shall be of 2.5 sqmm nominal cross section area and 4 sqmm nominal cross section area for 6 Amps and 16 Amps points respectively. Total points on a circuit shall be not more than 6 points or load of 800 watts, whichever is less and one circuit for each power point. Wiring diagram/ layout may be ascertained from EIC before quoting the tender. Where the total load to be fed is exceeding 1 KW, it shall be controlled by a MCB and less than IC Switches shall control 1KW.

27.2 Materials:-

27.2.1 Cables for internal wiring for light ,power and sub mains:-

27.2.2 Wiring in conduits (surface/ concealed):- Irrespective whatever is specified in Schedule "A" Cable for wiring in conduits shall be insulated unsheathed, Cable shall be of copper/Aluminium conductors (as mentioned in schedule 'A') , multi- stranded and an approved product of Bureau of Indian standard , duly marked with ISI-1596 or PVC insulated cables ISI:694. Cables shall be PVC insulated, unsheathed.

27.2.3 The rates quoted by contractor in Sch'A' or given in SSR for sheathed cable shall be deemed for the unsheathed cable. No price adjustment shall be made on this account. The contractor shall keep this aspect in mind while quoting the item rates or percentage above/ below SSR.

27.2.4 Wherever in Schedule 'A' , Conduit fittings / point wiring mentioned as 'Concealed " the rates of Schedule 'A' items shall be deemed included for cutting chases for conduits, conduit fittings and sunk Pressed steel boxes.

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27.2.5 Unless specified otherwise specifically, all the point wiring shall be done in PVC surface conduits and fittings with sunk pressed steel boxes and plastic laminated sheets.

27.2.6 Weather Proof Cables:- Refer Para 19.26 of SSR Part I. These shall be single core or flat twin core with Aluminium conductor, an approved product of BIS, duly ISI marked.

27.2.7 Flexible cords, twisted with copper conductors:- Refer Para 19.27 of SSR Part I. These cords shall be twin core with tinned annealed stranded copper conductor conforming to IS 9968.

27.2.8 Wooden boards and blocks:- Refer Para 19.28.2 and 19.28.3 of SSR Part I. Wooden boards / blocks shall be of seasoned shisham wood. These shall be of dovetailed joints. Thickness of back and face shall be not less than 6 mm and sides not less than 20 mm. The boards shall be finished with 2 coats of shellac varnish. Round blocks shall be single, varnished, 40 to 50 mm thick and 80 to 100 mm in dia. The rates given in SSR shall also be deemed for the shisham wood.

27.2.9 Conduit and conduit fittings:- Refer Para 19.29 of SSR Part I. The conduit and conduit fittings shall be of rigid steel conduit or PVC conduit as specified in Schedule "A". Steel conduit shall be stove enameled. Conduit pipes shall be ISI marked. PVC conduit shall be of medium grade. The dia. Of pipe/conduits shall be not less than that specified in clause 19.125 of SSR part I.

27.2.10 Wooden gutties:- Refer Para 19.30 of SSR Part I. All wooden gutties shall be of 1st class HW shisham, well seasoned. The rates given in SSR/ Schedule "A" shall be deemed of the shisham wood.

27.2.11 Screws and fasteners:- All screws shall be of Cadmium plated Iron.

27.2.12 Ceiling Roses:- It shall be surface type, an approved product of BIS and shall bear ISI certification mark. It shall consist of 3 terminals and out side dia. shall be not less than 63.5 mm.

27.2.13 Conical shades:- Refer Para 19.33 of SSR Part I. The dia. of shades shall be as indicated. It shall be stove enameled white inside and gray outside.

27.2.14 Bulk Head fittings:- These shall be of die cast aluminium and fitted with porcelain lamp holder, vitreous enameled reflector with hinged water tight front with key and wire metal guards as per electrical plate No.6. In case hinged front is not available, it shall be as approved by EIC without any price adjustment.

27.2.14 Sunk type boxes:- Refer Para 19.38 of SSR Part I. Boxes for housing electrical accessories and recessed into a wall shall be of sheet metal, conforming to IS 5133. Boxes shall be painted with two coats of red oxide internally and externally.

27.2.15 Water tight switches:- Refer Para 19.39 of SSR Part I. It shall be an approved product of BIS, having ISI marking as 3854. It shall be mounted in a Galvanised CI box provided with lever and handle.

27.2.16 Sockets outlets, lamp holders, flush type switches etc.:- These shall be an approved product of BIS and shall have ISI markings (except 5 pin socket outlets).

27.2.17 Distribution fuse Boards and cutouts:- Refer Para 19.44 of SSR Part I. These shall be of make specified. DB when used with MCB shall be of make as that of MCB.

27.2.18 Switch fuses:- Refer Para 19.45 of SSR Part I. These shall be of sheet metal an approved product of BIS having ISI certification mark a4064. Fuse used in switch shall be ISI 2086.

27.2.19 MCBs:- MCB s shall be a product of BIS and shall have ISI marking –8823. Current rating shall be as specified. Rupturing capacity shall be not less than 6 KA.

27.2.20 CFL (Compact Florescent lamp) Retrofit:- These shall be of capacity as specified in Schedule 'A'. These shall be ISI marked an approved product of BIS. ISI-15111. PF shall be more than 0.50

27.3 Workmanship:-

27.3.1 Refer Para 19.101 to 19.147 of SSR Part I including appendix.

27.3.2 Wiring shall be carried out on battens/ in conduit wiring system recessed or surface type as specified in Schedule "A".

27.3.3 The numbers of cables allowed in the PVC rigid conduits shall not exceeds than specified in clause 19.125 of SSR Part I.

27.3.4 Colour coding shall be strictly followed as per clause 19.123 of SSR Part I and work shall be got done by licensed electricians, necessary proof thereof shall be produced to EIC by contractor.

27.3.5 Before commencement of work, EIC and contractor shall prepare a line diagram showing system and locations of all fittings and fixtures jointly. These shall be kept on record.

27.3.6 In case of work of rewiring, existing point wiring / accessories shall be dismantled/ taking down only when new wiring has been completed in all respects.

**PARTICULARS SPECIFICATIONS (CONTD...)
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28.1 EARTHING.

28.1.1. Unless otherwise specified in Schedule 'A' or elsewhere provide earthing all as per electrical plate No. 5 on Srl page 19.49 of MES schedule Part I including PCC chamber, GI light grade pipe protection funnel, RCC cover 1:2:4 type B-1 reinforced with MS round bars 6 mm dia 150 mm c/c both ways and earth work and excavation. GI pipe protection to earth wire shall be provided upto main board. Excavation shall be in any type of soil and surplus soil shall be removed away to a distance of 50 metre and site left clean and tidy. PCC for chamber shall be 1:2:4 type B-1. The unit rate shall include for the above provisions.

28.1.2 The cable lugs at the end of earth wire shall be connected with 03 Nos 6 mm dia galvanised iron rivets spaced not more than 20 mm centre to centre. Excavation and earth pit shall be passed by Engineer-in-Charge. The contractor shall then execute installation of earth plate in the presence of authorised MES representative. Charcoal dust and return filling shall be done in layers not exceeding 150 mm thick each layer properly watered and rammed. Surplus spoil shall be removed away and site left clean and tidy.

29. TESTING.

Refer para 19.14.6 of MES Schedule Part I. The work will be completed only after such testing and the record of such testing will be certified as satisfactory by Engineer-in-Charge and shall be signed by the Engineer-in-Charge and contractor jointly. The contractor at his own cost shall provide all instruments, material and labour required for testing.

30. EXTERNAL ELECTRIFICATION:-

30.1. **General:-** Refer relevant clauses of excavation and earth work, concrete, brick work, plastering, painting, white washing, steel, form work, electrical fittings/ accessories of particular specifications for civil works and internal electrification generally and shall be followed for specific provisions given in Schedule "A" or specified herein these documents.

30.2. Materials and general specifications:-

All equipment's and materials shall be new and shall conform to the relevant "Indian Standard" (Latest edition) / British Standards in case, Indian Standards are not available for a particular equipment or item(s), unless specified otherwise in Schedule "A".

30.3. The contractor shall produce test certificates from the manufacturer's that the equipment's/ items conform to the relevant IS/BS.

30.4. The various equipment's shall be any of the makes indicated hereinafter at the contractor's option subject to the approval of EIC/GE. However the equipment's/ articles shall be conforming to relevant IS/BS.

30.5. The contractor shall submit complete technical details and specifications, illustrated technical pamphlet/ literature of the equipment offered by him under this contract at the time of submission of his tender.

30.6. All similar equipment and similar materials shall be of same make and shall be interchangeable.

30.7. The particular given hereinafter are only to particularize, amend and/ or emphasis the aforesaid specifications in the MES SSR Part I, which are not repeated here.

30.8. The contractor shall be responsible for supplying, installing, connecting, commissioning and testing all equipment and fittings according to sound Engineering practice and tested as specified hereinafter to the entire satisfaction of EIC.

**PARTICULARS SPECIFICATIONS (CONTD...)
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30.9. Standard of quality and workmanship:-

30.9.1 All electrical works shall be carried out by properly skilled and licensed electricians under the supervision of qualified electrical supervisors /Engineers. The contractor shall on demand by EIC produce such evidences of qualifications of his workmen/ supervisors /Engineers, either at commencement of the work or at any time thereafter during the currency of the contract. The entire work shall be of high class with best workmanship and to the entire satisfaction of EIC/GE.

30.9.2. The equipment specified in the contract is for important installation, where continuity of services is very essential. The design and manufacture of the equipment and the materials used, therefore, should be of the highest standard to ensure continuous and trouble free services.

30.10. POWER CABLES

(i) The tenderer shall submit complete test certificates giving full particulars/specifications of cable, constructional details and maximum continuous current carrying capacities in ground, air and induct along with standard design conditions. It shall be guaranteed by the tenderers that the cable offered by them shall be of specified sizes/makes and shall carry normal load as well as short circuit current in the event of any fault at ground conditions of Bareilly.

(ii) All cables shall be as mentioned in Schedule 'A' and shall be laid/fixed and tested strictly in accordance as specified in clause 19.73 to 19.86, 19.92 to 19.94 of SSR Part-I(2009).

(iii) Wherever cables will be crossing or running parallel to existing telephone cables, suitable measures will be taken and distance will be observed as per latest IS on the subject.

(iv) LAYING OF CABLES : Cable shall be laid all as specified in respective clauses of SSR Part-I.

(v) SAND CUSHIONING / FILLING : Sand cushioning / filling in trenches shall be carried out all as specified and as directed by the Engineer-in-Charge. Sand shall be free from foreign matters and shall be river fine sand.

(vi) HT CABLE END BOXES : HT cable end boxes outdoor type, of specifications shall be provided as specified in Schedule 'A' and as mentioned in clause 19.87 of SSR Part-I at locations as directed by the Engineer-in-Charge.

(vii) TESTING OF CABLE : The cable shall be tested before laying and after laying and jointing as specified in clause 19.93 of SSR Part-I.

(viii) All the expenses incurred on tests including provision of testing apparatus required to carry out shall be borne by the contractor without any extra cost to the Government. If the result of tests are not found satisfactory, the contractor shall at his own expense, rectify/replace the defective /materials or any part thereof as directed by the Engineer-in-Charge. The decision of the Accepting Officer shall be final, conclusive and binding in this respect.

30.11. TESTING OF CABLES :

(a) The new cables shall be tested before jointing as well as after completion of jointing cables. Cables shall be tested for:-

- (i) Continuity
- (ii) Insulation resistance to earth.
- (iii) Insulation resistance between conductors.
- (iv) Absence of cross phasing.

(b) On completion of laying/fixing and jointing the following tests shall be carried out :-

- (i) Insulation resistance test of sectional overall.
- (ii) Continuity test of overall conductor & earthing.
- (iii) Earth test.
- (iv) Full load Test.

**PARTICULARS SPECIFICATIONS (CONTD...)
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30.11.1 The results of aforesaid tests shall be recorded jointly and signed (in triplicate) by the contractor and the Engineer-in-Charge.

30.11.2 The new cable shall be tested before jointing as well as after completion of jointing of cables. Cables shall be tested as per details given in Para 19.93 & 19.94 of SSR Part I.

30.11.3 The results of aforesaid tests shall be recorded joints and signed (in triplicate) by the contractor and the Engineer-in-Charge.

31. GENERAL

If any of the specifications are not given above, the materials used shall either bear ISI marking or conforming to ISS (If ISI marked not manufactured) and workmanship shall be all as per MES Schedule 2009. Part-I.

Note : Para Numbers/clause Nos SSR part I wherever written wrongly shall be read as respective correct clause/paras of SSR Part I (2009) only.

SIGNATURE OF CONTRACTOR

DATED : _____

AGE (Contracts)
For Accepting Officer

**PARTICULARS SPECIFICATIONS (CONTD...)
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Appendix 'A'

**LISTS OF ISI/BIS/CERTIFIED (MARKED) PRODUCTS
TO BE INCORPORATED IN WORKS**

SI No.	Material	IS No.
1	Concrete	
	(a) Integral water proofing compounds	IS:2645
	(b) Cement Ordinary Portland	IS:8112
2	Joinery	
	(a) Wooden flush door shutters, solid core type	IS:2202 Part 1
3	Builders Hardware	
	(a) Steel Butt Hinges	IS:1341
	(b) Non ferrous butt Hinges	IS:205-1978
	(c) Ferrous Tower/ Barrel Bolts	IS:204 Part I
	(d) Non- Ferrous Tower/ Barrel Bolts	IS:204 Part II
	(e) Door Handles (Non Ferrous)	IS:208
	(f) Parliament Hinges , Ferrous	IS:362
	(g) Hydraulically operated Door closers	IS:3564
	(h) Continuous piano hinges	IS:3818
	(j) Non Ferrous metal sliding door bolts	IS:2681
	(k) Tee and strap Hinges	IS:206
	(l) Mild steel sliding door bolts	IS:281
	(m) Hasp and staples	IS:363
4	Roof covering	
	(a) Bitumen felts of water proofing and damp proofing (Hessian felt)	IS:1322
	(b) Glass fiber base felts type 2 grade II	IS:7193
5	Ceiling and lining	
	(a) Plywood for General purposes	IS:303
	(b) Untenured particle Board	IS:3087
	(c) Fiber Hardboard	IS:1658
	(d) Medium density fiber board	IS:12406
	(e) Veneered Particle Board	IS:3097
6	Flooring	
	(a) White Portland cement	IS:8042
	(b) Cement concrete flooring tiles	IS:1237
7	Water supply, plumbing , drains and sanitary appliances	
	(a) Concrete pipes with or without reinforcement	IS:458
	(b) Salt Glazed stone ware pipes	IS:651
	(c) Centrifugal cast (spun) iron spigot and socket soil, waste and vent pipes, pipes fittings and accessories	IS:3989
	(d) UPVC soil , waste and rain water pipes	IS:4985
	(e) Sand cast iron Spigot and socket soil waste and ventilating pipes and fittings	IS:1729
	(f) Galvanised Mild steel tubes and fittings	IS:1239 (Part I and Part II)
	(g) Vitreous China Sanitary appliances	IS:2556
	(h) Plastic water closet seats and covers	IS:2548

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**LISTS OF ISI/BIS/CERTIFIED (MARKED) PRODUCTS
TO BE INCORPORATED IN WORKS**

	(j) Flushing cistern for water closets and urinals other than plastic	IS:774 , IS:2326 (automating flushing cistern)
	(k) Ball valves (Horizontal plunger type) including floats for water supply purposes	IS:1703
	(l) Cast copper alloy screw down bib taps and stop valves	IS:781
	(m) Pillar taps	IS:1795
8	Electrical works	
	(a) Ceiling rose	IS:371
	(b) Tumbler switches	IS:3854
	(c) Socket outlet- 3 pin plug and socket	IS:1293
	(d) Switch fuses (Main & Switch)	IS:4064
	(e) Rigid steel conduit	IS:1653
	(f) Rigid non metallic Conduit	IS:2509
	(g) Single core cable polyethylene insulated and PVC sheathed cable	IS:1596
	(h) Starter for tube light	IS:2215
	(j) Fluorescent lamps	IS:2418
	(k) Aluminum and steel cored stranded conductor	IS:398
	(l) MCB	IS:8828
	(m) PVC insulated cables	IS:694
	(n) paper insulated lead sheathed cables	IS:692

SIGNATURE OF CONTRACTOR
DATED : _____

AGE (Contracts)
for Accepting Officer

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LIST OF PRODUCT APPROVAL

S. No.	PRODUCT NAME	NAME OF MAKE / MANUFACTURER
1	Floor and Wall tiles	
	Ceramic & Vitrified tiles	Kajaria, Somany Ceramics Ltd, Johnson, Orient Bell Ltd (Brand-Orient bell), Aparna Enterprises Ltd. Sunshine Tiles Company Pvt Ltd (Brand Sun heart)
	Cement Concrete Inter locking Paver Blocks	Maa Vineeta Industries, Mehtab Tile, Tulip Infrastructure Saharanpur.
	Chequered Tiles any type	Mehtab Tile.
	Wooden Flooring	Green panel Industries Ltd, Action Tesa, Armstrong
2	Paints	
	Exterior Paint	Asian Paints Ltd, Berger Paints India Ltd, Kansai Nerolac Paints Ltd, JK MAXX PAINTS
	Interior Paint	Asian Paints Ltd, Berger Paints India Ltd, Kansai Nerolac Paints Ltd, JK MAXX PAINTS
	Enamel/Oil Paint	Asian Paints Ltd (<i>Apcolite Premium Gloss</i>), Berger Paints India Ltd (<i>Luxol Hi-Gloss</i>), Kansai Nerolac Paints Ltd (<i>Synthetic Hi-Gloss Enamel</i>)
	Cement based Paints	Super Snowcem, Duracem Plus,
	High Build Textured & Wall finishes	Asian Paints Ltd, Berger Paints India Ltd, Kansai Nerolac Paints Ltd, JK MAXX Paints
	Plastic Emulsion Paint/Synthetic Enamel Paint	JK MAXX PAINTS
3	Factory made Doors/Windows, Ventilators & Frames	
	UPVC shutters for doors, windows, ventilators & Frames	Fenesta, Koemmerling, Aluplast, Prominace,
	PVC Doors & Choukhats & Windows	Rajshri Plastiwood, JKD
	UPVC Doors & Windows	Aparna Enterprises Ltd
	PVC shutters for doors, windows, ventilators & Frames	Yamuna Interiors Pvt Ltd (Brand - Black Cobra)Yamuna Nagar, Kumar Arch Tech Pvt Ltd (Brand - Eeocell), JKD
	WPC doors, windows & Frames	Rajshri Plastiwood, GIZA, Ecocell, Yamuna Interiors Pvt Ltd (Brand - Black Cobra)Yamuna Nagar, JKD
	Pre-painted steel steel windows, ventilators & door frames	Tata Praves, Elixir Met Form Pvt Ltd, Ashish Industries
	Wooden flush door/ panel/gauge shutters for door & window	Century Plyboard (India)Ltd. (Brand-Century Doors), Greenply Industries Ltd., Green panel Industries Ltd, Jain Doors Pvt Ltd, Punjab Plywood industries (Brand Black Cobra) Yamuna Nagar, A-1Teak Products PvtLtd

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	Steel windows, Ventilators, Door frames Shutters, Z Section steel windows. (Fabricators)	Trisul Industries Meerut, Chandni Industries Gaziabad, Ashwani & Sons,
	Aluminum shutters, frames for doors, windows& ventilators. (Fabricators)	Trisul Industries Meerut, Chandni Industries Ghaziabad, Deco Grill , sterlite
4	Ply/veneers, laminates/false ceilings/particle boards/wooden board for wooden flooring etc.	
	UPVC/PVC False Ceiling	Rajshri Plastiwood, Accura Polytech Pvt Ltd, Kumar Arch Tech Pvt Ltd (Brand - Ecocell)
	Metal false ceiling	Manraj Ceiling Products Pvt Ltd (Brand –Rayal Kraft), Armstrong
	Gypsum Board False Ceilling/Gypsum Plaster	Everest Industries Ltd, Gyproc, Armstrong, USG Boral, JK Cement Ltd
	Non-Asbestos Fiber Cement Board (Confirming to IS: 14862/ 2000) False Ceiling	Everest Industries Ltd, Shera, Kalsi, Prima, Dura green
	Glass Reinforced Gypsum (GRG) False Ceiling	Gyproc, Armstrong, USG Boral
5	Wall Paneling/ Partitions/ Ply Board/ MDF & HOF Boards	
	Plain Particle Board/ Particle board ceiling, Pre-laminated Particle Board	Green panel Industries Ltd, Century Ply, Green ply, Kitply, Panjab Plywood industries (Brand-Black Cobra) Yamuna Nagar
	Aluminum Composite Panels (ACP)	Aludecor, Alstrong, Alucobond, Alstone, Viva, Virgo Laminates Ltd, Alexia Panels Dehradun (Brand – Alex Panels)
	Laminates/Veneers	Merino Laminates, Royale Touche Laminates, Greenlam Laminates, Century Laminates, Kitply, Sunmica, Formica, Green panel Industries Ltd, Virgo Laminates Ltd
	Plywood for general purpose	Century Ply, Green ply, Kitply, Green panel Industries Ltd, Panjab Plywood industries (Brand-Black Cobra) Yamuna Nagar
6	Sanitary fittings/ fixtures	
	CP Bath fittings & accessories	Jaquar & Co Pvt Ltd (Brand - Jaquar), Somany Ceramics Ltd, CERA Sanitary ware Ltd, Asian Paints Ltd (Brand-Royale, Essess & Bathsense), Jain Enterprises (Brand –Jaynam), Sparsh Pearl
	PTMT Taps, bath fittings & accessories	Jaynam, Sparsh Pearl, R S Industries Delhi Bluestar Sanitary Industries (Brand- Silvershine)
	Stop Valves/Gate Valves/ Bib Taps (Brass)	Sparsh Pearl, Somany Ceramics Ltd Dhawan Sanitary Udyog New Delhi (Brand - Prima), K S Engg Works Gaziabad, Blue Star Sanitary Industries Pvt Ltd.
	WHB/WC/Urinal	Somany Ceramics Ltd, Jaquar & CoPvt Ltd (Brand - Jaquar), CERA Sanitary ware Ltd, Asian Paints Ltd, (Brand –Royale Essess & Bath sense), Jaquar & CoPvt Ltd (Brand - Essco)
	Metro pole & Concealed Flushing Cistern	Jaquar, Grohe, Kohler, Somany Ceramics Ltd Sparsh Pearl.

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	Seat Cover for EWC	Jaquar, Grohe, Kohler, CERA Sanitary ware Ltd, Asian Paints Ltd, (Brand –Royale Essess & Bath sense) Sparsh Pearl, Somany Ceramics Ltd
	PVC Flushing Cistern (IS-774/1984)	CERA Sanitary ware Ltd, Parryware, Silvershine Jaynam, Sparsh Pearl, Somany Ceramics Ltd
	Stainless Steel Sinks/Plate Rack	Johnson, Nirali, Hindware, Blustar Sanitary Industries Pvt Ltd (Brand- Silvershine), Plastocraft Sanitary India Pvt Ltd. , Jaynam
7	Pipes	
	CPVC Pipes & fittings	Astral Ltd (Brand-CPVC Pro), Ashirvad, Supreme, Sentini Flopipes India Pvt Ltd, Surya Roshni Ltd, Savour Fair manufacturing Co Pvt Ltd, Ganesh Gauri Industries
	UPVC Pipes & fittings	Astral Ltd (Brand-Aqua Safe & Aquarius), Ashirvad, Supreme, Sentini Flopipes India Pvt Ltd, Surya Roshni Ltd, Savour Fair manufacturing Co Pvt Ltd, Ganesh Gauri Industries
	HDPE Pipe/MDPE Pipe/PVC Pipe	Astral Ltd (Brand- HDPE Pro, Finolex, Venkatesh Indigenous Pipes Pvt Ltd (Brand – VIP), Ganesh Gauri Industries
	PPR Pipes & fittings	Prince, Finolex, Supreme, SFMC, KPT
	ERW MS Pipe	Surya Roshni Ltd, Tata, Jindal
	Centrifugally cast (Spun) iron S/S soil, waste, ventilating & rain water pipes, fittings & accessories confirming to IS : 3989/2009	Singhal Iron Foundry Pvt Ltd (Brand-SKF)
	Hub less centrifugally cast (Spun) iron Pipes, fittings & accessories confirming to IS : 15905/2011	Singhal Iron Foundry Pvt Ltd, Mathura
	SWR pipes as per IS 13592 for pipes & IS-14735 for fittings	Astral Ltd (Brand- Drain Master, Supreme, Sentini Flopipes India Pvt Ltd, Surya Roshni Ltd, Savour Fair Manufacturing Co Pvt Ltd, Ganesh Gauri Industries
	PVC UG drainage pipe as per IS : 16098 Part-1	Astral Ltd (Brand-Foam core) ,Prince, Finolex
	DWC UG drainage pipes as per IS:16098 Part-2	Astral Ltd (Brand-D-rex), Prince, Finolex
	PP low noise SWR Pipe	Astral Ltd (Brand- Astral Silencio)
	DI Pipe & fittings	Tata, Electro steel, Jindal Saw Ltd
	GI Pipes	Tata, Jindal Hisar,
8	Lifts Elevators	
	Lifts/Elevators	Otis Elevator Company India Ltd, Kone India, Schindler India Pvt Ltd, Hitachi Lifts India Pvt Ltd

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9	DG Sets	
	Engine	Kirloskar, Cummins, Greeves Cotton, Caterpillar, Eicher Engines
	Alternator	Kirloskar, Stemford, Crompton, Jyoti
	DG Set assembled with sound proof canopy	Kirloskar, Cummins, Greeves Cotton, Sudhir
10	Crane	
	(a) Single/Double Girder EOT Cranes (b) Goliath Cranes (c) Jib Cranes	Safex Energy Pvt Ltd, Pioneer Cranes & Elevators Pvt Ltd, Hercules Hoists Ltd
11	Transformers	
	Compact Sub Station	Seimens, Schneider, Crompton, ABB, Indian Transformers & Electricals Pvt Ltd, (Brand – ITE Gurgaon)
	Transformers 11/ 0.433KV upto 500 KVA copper wound	Crompton, Schneider, Indian Transformers & Electricals Pvt Ltd (Brand - ITE Gurgaon), Mahendra Engineering Ltd.
	Distribution Transformer Copper Wound upto and incl 33 KV, 2500 KVA	Guru Teg Bahadur Metal Works, Esennar Transformers Pvt Ltd.
	Transformers step down 11/0.433KV above 500 KVA copper wound	Crompton, Schneider, Indian Transformers & Electricals Pvt Ltd (Brand - ITE Gurgaon)
	Transformers step down 33/11KV upto 10MVA & 33/0.433 KV copper wound	Crompton, Schneider, Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)
	HT CT & PT	Schneider, Crompton Greeves, ABB, Seimens
12	Central AC Plants	
	Chiller for Central AC Plant	Voltas, Carrier, Bluster
	Cooling Coil/AHU	Voltas, Bluster, Zeco, Accel, Carrier
	Cooling Tower	Paharpur, Miir, Delta, Advance
	Expansion Valves/Thermostat/Humidistat	Danfoss, Sporlon, Indfoss, Honeywell
	Package type AC (Compressor/ Condenser/Cooling Coil)	Bluestar, Hitachi, Daikin, Voltas
	Window/ Split ACs	O General, Mitsubishi Electric, Hitachi, Daikin, Voltas, Carrier
13	HT Circuit Breakers	
	VCB/SF6 Circuit Breakers/HT Switchgear Panels up to 36 KV	Schneider, ABB, Siemens, Crompton, Indian Transformers & Electricals Pvt Ltd (Brand-ITE Gurgaon)
	HT Ring main Units, (RMU) upto 36 KV	Schneider, ABB, Siemens, Crompton
	Automatic Power Factor Correction Panels	
	APFC Panel (Fabricators)	Indo Asian , Shalabh (India) Industries Roorkee, Power Systems & Control Jalandhar, REW Constructions Pvt Ltd, Prestige Switchgears Pvt. Ltd.
	Power Factor Improvement Capacitor	L&T MDXL Ultra, Schneider, Siemens, ABB, Neptune, Havells
	APFC Relay	L&T, Schneider, Ducati

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

14	HT Cables	
	HT XLPE cable upto 33KV as per relevant IS	Polycab Wires PvtLtd., Gloster Cables Ltd, KEI Industries Ltd, Diamond Power Infrastructure Ltd
15	Switch, Socket, Fan Regulator, Lamp holder, Ceiling rose, Bell push, Call Bell	
	Modular Switch/ Socket/Fan regulator/ Bell push & mounting accessories	Legrand, Anchor Electrical Pvt Ltd, Pressfit Pipes and Profiles, BENLO
	Non modular Switch/ Socket/ Fan regulator/ Lamp holder/ Ceiling rose/ Bell push/ Call bell (Polycarbonate body only)	SSK, Anchor penta PC, Pressfit Pipes and Profiles, HPL Electric & Power Ltd, Eveready
16	Fans	
	Ceiling fans ISI marked & not less than 4 BEE Star rating	Crompton, Orient, Khaitan, Usha, Bajaj, HPL Electric & Power Ltd.
	BLDC Ceiling fan	Atomberg, Havells, Bajaj, Crompton, Orient, Khaitan
	Air Circulator I Wall mounted fan/Cabin fan/ Fresh air fan	Havells, Bajaj, Crompton, Orient, Usha, Khaitan
	Exhaust fans	Crompton, Orient, Khaitan, Usha, Bajaj, HPL Electric & Power Ltd.
17	Lights and light fittings	
	LED light fittings for indoor/outdoor use LED Lamp	Crompton, Orient, Bajaj, Polycab India Ltd, Mepro Electrical Aligarh, HPL Electric & Power Ltd, Eveready, BENLO
	Solar based LED street light fittings	Philips, Bajaj, Crompton, BENLO
18	Fire Protection, Fire Alarm & medical gas networks	
	Fire Alarm system with PA System including fire control panels	ASES Security Pvt Ltd
	Fire rated smoke check doors	ASES Security Pvt Ltd
19	Voltage Stabilizer Servo Type	
20	Wire & Cables	
	PVC insulated 1100 V grade ISI marked FRLS internal wiring cables with copper conductor	Seewel, BENLO, Havel's, Polycab, Anchor, Grandlay, Shalabh (India) Industries Roorkee, SCI,
	LT Cable UG(XLPE/PVC)Heavy duty armored/ unarmoured upto 1100V Aluminum & Copper conductor solid/stranded as per IS 1554 Ptl&7098/ 1988 Ptl	Seewel, BENLO, Havells, Polycab, Gloster, Grandlay, SCI, Shalabh (India) Industries Roorkee,
21	Change Over Switch /Switch Fuse	
	Change Over Switch /Switch Fuse	Indo Asian, HPL, Crystal Electric Co Pvt Ltd, Shalabh (India) Industries Roorkee, BENLO

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

22	LT Switchgears	
	MCBs & MCB DBs	Indo Asian, HPL, Crystal Electric Co Pvt Ltd, Shalabh (India) Industries Roorkee, BENLO
	ACB	Indo Asian, Havells, HPL(Fixed type 400A to 2000A only)
	All types of LT Panels, Feeder pillars & LT Control Panels	
	LT Panels, Feeder Pillars & LT Control Panels	Shalabh (India) Industries Roorkee, Indo Asian, Power Systems & Control Jalandhar, REW Constructions Pvt Ltd, Prestige Switchgears Pvt. Ltd.
23	Voltage Stabilizers	
	Voltage Stabilizers upto 5 KVA	Chime Electronics Delhi (Brand- Sinetrac), Vintek Electronics New Delhi (Brand -Volina), Vintec Electronic Laboratory New Delhi
	Voltage Stabilizers above 5 KVA	ITE Gurgaon, Chime Electronics Delhi (Brand- Sinetrac), Vintek Electronics New Delhi (Brand- Volina), Vintec Electronic Laboratory New Delhi
24	Air Insulated Trunk Bus Bars /Bus Ducts/ Bus Bar Trunking Systems	
	HT & LT Bus ducts	Nitya Electro controls Pvt Ltd, United Engg Services Kanpur, Sparsh Engg Corpn Lucknow
25	Jointing Kits for Power Cables	
	Heat and Cold Shrinkable Cable jointing kits 1.1KV to 66 KV	Hari Consolidated Pvt Ltd (Brand - Cabseal), M-Seal, Raychem
26	Misc E/M Items	
	Hollow Section Pipe	Surya Roshni Ltd
	HRC Fuse	BENLO
	Smart/Temper proof Energy Meter single phase/Three phase	BENLO
27	Misc B/R Items	
	White Cement and Wall Putty	JK Cement Ltd (Brand- JK White), Birla, Nerolac, Asian, Berger
	Rotational moulded Water Storage Tanks	JS Polyplast

(Signature of Contractor)
Dated.....

AGE (Contracts)
For Accepting Office

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

APPENDIX 'C'**SOURCES OF MATERIALS**

Sl No	Name of Materials	Source of materials for works at station
1	2	3
1.	Bricks and brick tiles	As specified here-in-before
2.	(a) Coarse aggregate for all RCC.	Crushed stone from Haldwani, river boulder/River single as specified
	(b) ~do~ for all PCC work	--- do ----
3.	Coarse sand for RCC/PCC lime concrete, DPC and seal coat.	Haldwani River coarse sand
4.	Sand for plastering, pointing, brick work, sand for cushioning filling and covering for LT cables.	Haldwani River sand
5.	Broken/ crushed stone/ boulders for hard core and road work, stone chipping for premix carpeting.	Haldwani River boulders/broken boulders stones/crushed stones
6.	Stone for WBM and Sub base / soling, screening and Premix carpet	---do----

NOTE :

1. Sources indicated are for guidance only. The tenderer shall ascertain the actual position/exact location of source before submitting his tender and no additional payment shall be made on account of misunderstanding or its distance from site of works.
2. The tenderer shall amongst other things also ascertain all information such as royalties, taxes, duties and other charges etc, on the materials and no additional payment shall be made on account of the foregoing.
3. However, if due to any ban imposed by the State Government/any other authority on obtaining the materials from the sources specified hereinabove and /or specified materials is not available at the source mentioned above, these materials conforming to specifications as given in particular specifications will be obtained by the contractor from any other place/source without any extra cost to the Government.

SIGNATURE OF CONTRACTOR
DATED : _____

AGE (Contracts)
for Accepting Officer

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

APPENDIX 'D'**MATERIALS AND TESTS**

S No	Material	Tests	Method of Test-ing	Frequency of tests	Level of test	Rate	Remarks
1	2	3	4	5	6	7	8
1.	Brick	(i) Compressive strength	IS-3495 (Pt-I)	As per IS:5454 as given under :- <u>Lot</u> Permissible <u>Size</u> <u>Sample</u> Nos of defective bricks	A	330/-	Checks for visual and dimensional characteristics shall also be carried out as per IS : 5454.
		(ii) Water absorption	-do-		A	330/-	
		(iii) Efflorescence	-do-	1001 to 10000 5 0 10001 to 35000 10 0 35001 to 50000 15 1	A	330/-	LEGEND A-Site lab B- CTL/Zonal lab. C- National test House/SEMT Wing/Engg College
2.	Coarse aggregate	(i) Sieve analysis	IS-2386 (Pt-I)	One test for every 15 cub metre of aggregates or part thereof brought at site.	A	660/-	
		(ii) Flakiness Index	-do-	-do-	A	250/-	
		(iii) Estimation of deleterious material	-do-	One test for every 100 cub metre of aggregate or part thereof.	A	600/-	
		(iv) Organic impurities	-do-	One test per source of supply	C	275/-	
		(v) Moisture content.	IS-2386 (Pt-II)	Regularly as reqd.	A	330/-	
		(vi) Specific gravity	-do-	One test for each source of supply	B	330/-	
3.	Fine aggregate	(i) Sieve analysis	IS-2386 (Pt-I)	One test for every 100 Cub metre of aggregates or part thereof when brought to site.	A	660/-	
		(ii) Test for clay, silt and impurities	-do-	-do-	A	500/-	

(Contd...)

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

APPENDIX 'D'**MATERIALS AND TESTS**

1	2	3	4	5	6	7	8
		(iii) Specific gravity	-do-	One for each source of supply.	B	330/-	
		(iv) Moisture content	-do- (Pt-II)	Regularly as required subject to 2 tests/day when being used.	A	330/-	
		(v) Test for organic impurities	-do- (Pt-I)	One test for each source of supply.	C	275/-	
4.	Structural concrete (M-15 grade & above	(i) Slump test or compacting factor or VEE-BEE time.	IS-1199	The minimum frequency of sampling of concrete of each grade shall be as under:- Qty of concrete No of in the work (Cub M) sample.	A	300/-	(i) Random sample shall be carried out to cover mixing units.
		(ii) Compressive Strength	IS-516	1-5, 6-15 16-30 31-50 51 and Above 1 2 3 4 4 + 1 for each addl 50 cub M part thereof	A	90 0/-	(ii) Refer IS-456-1978 cl 14.2 for frequency of sampling.
5.	(a) PCC block walling hollow block.	(i) Compressive strength	IS: 2185-1997 (Appx 'B')	8 Block of 14.	A	900/-	Samples-14 Blocks consignment of every 5000 blocks or part thereof.
		(ii) Water absorption	-do- (Appx 'E')	3 blocks out of 14.	B	330/-	
		(iii) Density	-do- (Appx 'A')	3 block out of 14.	B	330/-	

(Contd...)

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

APPENDIX 'D'**MATERIALS AND TESTS**

1	2	3	4	5	6	7	8
	(b) PCC solid block for walling.	(i) Compressive Strength	IS: 2185	12 Blocks of 18.	A	900/-	Samples-18 Blocks consignment of every 01 blocks shall be checked.
		(ii) Water absorption	-do-	3 block out of 18.	B	330/-	
		(iii) Density	-do-	3 block out of 18.	B	330/-	
6.	Cement flooring tiles/ Terra-zzo tiles	(i) Water absorption	IS: 1237-1980 (Appx 'D')	6 Tiles out of 18.	B	330/-	Samples : 18 tiles from each source of supply selected.
		(ii) Wet transverse strength.	-do-	-do	B	660/-* At Random	
		(ii) Resistance to wear.	-do-	-do-	C	1000/-	
7.	Burnt clay roofing tiles (hand made IS-2690 (Pt-II)).	(i) Water absorption	IS-3495 (Part-II)	6 tiles out of 12.	B	216/-	SAMPLES : 12 tiles from each source of supply selected at random.
	LENGTH 150mm to 250mm	(ii) Compressive strength	-do-(Part-I)	-do-	A	180/-	
	WIDTH 100mm to 200mm						
	THICKNESS 35mm to 50mm						

(Contd...)

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

APPENDIX 'D'**MATERIALS AND TESTS**

1	2	3	4	5	6	7	8
8.	Mangalore pattern roofing tiles.	(i) Water absorption	IS:654 (Appx 'A')	6 tiles out of 32	B	180/-	SAMPLES : 32 tiles from each consignment of 3000 tiles or part thereof these tiles shall be checked for dimensions and weight.
		(ii) Breaking load.	-do- (Appx 'C')	-do-	B	120/-	
9.	Timber	(i) Specific gravity.	IS: 1708-1960	Minimum 3 samples from A lot of 4 cum or 250 pieces of seasoned timber.	B	120/-	
		(ii) Moisture content.	-do-		A	120/-	
10.	Water for constn. Purposes.	(i) Test for acidity	IS-456 & 3025	Once at the stage of approval of source of water	B	500/-	
		(ii) Test of alkalinity.	-do-		B	500/-	
		(iii) Test for total dissolved solid (TDS) content.	-do-		C	500/-	
11.	Welding of steel work.	Visual inspection test.	IS-822-1970 clause 7.1	100 visual inspection.	Work Site.	360/-	Specialised test, their method and frequency to be decided on consideration of their importance by the Accepting Officer.

(Contd...)

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

APPENDIX 'D'**MATERIALS AND TESTS**

1	2	3	4	5	6	7	8																
12.	Timber paneled glazed door window shutters (include-ing factory made shutters)	(a) Dimensions sizes workmanship and finish	IS: 1003-1977 (Pt-I)	Frequency of sampling From each lot shall be as under :- <table><tr><td><u>Lot Size</u></td><td><u>Sample Size</u></td></tr><tr><td>26 to 50</td><td>5</td></tr><tr><td>51 to 100</td><td>8</td></tr><tr><td>101 to 150</td><td>13</td></tr><tr><td>151 to 300</td><td>20</td></tr><tr><td>301 to 500</td><td>32</td></tr><tr><td>501 to 1000</td><td>50</td></tr><tr><td>1001 to above</td><td>80</td></tr></table>	<u>Lot Size</u>	<u>Sample Size</u>	26 to 50	5	51 to 100	8	101 to 150	13	151 to 300	20	301 to 500	32	501 to 1000	50	1001 to above	80	A	180/-	
<u>Lot Size</u>	<u>Sample Size</u>																						
26 to 50	5																						
51 to 100	8																						
101 to 150	13																						
151 to 300	20																						
301 to 500	32																						
501 to 1000	50																						
1001 to above	80																						
		(b) STRENGTH TEST																					
		(i) Slamming	IS: 3034-1990	From each 5% lot of the factory made shutter shall be tested for strength test.																			
		(ii) Impact Indentation	-do-																				
		(iii) Shock resistance	-do-																				
		(iv) Edge lodging.	-do-																				
13.	Ply wood (IS:303-1989)	(a) Moisture content	IS: 1734-1983 (Part-I)	Six test pieces cut from each of the boards selected as per table I shall be subjected to tests.	B	240/-	Sampling shall be as per IS:7635-1975 Table 2.																
		(b) Water resistance test.	-do- (Part-VI)		C	60/-																	

(Contd...)

**PARTICULARS SPECIFICATIONS (CONTD...)
(MATERIALS AND WORKMANSHIP)**

<u>MATERIALS AND TESTS</u>							
1	2	3	4	5	6	7	8
14.	Wood particle board (medium) density IS:3087-1985	(a) Density	IS-2380-1977 (Part-III)	Three test specimens from each sample (size 150mm x 75mm).	A	60/-	Sampling shall be as per IS:3087-83 clause 2 with moisture meter.
		(b) Moisture content.	-do-	-do-	A&B	60/-	
		(c) Water absorption.	(Part-XVI)	-do- (Size 300mmx30mm)	A	60/-	
		(d) Swelling due to surface absorption.	-do- (Part-XVII)	-do- (Size 125mm x 100mm)	A	60/-	
		(e) Swelling in water.	-do-	(Size 200mm x 100mm)	A	60/-	
		(f) Modulus of rupture	-do- (Part-IV)	Three test specimens as per IS:2380-1977	B	90/-	
		(g) Screw with drawl strength.	-do- (Part-XIV)	-do- as per IS:2385	C	120/-	
15	Cement	Setting time			B	500/-	
		Soundness			B	550/-	
		Compressive strength			B	550/-	
		Fineness			B	275/-	
16	Reinforce-ment steel	Physical test up to 16mm dia(Nominal mass, tensile elongation, bend & rebend			B	2500/-	
		More than 16mm dia			B	2750/-	

Signature of Contractor
Dated:

AGE (Contracts)
for Accepting Officer

GUARANTEE BOND**(To be used by approved Scheduled Banks)**

1. In consideration of the President of India (hereinafter called "the Government") having agreed to exempt _____ (hereinafter called "the said Contractor(s)") from the demand, under the terms and conditions of an Agreement dated _____ made between _____ and _____ for _____ (hereinafter called "the said Agreement"), of performance security for the due fulfillment by the said Contractor(s) of the terms and conditions contained in the said Agreement, on production of a Bank Guarantee for Rs _____ (Rupees _____ only), we, _____ Bank Ltd (hereinafter referred to as "the Bank") do hereby undertake to pay to the Government an amount not exceeding Rs _____ against any loss or damage caused to or suffered or would be caused to or suffered by the Government by reason of any breach by the said Contractor(s) of any of the terms or conditions contained in the said Agreement.
2. We _____ Bank Ltd do hereby undertake to pay the amounts due and payable under this guarantee without any demur, merely on a demand from the Government stating that the amount claimed is due by way of loss or damage caused to or would be caused to or suffered by the Government by reason of any breach by the said Contractor (s) of any of the terms or conditions contained in the said Agreement or by reason of the contractor's (s) failure to perform the said Agreement. Any such demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this guarantee. However, our liability under this guarantee shall be restricted to an amount not exceeding Rs _____.
3. We _____ Bank Ltd further agree that guarantee herein contained shall remain in full force and effect during the period that would be taken for the performance of the said Agreement and that it shall continue to be enforceable till all the dues of the Government under or by virtue of the said Agreement have been fully paid and its claims satisfied or discharged or till _____ (Office/Department), Ministry of _____ certifies that the terms and conditions of the said Agreement have been fully and properly carried out by the said Contractor(s) and accordingly discharges the guarantee. Unless a demand or claim under this guarantee is made to us in writing on or before the _____ we shall be discharged from all liability under this guarantee thereafter.
4. We _____ Bank Ltd further agree with the Government that the Government shall have the fullest liberty without our consent and without affecting in any manner our obligations hereunder to vary any of the terms and conditions of the said Agreement or to extend time of performance by the said Contractor(s) from time to time or to postpone for any time or from time to time any of the powers exercisable by the Government against the said Contractor(s) and to forbear or enforce any of the terms and conditions relating to the said agreement and we shall not be relieved from our liability by reason of any such variation, or extension being granted to the said Contractor (s) or for any forbearance, act or omission on the part of the Government or any indulgence by the Government to the said Contractor (s) or any such matter or thing whatsoever which under the law relating to sureties would but for this provision have effect of so relieving us.
5. We _____ Bank Ltd lastly under take not to revoke this guarantee during its currency except with previous consent of the Government in writing.

Dated the _____ day of _____ 2026 for _____ Bank Ltd.